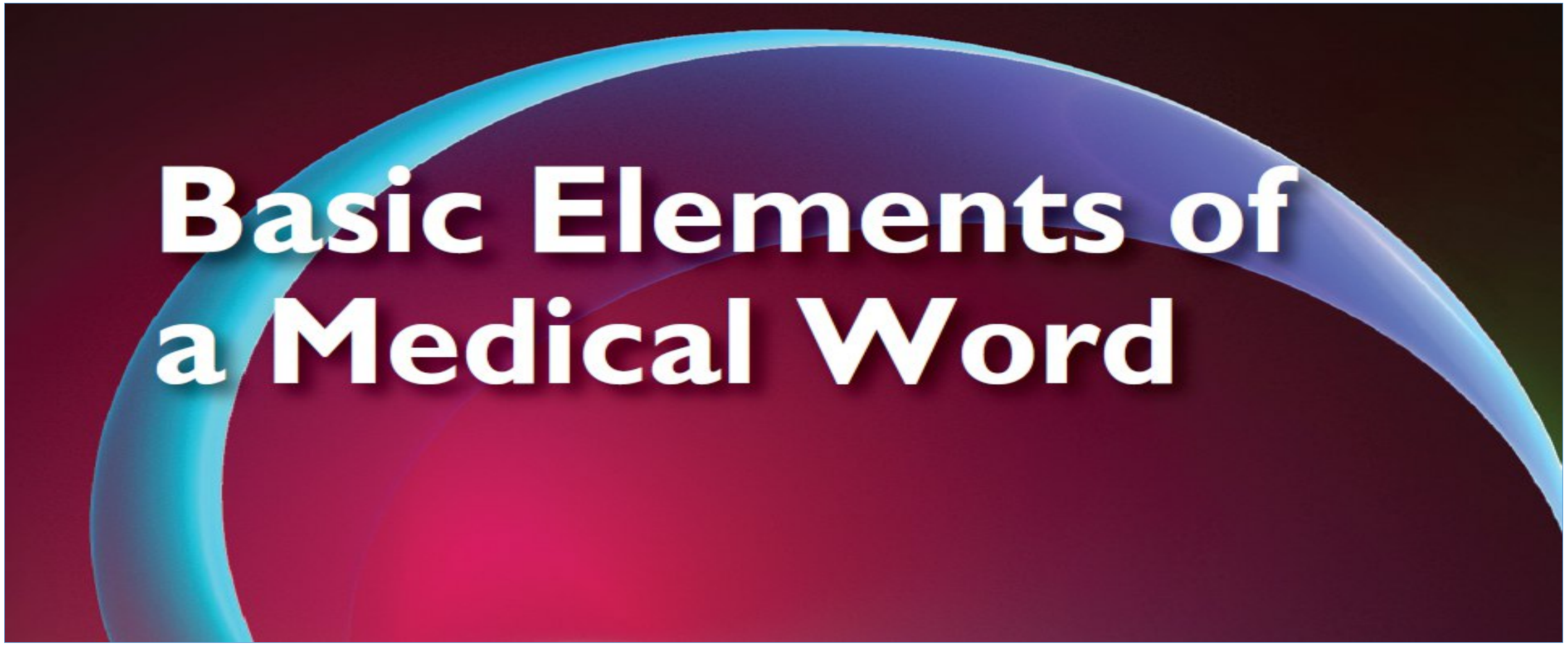

Medical Terminology Systems
A Body Systems Approach
8TH EDITION
Barbara A. Gylys, MEd, CMA-A (AAMA)
Professor Emerita
University of Toledo
Toledo, OH
Mary Ellen Wedding, MEd, MT(ASCP), CMA (AAMA), CPC (AAPC)
Professor Emerita
University of Toledo
Toledo, OH

CHAPTER1



Chapter Outline

- Objectives
- Medical Word Elements
  - Word Roots
  - Combining Forms
  - Suffixes
  - Prefixes
- Basic Guidelines
  - Defining Medical Words
  - Building Medical Words
- Medical Word Building Summary

Objectives

Upon completion of this chapter, you will be able to:

- Identify the four word elements used to build medical words.
- Divide medical words into their component parts.
- Apply the basic rules to define and build medical words.
- Pronounce medical terms presented in this chapter.
- Demonstrate your knowledge of this chapter by completing the learning activities.

Medical Word Elements

The language of medicine is a specialized vocabulary used by health-care providers.

Many current medical word elements originated as early as the 4th century B.C. when Hippocrates practiced medicine.

With technological and scientific advancements in medicine, new terms have evolved to reflect these innovations.

For example, radiographic terms, such as magnetic resonance imaging (MRI) and ultrasound (US), are now commonly used to describe current diagnostic procedures.

A medical word consists of some or all of the following elements:

- Word root (WR)
- Combining form (CF)
- Suffix (S)

Medical Word Elements (cont.)

How these elements are combined and whether all or some of them are present in a medical term determines the meaning of

a word.

To understand the meaning of medical words, it is important to learn how to divide them into their basic elements.

The purpose of this chapter is to cover the basic principles of medical word building and learn how to pronounce the terms correctly. Thus, pronunciations of medical terms are provided throughout the lectures.

Word Roots (WR)

A word root (WR)

It is the foundation of a medical term and contains its primary meaning. All medical terms have at least one word root. Most word roots are derived from the Greek or Latin language; thus, two different roots may have the same meaning.

For example, the Greek word dermatos and the Latin word cutane both refer to the skin.

(See Table 1-1)

Table 1-1 Examples of Word Roots

This table lists examples of English terms with their Greek and Latin origins as well as word analyses of corresponding medical terms. Phonetic pronunciations are provided to help you practice pronouncing the medical terms.

English Term	Greek or Latin Term*	Word Root	Word Analysis
skin	dermatos (Gr)	dermat	**dermat/**itis (dĕr-mă-TĪ-tĭs): inflammation of the skin
			Dermatitis *is a general term used to describe an inflammatory condition of the skin.*
	cutis (L)	cutane	**cutane/**ous (kū-TĀ-nē-ŭs): pertaining to the skin
			Cutaneous *is a term that identifies an anatomical structure.*
kidney	nephros (Gr)	nephr	**nephr/**oma (nĕ-FRŌ-mă): tumor of the kidney
			Nephroma *is a tumor (benign or malignant) of kidney tissue.*
	renes (L)	ren	**ren/**al (RĒ-năl): pertains to the kidney
			Renal *is a term that identifies an anatomical structure.*

Combining Forms (CF)

A combining form (CF)

It is created when a word root is combined with a vowel. The vowel, known as a combining vowel, is usually an o but is sometimes an i or e.

The combining vowel has no meaning of its own but enables the connection of two or more word elements.

Like a word root, a combining form is the basic foundation to which other word elements are added to build a complete medical word.

In this text, a combining form will be listed as word root/vowel (such as gastr/o), as illustrated in Table 1-2.

Table 1-1 Examples of Word Roots—cont'd

English Term	Greek or Latin Term*	Word Root	Word Analysis
mouth	stomatos (Gr)	stomat	**stomat**/itis (stō-mă-TĪ-tĭs): inflammation of the mouth *The word root* stomat *is commonly confused with the English term* stomach. *However,* stomat *is derived from the Greek word for* mouth. *The word root for the stomach is* gastr, *derived from the Greek word* gastros.
	oris (L)	or	**or**/al (OR-ăl): pertaining to the mouth Oral *is a term that identifies an anatomical structure.*

*It is not important to know the origin of a medical word. This information is provided here to clarify and illustrate that there may be two different word roots for a single term.

Suffixes (S)

A suffix (S)

It is a word element placed at the end of a medical word that alters its meaning.

All medical terms have a suffix. In the terms pancreat/itis (inflammation of the pancreas) and pancreat/o/pathy (disease of the pancreas), the suffixes are -itis (inflammation) and -pathy (disease).

In medical terminology, a suffix usually describes a pathology (disease or abnormality), symptom, surgical or diagnostic procedure, or part of speech.

To link a suffix that begins with a vowel, use a word root.

To link a suffix that begins with a consonant, use a combining form.

(Review Table 1-3, which illustrates this principle.)

Table 1-3 Examples of Suffixes

This table lists examples of pathological suffixes linked with a word root (WR) and a combining form (CF). Phonetic pronunciations are provided to help you practice pronouncing the medical terms.

Suffix	=	Medical Word	Meaning	Rationale
-itis (inflammation)	=	gastr/**itis** găs-TRĪ-tĭs	inflammation of the stomach*	The suffix *-itis* begins with a vowel and requires a WR.
-megaly (enlargement)	=	gastr/o/**megaly** găs-trō-MĔG-ă-lē	enlargement of the stomach	The suffix *-megaly* begins with a consonant and requires a CF.
-oma (tumor)	=	hepat/**oma** hĕp-ă-TŌ-mă	tumor of the liver	The suffix *-oma* begins with a vowel and requires a WR.
-cyte (cell)	=	hepat/o/**cytes** HĔP-ă-tō-sīts	cells in the liver	The suffix *-cytes* begins with a consonant and requires a CF.

* To define a medical term, first define the suffix and then the first part of the word.

Prefixes (P)

A prefix (P)

It is a word element attached to the beginning of a medical word or word root. However, not all medical terms have a prefix.

Adding or changing a prefix changes the meaning of the word.

Prefixes usually indicate a number (measurement), time, position, direction, or negation.

Many of the same prefixes used in medical terminology are also used in the English language.

(Review Table 1-4 to reinforce the principles of linking a prefix to other word elements.)

Table 1-4 Examples of Prefixes

This table lists examples of prefixes linked to a word root and a suffix. Note that the suffixes begin with a vowel and are linked to a word root. Phonetic pronunciations of the constructed medical words are provided to help you practice pronouncing the medical terms.

Prefix	+	Word Root	+	Suffix	=	Medical Word	Meaning
an- (without, not)	+	**esthes** (feeling)	+	**-ia** (condition)	=	**an/esthes/ia** ăn-ĕs-THĒ-zē-ă	condition of not feeling
hyper- (excessive, above normal)	+	**therm** (heat)	+	**-ia** (condition)	=	**hyper/therm/ia** hī-pĕr-THĔR-mē-ă	condition of excessive heat
intra- (in, within)	+	**muscul** (muscle)	+	**-ar** (pertaining to)	=	**intra/muscul/ar** ĭn-tră-MŬS-kū-lăr	pertaining to within the muscle
para- (near, beside; beyond)	+	**nas** (nose)	+	**-al** (pertaining to)	=	**para/nas/al** păr-ă-NĀ-săl	pertaining to (area) near the nose
poly (many, much)	+	**ur** (urine)	+	**-ia** (condition)	=	**poly/ur/ia** pŏl-ē-Ū-rē-ă	condition of much urine
pre- (before)	+	**nat** (birth)	+	**-al** (pertaining to)	=	**pre/nat/al** prē-NĀ-tăl	pertaining to before birth

Basic Guidelines

Defining and building medical words are crucial skills in understanding the meaning of a medical word. Following the basic guidelines will help you develop these skills.

Defining Medical Words

Here are three steps for defining medical words using gastr/o/enter/itis as an example.

- Step 1. Define the suffix, or last part of the word. In this case, the suffix -itis means inflammation.
- Step 2. Define the first part of the word (which may be a word root, combining form, or prefix). In this case, the combining form gastr/o means stomach.
- Step 3. Define the middle parts of the word. In this case, the word root enter means small intestine.

Defining Medical Words (Cont.)

Example:

When you analyze gastr/o/enter/itis following the three steps, the meaning is:

1. inflammation (of)
2. stomach (and)
3. small intestine.

Thus, the meaning of gastr/o/enter/itis is inflammation (of) stomach (and) small intestine.

Table 1-5 further illustrates this process.

Table 1-5 Defining Gastroenteritis

This table illustrates the three steps of defining a medical word using the example gastroenteritis.

Combining Form	Middle	Suffix
gastr/o	enter/	-itis
stomach	intestine	inflammation
(step 2)	(step 3)	(step 1)

Building Medical Words

There are three basic rules for building medical words.

Rule 1:

A word root links a suffix that begins with a vowel.

Word Root	+	Suffix	=	Medical Word	Meaning
hepat (liver)	+	-itis (inflammation)	=	hepatitis hĕp-ă-TĪ-tĭs	inflammation of the liver

Building Medical Words (Cont.)

Rule 2:

A combining form (root + o) links a suffix that begins with a consonant

Combining Form	+	Suffix	=	Medical Word	Meaning
hepat/o	+	-cyte	=	hepatocyte	liver cell
(liver)		(cell)		HĔP-ă-tō-sīt	

(Cont.)

Rule 3:

A combining form links one root to another root to form a compound word. This rule holds true even if the second root begins with a vowel, as in oste/o/arthr/itis .

Keep in mind that the rules for linking multiple roots to each other are slightly

Combining Form	+	Word Root	+	Suffix	=	Medical Word	Meaning
oste/o (bone)	+	chondr (cartilage)	+	-itis (inflammation)	=	osteochondritis ŏs-tē-ō-kŏn-DRĪ-tĭs	inflammation of bone and cartilage
	+	arthr (joint)	+	-itis (inflammation)	=	osteoarthritis ŏs-tē-ō-ăr-THRĪ-tĭs	inflammation of bone and joint

It is time to review medical word elements by completing Learning Activities 1-1 and 1-2 on pages 8–9.

Medical Word Building Summary

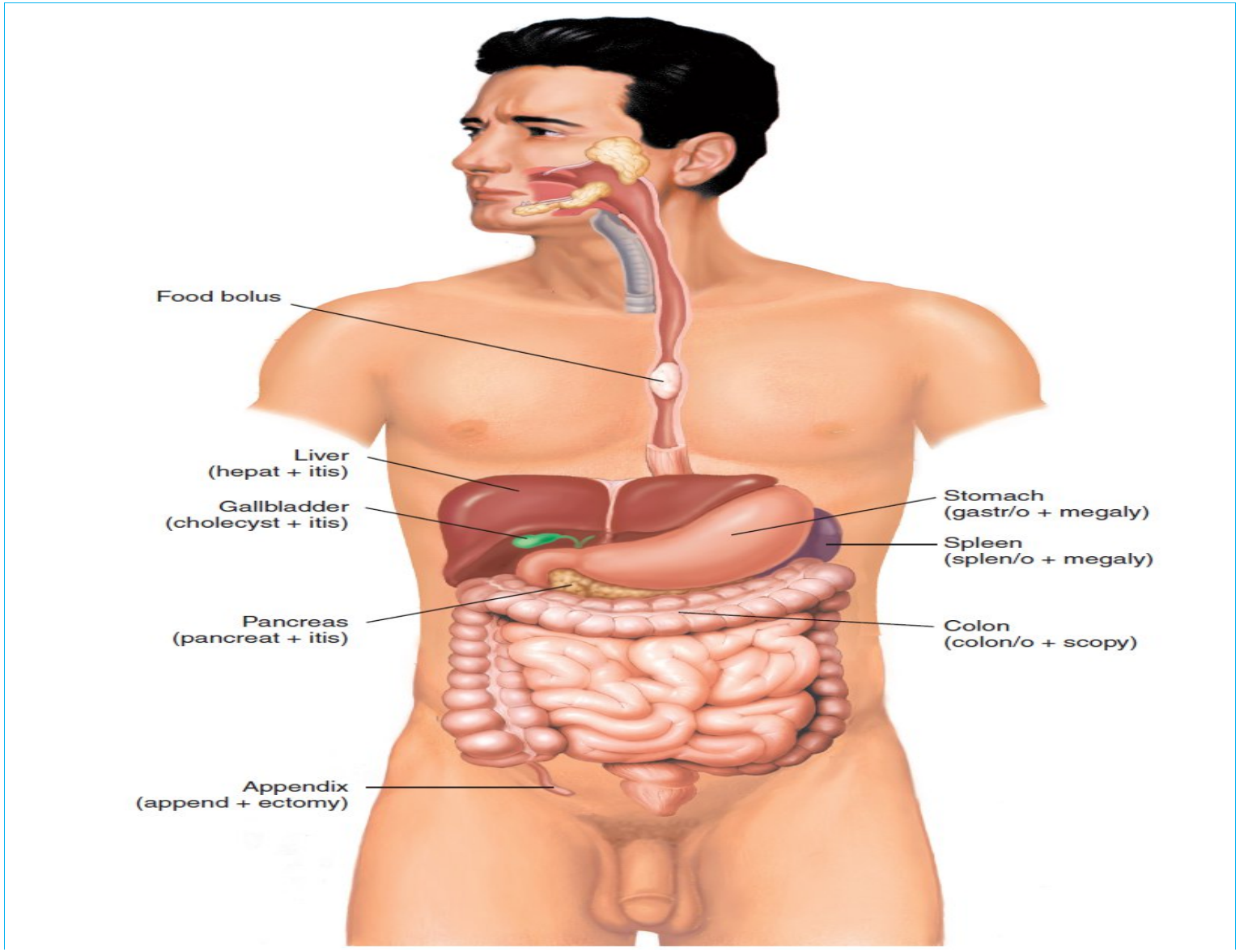
The illustration that follows demonstrates medical word building.

Building a medical word that means enlargement of the stomach requires knowledge of the word element for stomach

(gastr or gastr/o) and the suffix for enlargement (-megaly).

The medical word for enlargement of the stomach is gastromegaly.

To develop medical word building skills, study the combinations of word building elements in the digestive system illustration that



It is time to review surgical suffixes by completing Learning Activities 1-6 on page 14.

Learning Activities

The learning activities that follow provide a review of the basic medical word elements introduced in this chapter.

Complete each activity from 1-1 to 1-6 and review your answers to evaluate your understanding of the chapter 1.



CHAPTER2



Chapter Outline

- Objectives
- Suffix Linking
- Suffix Types
  - Surgical, Diagnostic, Pathological, and Related Suffixes
  - Grammatical Suffixes
  - Plural Suffixes

Objectives

Upon completion of this chapter, you will be able to:

- Identify examples of surgical, diagnostic, pathological, and related suffixes.
- Link combining forms and word roots to suffixes.
- Define and provide surgical, diagnostic, pathological, and related suffixes.
- Define and provide adjective, noun, and diminutive suffixes.
- Locate and apply guidelines for pluralizing terms.
- Pronounce medical terms presented in this chapter.
- Demonstrate your knowledge of the chapter by completing the learning activities.

Suffix Linking

In medical words, a suffix is added to the end of a word root or combining form to change its meaning.

Recall Rule 1 and Rule 2 on slides 17-18 for linking suffixes.

- When a suffix begins with a vowel, use the root word for linking the two word elements.

- When the suffix begins with a consonant, use the combining form for linking the two word elements.

For example:

the word root hemat means blood.

The suffix -emesis means vomiting

and -logy means study of.

Hemat/emesis means vomiting blood

hemat/o/logy is the study of blood.

Review Table 2-1, which illustrates examples of word roots linked with suffixes that begin with a vowel and combining forms linked with suffixes that begin with consonant.

Table 2-1 Word Roots and Combining Forms With Suffixes

This table provides examples of word roots linking a suffix that begins with a vowel. It also provides examples of combining forms (root + o) linking a suffix that begins with a consonant.

Element	+	Suffix	=	Medical Word	Meaning
Word Roots					
hemat (blood)	+	**-emesis** (vomiting)	=	**hemat/emesis** hĕm-ăt-ĔM-ĕ-sĭs	vomiting blood
arthr (joint)	+	**-itis** (inflammation)	=	**arthr/itis** ăr-THRĪ-tĭs	inflammation of a joint
oste (bone)	+	**-oma** (tumor)	=	**oste/oma** ŏs-tē-Ō-mă	tumor of bone
Combining Forms					
hemat/o (blood)	+	**-logy** (study of)	=	**hemat/o/logy** hē-mă-TŎL-ō-jē	study of blood
arthr/o (joint)	+	**-centesis** (surgical puncture)	=	**arthr/o/centesis** ăr-thrō-sĕn-TĒ-sĭs	surgical puncture of a joint
oste/o (bone)	+	**-dynia** (pain)	=	**oste/o/dynia** ŏs-tē-ō-DĬN-ē-ă	pain in bone

compound words (CW)

Medical words that contain more than one word root are known as compound words CW).

Multiple word roots within a compound word are joined together with a vowel, regardless of whether the second root begins with a vowel or a consonant.

Example:

Notice that a vowel is used in Table 2-2 between oste and arthr, even though the second root, arthr, begins with a vowel

(Keep in mind that the rule for linking multiple roots is slightly different from the rules for linking roots to suffixes. To reinforce your understanding of building compound words, refer to Rule 3 on slide 19. Use the steps when in doubt about building a medical word).

Table 2-2 Compound Words With Suffixes

This table provides examples of medical terms with more than one word root, also known as compound words. The table lists suffixes linked with roots when the suffix begins with a vowel, and it lists combining forms when the suffix begins with a consonant.

Combining Form	+	Word Root	+	Suffix	=	Medical Word	Meaning
oste/o (bone)	+	**arthr** (joint)	+	**-itis** (inflammation)	=	**oste/o/arthr/itis** ŏs-tē-ō-ăr-THRĪ-tĭs	inflammation of the bone and joint
encephal/o (brain)	+	**mening** (meninges)	+	**-itis** (inflammation)	=	**encephal/o/mening/itis** ĕn-sĕf-ă-lō-mĕn-ĭn-JĪ-tĭs	inflammation of the brain and meninges
oste/o (bone)	+	**arthr/o** (joint)	+	**-pathy** (disease)	=	**oste/o/arthr/o/pathy** ŏs-tē-ō-ăr-THRŎP-ă-thē	disease of the bone and joint
encephal/o (brain)	+	**mening/o** (meninges)	+	**-pathy** (disease)	=	**encephal/o/mening/o/pathy** ĕn-sĕf-ă-lō-mĕn-ĭn-GŎP-ă-thē	disease of the brain and meninges

Suffix Types

An effective method in mastering medical terminology is to learn the major types of suffixes in categories. Grouping the surgical, diagnostic, pathological, related, and grammatical suffixes makes them easier to remember.

** Surgical, Diagnostic, Pathological, and Related Suffixes

- Surgical suffixes describe a type of invasive procedure performed on a body part.

(See Table 2-3.)

- Diagnostic suffixes describe a procedure performed to identify the cause and nature of an illness.

- Pathological suffixes describe an abnormal condition or disease. (See Table 2-4.)

Table 2-3 Surgical Suffixes

This table lists commonly used surgical suffixes along with their meanings and word analyses.

Suffix	Meaning	Word Analysis
-centesis	surgical puncture	arthr/o/**centesis** (ăr-thrō-sĕn-TĒ-sĭs): surgical puncture of a joint *arthr/o:* joint *Arthrocentesis is a surgical puncture to remove fluid in a joint by using a needle inserted into the joint space. It may also help to obtain samples of synovial fluid for diagnostic purposes and to instill medications. (See Fig. 2-1.)*

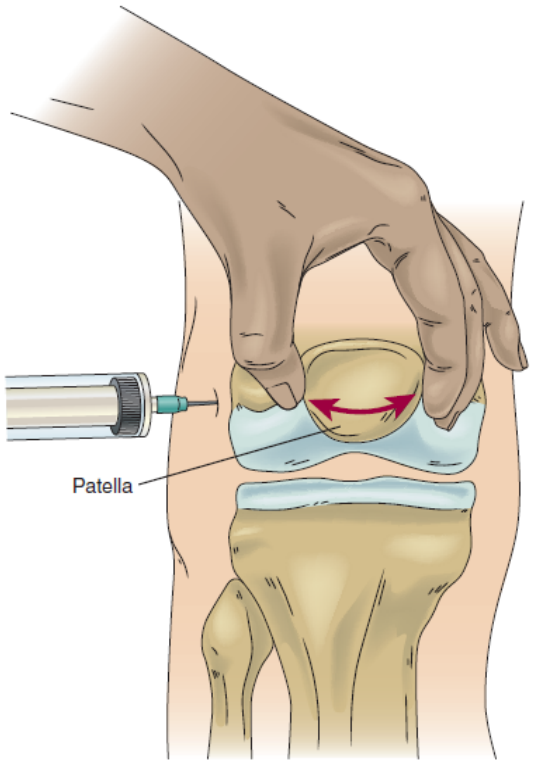


Figure 2-1 Arthrocentesis of the knee.

(continued)

Table 2-3 Surgical Suffixes—cont'd

Suffix	Meaning	Word Analysis
-clasis	to break; surgical fracture	oste/o/**clasis** (ŏs-tē-ŎK-lă-sĭs): surgical fracture of a bone *oste/o:* bone ***Osteoclasis is performed to correct a deformity of a bone.***
-desis	binding, fixation (of a bone or joint)	arthr/o/**desis** (ăr-thrō-DĒ-sĭs): binding or fixation of a joint *arthr/o:* joint ***Arthrodesis fuses bones across the joint space in a degenerated, unstable joint.***
-ectomy	excision, removal	append/**ectomy** (ăp-ĕn-DĔK-tō-mē): excision of the appendix *append:* appendix
-lysis	separation; destruction; loosening	thromb/o/**lysis** (thrŏm-BŎL-ĭ-sĭs): destruction or separation of a blood clot *thromb/o:* blood clot ***Drug therapy is usually used to dissolve a blood clot.***

Table 2-3 Surgical Suffixes—cont'd

Suffix	Meaning	Word Analysis
-pexy	fixation (of an organ)	mast/o/**pexy** (MĂS-tō-pĕks-ē): fixation of the breast(s) *mast/o:* breast ***Mastopexy, an elective surgery, affixes sagging breasts in a more elevated position, commonly improving their shape.***
-plasty	surgical repair	rhin/o/**plasty** (RĪ-nō-plăs-tē): surgical repair of the nose *rhin/o:* nose ***Rhinoplasty is a type of surgery that changes the size or shape of the nose.***
-rrhaphy	suture	my/o/**rrhaphy** (mī-OR-ă-fē): suture of a muscle *my/o:* muscle
-stomy	forming an opening (mouth)	trache/o/**stomy** (tră-kē-ŎS-tō-mē): forming an opening (mouth) into the trachea *trache/o:* trachea (windpipe) ***A tracheostomy is an artificial opening created to bypass an obstructed upper airway.***

Table 2-3 Surgical Suffixes—cont'd

Suffix	Meaning	Word Analysis
-tome	instrument to cut	oste/o/**tome** (ŎS-tē-ō-tōm): instrument to cut bone *oste/o:* bone ***An osteotome is a surgical chisel used to cut through bone.***
-tomy	incision	trache/o/**tomy** (trā-kē-ŎT-ō-mē): incision of the trachea *trache/o:* trachea (windpipe) ***Tracheotomy opens a direct airway through the neck and into the trachea (the windpipe).***
-tripsy	crushing	lith/o/**tripsy** (LĬTH-ō-trĭp-sē): crushing a stone *lith/o:* stone, calculus ***Lithotripsy is a surgical procedure to remove a stone or calculus in the kidney, ureter, bladder, or gallbladder.***



It is time to review surgical suffixes by completing Learning Activities 2-1, 2-2, and 2-3.

Table 2-4 Diagnostic, Pathological, and Related Suffixes

This table lists commonly used diagnostic, pathological, and related suffixes, along with their meanings and word analyses.

Suffix	Meaning	Word Analysis
Diagnostic		
-gram	record, writing	electr/o/cardi/o/**gram** (ē-lĕk-trō-KĂR-dē-ō-grăm): record of electrical activity of the heart *electr/o:* electricity *cardi/o:* heart
-graph	instrument for recording	electr/o/cardi/o/**graph** (ē-lĕk-trō-KĂR-dē-ō-grăf): instrument for recording electrical activity of the heart *electr/o:* electricity *cardi/o:* heart
-graphy	process of recording	electr/o/cardi/o/**graphy** (ē-lĕk-trō-kăr-dē-ŎG-ră-fē): process of recording electrical activity of the heart (see Fig. 2-2.) *electr/o:* electricity *cardi/o:* heart

Table 2-4 Diagnostic, Pathological, and Related Suffixes

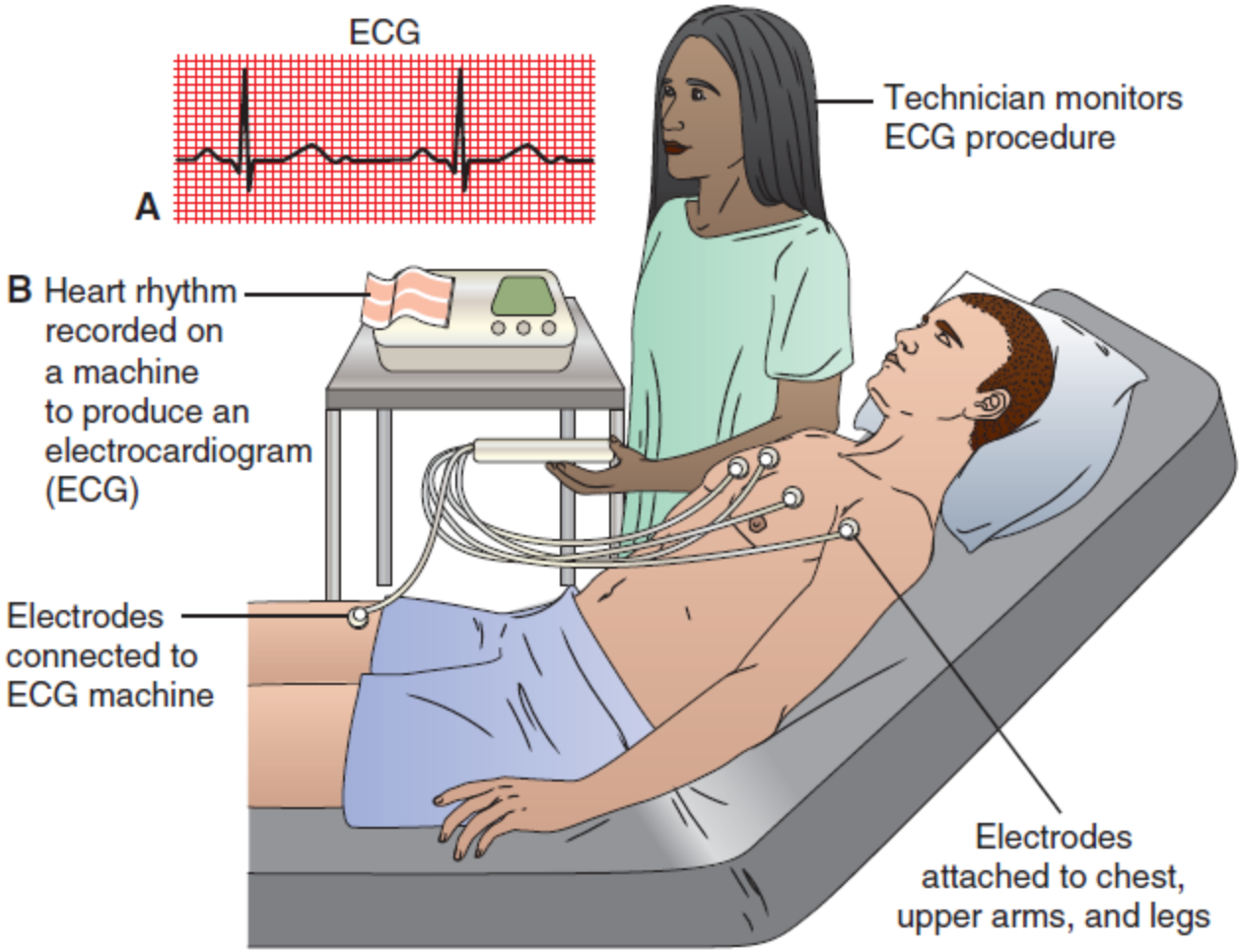


Figure 2-2 Electrocardio**graphy** (ECG) is the ***process of recording*** electrical activity of heart muscle. (A) An electrocardio**gram** is ***a record*** taken during the procedure that shows as a line tracing on a scrolling graph paper. The dips and peaks of the tracing are labeled with the letters P, Q, R. S, and T, which correspond to events of the cardiac cycle. (B) An electrocardio**graph** is the ***instrument for recording*** the electrical impulses of the heart.

Table 2-4 Diagnostic, Pathological, and Related Suffixes

Suffix	Meaning	Word Analysis
-scope	instrument for examining	endo/**scope** (ĔN-dō-skōp): instrument for examining within *endo-*: in, within *An endoscope is a flexible or rigid instrument consisting of a tube and optical system for observing the inside of a hollow organ or cavity.*
-scopy	visual examination	endo/**scopy** (ĕn-DŎS-kō-pē): visual examination within *endo-*: in, within *Endoscopy is performed to visualize a body cavity or canal using a specialized lighted instrument called an endoscope.*

Table 2-4 Diagnostic, Pathological, and Related Suffixes—cont'd

Suffix	Meaning	Word Analysis
Pathological and Related		
-algia	pain	neur/**algia** (nū-RĂL-jē-ă): pain of a nerve *neur:* nerve ***Neuralgic pain usually occurs along the path of a nerve.***
-dynia		ot/o/**dynia** (ō-tō-DĬN-ē-ă): pain in the ear *ot/o:* ear ***Otodynia, also called otalgia, is commonly known as an earache.***
-cele	hernia, swelling	hepat/o/**cele** (hĕ-PĂT-ō-sēl): hernia or swelling of the liver *hepat/o:* liver
-ectasis	dilation, expansion	bronchi/**ectasis** (brŏng-kē-ĔK-tă-sĭs): dilation or expansion of the bronchi *bronchi:* bronchus (plural, *bronchi*)
-emesis	vomiting	hyper/**emesis** (hī-pĕr-ĔM-ĕ-sĭs): excessive vomiting *hyper-:* excessive, above normal

Table 2-4 Diagnostic, Pathological, and Related Suffixes—cont'd

Suffix	Meaning	Word Analysis
Pathological and Related		
-emia	blood condition	leuk/**emia** (ă-NĒ-mē-ă): white blood *an-:* without, not ***Leukemia is a cancer of the white blood cells (leukocytes).***
-gen	forming, producing, origin	carcin/o/**gen** (kăr-SĬN-ō-jĕn): forming, producing, or origin of cancer *carcin/o:* cancer ***A carcinogen is a substance or agent, such as a cigarette, that causes the development or increases the incidence of cancer.***
-genesis		carcin/o/**genesis** (kăr-sĭ-nō-JĔN-ĕ-sĭs): forming, producing, or origin of cancer *carcin/o:* cancer ***Carcinogenesis is the transformation of normal cells into cancer cells, commonly as a result of chemical, viral, or radioactive damage to genes.***
-itis	inflammation	gastr/**itis** (găs-TRĪ-tĭs): inflammation of the stomach *gastr:* stomach
-malacia	softening	chondr/o/**malacia** (kŏn-drō-măl-Ā-shē-ă): softening of cartilage *chondr/o:* cartilage

Table 2-4 Diagnostic, Pathological, and Related Suffixes—cont'd

Suffix	Meaning	Word Analysis
Pathological and Related		
-megaly	enlargement	cardi/o/**megaly** (kăr-dē-ō-MĔG-ă-lē): enlargement of the heart *cardi/o:* heart
-oma	tumor	neur/**oma** (nū-RŌ-mă): tumor of a nerve *neur:* nerve ***A neuroma is a benign tumor composed of nerve tissue.***
-osis	abnormal condition; increase (used primarily with blood cells)	cyan/**osis** (sī-ă-NŌ-sĭs): dark blue or purple discoloration of the skin and mucous membrane *cyan:* blue ***Cyanosis is a bluish discoloration of the skin that indicates a deficiency of oxygen in the blood.***
-pathy	disease	my/o/**pathy** (mī-ŎP-ă-thē): disease of muscle *my/o:* muscle

Table 2-4 Diagnostic, Pathological, and Related Suffixes—cont'd

Suffix	Meaning	Word Analysis
Pathological and Related		
Suffix	**Meaning**	**Word Analysis**
-penia	decrease, deficiency	oste/o/**penia** (ŏs-tē-ō-PĒ-nē-ă): decrease in bone mass *oste/o:* bone ***Osteopenia is characterized by bone loss that is not as severe as that in osteoporosis.***
-phobia	fear	hem/o/**phobia** (hē-mō-FŌ-bē-ă): fear of blood *hem/o:* blood ***Hemophobia is an abnormal aversion to the sight of blood.***
-plegia	paralysis	quadri/**plegia** (kwŏd-rĭ-PLĒ-jē-ă): paralysis of four *quadri:* four ***Quadriplegia is a paralysis of four extremities, both arms and legs. (See Fig. 2-3.)***

Table 2-4 Diagnostic, Pathological, and Related Suffixes—cont'd

Suffix	Meaning	Word Analysis
Pathological and Related		
-ptosis	prolapse, downward displacement	blephar/o/**ptosis** (blĕf-ă-rō-TŌ-sĭs): prolapse or downward displacement of the eyelid *blepharlo:* eyelid ***Blepharoptosis is a drooping of the upper eyelid(s).***
-rrhea	discharge, flow	dia/**rrhea** (dī-ă-RĒ-ă): discharge or flow through *dia-:* through, across ***Diarrhea is an abnormally frequent discharge or flow of fluid fecal matter from the bowel.***
-rrhexis	rupture	arteri/o/**rrhexis** (ăr-tē-rē-ō-RĔK-sĭs): rupture of an artery *arteri/o:* artery
-sclerosis	abnormal condition of hardening	arteri/o/**sclerosis** (ăr-tē-rē-ō-sklĕ-RŌ-sĭs): abnormal condition of hardening of an artery *arteri/o:* artery

Table 2-4 Diagnostic, Pathological, and Related Suffixes—cont'd

Suffix	Meaning	Word Analysis
Pathological and Related		
Suffix	**Meaning**	**Word Analysis**
-spasm	involuntary contraction, twitching	blephar/o/**spasm** (BLĔF-ă-rō-spăsm): involuntary contraction or twitching of the eyelid *blepharlo:* eyelid
-stenosis	narrowing, stricture	arteri/o/**stenosis** (ăr-tē-rē-ō-stĕ-NŌ-sĭs): abnormal narrowing or stricture of an artery *arterilo:* artery
-toxic	poison	hepat/o/**toxic** (HĔP-ă-tō-tŏk-sĭk): pertaining to poison in the liver *hepatlo:* liver ***Alcohol and drugs are examples of agents that have destructive effects on the liver.***



It is time to review surgical suffixes by completing Learning Activities 2-4, and 2-5.

Suffix Types (Cont.)

** Grammatical Suffixes

Short grammatical suffixes are attached to word roots to form parts of speech, such as adjectives and nouns. Many of these same suffixes are used in the English language.

(See Table 2-5)

Table 2-5 Adjective and Noun Suffixes

This table lists adjective and noun suffixes that are attached to word roots in a medical term, along with their meanings and word analyses.

Suffix	Meaning	Word Analysis
Adjective		
-ac	pertaining to	cardi/**ac** (KĂR-dē-ăk): pertaining to the heart *cardi:* heart
-al		neur/**al** (NŪ-răl): pertaining to a nerve

Table 2-5 Adjective and Noun Suffixes

This table lists adjective and noun suffixes that are attached to word roots in a medical term, along with their meanings and word analyses.

Suffix	Meaning	Word Analysis
Adjective		
-ar		muscul/**ar** (MŬS-kū-lăr): pertaining to muscle *muscul:* muscle
-ary		pulmon/**ary** (PŬL-mō-nĕr-ē): pertaining to the lungs *pulmon:* lung
-eal		esophag/**eal** (ē-sŏf-ă-JĒ-ăl): pertaining to the esophagus *esophag:* esophagus
-ic		thorac/**ic** (thō-RĂS-ĭk): pertaining to the chest *thorac:* chest
-ior		poster/**ior** (pŏs-TĒ-rē-or): pertaining to the back (of the body) *poster:* back (of body), behind, posterior
-ous		cutane/**ous** (kū-TĀ-nē-ŭs): pertaining to the skin *cutane:* skin
-tic		acous/**tic** (ă-KOOS-tĭk): pertaining to hearing *acous:* hearing

Table 2-5 Adjective and Noun Suffixes—cont'd

Suffix	Meaning	Word Analysis
Noun		
-ia	condition	pneumon/**ia** (nū-MŌ-nē-ă): condition of the lung(s) *pneumon:* air; lung ***Pneumonia is an infection of the lung, usually caused by bacteria, viruses, or diseases.***
-ism		thyroid/**ism** (THĪ-royd-ĭzm): condition of the thyroid gland *thyroid:* thyroid gland ***Thyroidism is a condition caused by overactivity of the thyroid gland.***
-iatry	medicine; treatment	psych/**iatry** (sī-KĪ-ă-trē): treatment of the mind *psych/o:* mind ***Psychiatry is the medical specialty concerned with treatment of mental illness, emotional disturbance, and abnormal behavior.***
-ist		hemat/o/log/**ist** (hē-mă-TŎL-ō-jĭst): specialist in the study of blood *hemat/o:* blood *log:* study of
-y	condition; process	neur/o/path/**y** (nū-RŎP-ă-thē): condition of nerve diseases *neur/o:* nerve *path:* disease ***Neuropathy is the study of disorders of the nerves.***

Table 2-5 Adjective and Noun Suffixes—cont'd

Suffix	Meaning	Word Analysis
Diminutive		
-icle	small, minute	ventr/**icle** (VĔN-trĭ-kl): small cavity, as of the brain or heart *ventr*: belly, belly side
-ole		arteri/**ole** (ăr-TĒ-rē-ōl): small or minute artery *arteri*: artery ***Arteries narrow to form arterioles (minute arteries), which branch into capillaries (microscopic blood vessels).***
-ule		ven/**ule** (VĔN-ūl): small or minute vein ven: vein ***A venule is a small vein that is continuous with a capillary.***



It is time to review surgical suffixes by completing Learning Activities 2-6.

Plural Suffixes

Suffixes are also used to denote singular and plural forms of a word.

English endings have also been adopted for commonly used medical terms.

When a word changes from a singular to a plural form, the suffix of the word is the part that changes.

Rules for Singular and Plural Suffixes

This table presents common singular suffixes, the rules for forming plurals, and examples of each.

Rule		Example	
Singular	**Plural**	**Singular**	**Plural**
-a	Retain *a* and add e.	pleur*a*	pleur*ae*
-ax	Drop *x* and add *ces.*	thor*ax*	thor*aces*
-en	Drop *en* and add *ina.*	lum*en*	lum*ina*
-is	Drop *is* and add es.	diagnos*is*	diagnos*es*
-ix	Drop *ix* and add *ices.*	append*ix*	append*ices*

Rules for Singular and Plural Suffixes

This table presents common singular suffixes, the rules for forming plurals, and examples of each.

Rule		Example	
-ex	Drop ex and add ices.	apex	apices
-ma	Retain ma and add ta.	carcinoma	carcinomata
-on	Drop on and add a.	ganglion	ganglia
-um	Drop um and add a.	bacterium	bacteria
-us	Drop us and add i.	bronchus	bronchi
-y	Drop y and add ies.	deformity	deformities

It is time to review surgical suffixes by completing Learning Activities 2-7.

Learning Activities

These activities provide review of the suffixes introduced in this chapter.

Complete each activity from 2-1 to 2-7 and review your answers to evaluate your understandir of the chapter 2.



CHAPTER3



Chapter Outline

- Objectives
- Prefix Linking
- Prefix Types
  - Prefixes of Position, Number, Measurement, and Direction
  - Other Common Prefixes

Objectives

Upon completion of this chapter, you will be able to:

- Define common prefixes used in medical terminology.
- Describe how a prefix changes the meaning of a medical word.
- Recognize and define prefixes of position, number and measurement, and direction.
- Pronounce medical terms presented in this chapter.
- Demonstrate your knowledge of this chapter by completing the learning activities.

Prefix Linking

Most medical words contain a root or combining form with a suffix. Some of them also contain prefixes.

A prefix is a word element located at the beginning of a medical word.

Substituting one prefix for another alters the meaning of the word.

For example:

In the term macro/cyte, macro- is a prefix meaning large;

-cyte is a suffix meaning cell.

A macr/o/cyte is a large cell.

Changing the prefix macro- (large) to micro- (small) changes the meaning of the word.

A micr/o/cyte is a small cell.

(See Table 3-1 for other examples of how a prefix changes the meaning of a word).

Table 3-1 Changing Prefixes and Meanings

In this table, each word has the same root, nat *(birth), and suffix,* -al *(pertaining to). By substituting different prefixes, new words with different meanings are formed.*

Prefix	+	Word Root	+	Suffix	=	Medical Word	Meaning
pre- (before)	+	**nat** (birth)	+	**-al** (pertaining to)	=	**pre/nat/al** prē-NĀ-tăl	pertaining to (the period) before birth
peri- (around)	+	**nat** (birth)	+	**-al** (pertaining to)		**peri/nat/al** pĕr-ĭ-NĀ-tăl	pertaining to (the period) around birth
post- (after)	+	**nat** (birth)	+	**-al** (pertaining to)		**post/nat/al** pōst-NĀ-tăl	pertaining to (the period) after birth

Prefix Types

Learning the major types of prefixes, such as prefixes of position, number, measurement, and direction, as well as some others, will help you master medical terminology.

** Prefixes of Position, Number or measurement, and Direction

Prefixes used in medical terms denote position, number and measurement, and direction.

- Prefixes of position describe a place or location. (See Table 3-2.)

- Prefixes of number and measurement describe an amount, size, or degree of involvement. (See Table 3-3.)

- Prefixes of direction indicate a pathway or route. (See Table 3-4.)

Table 3-2 Prefixes of Position

This table lists commonly used prefixes of position, along with their meanings and word analyses.

Prefix	Meaning	Word Analysis
endo-	in, within	**endo**/crine (ĔN-dō-krĭn): secrete within *-crine:* secrete ***Endocrine describes a gland that secretes directly into the bloodstream.***
intra-		**intra**/muscul/ar (ĭn-tră-MŬS-kū-lăr): within the muscle *muscul:* muscle *-ar:* pertaining to
epi-	above, upon	**epi**/derm/is (ĕp-ĭ-DĔR-mĭs): upon the skin *derm:* skin *-is:* noun ending ***The epidermis is the outer layer of the skin.***

Table 3-2 Prefixes of Position—cont'd

Prefix	Meaning	Word Analysis
hypo-	under, below; deficient	**hypo**/derm/ic (hī-pō-DĔR-mĭk): pertaining to under the skin *derm:* skin *-ic:* pertaining to ***Hypodermic needles are used for subcutaneous injections and to take fluid samples from the body, for example, taking blood from a vein in venipuncture. (See Fig. 3-1.)***

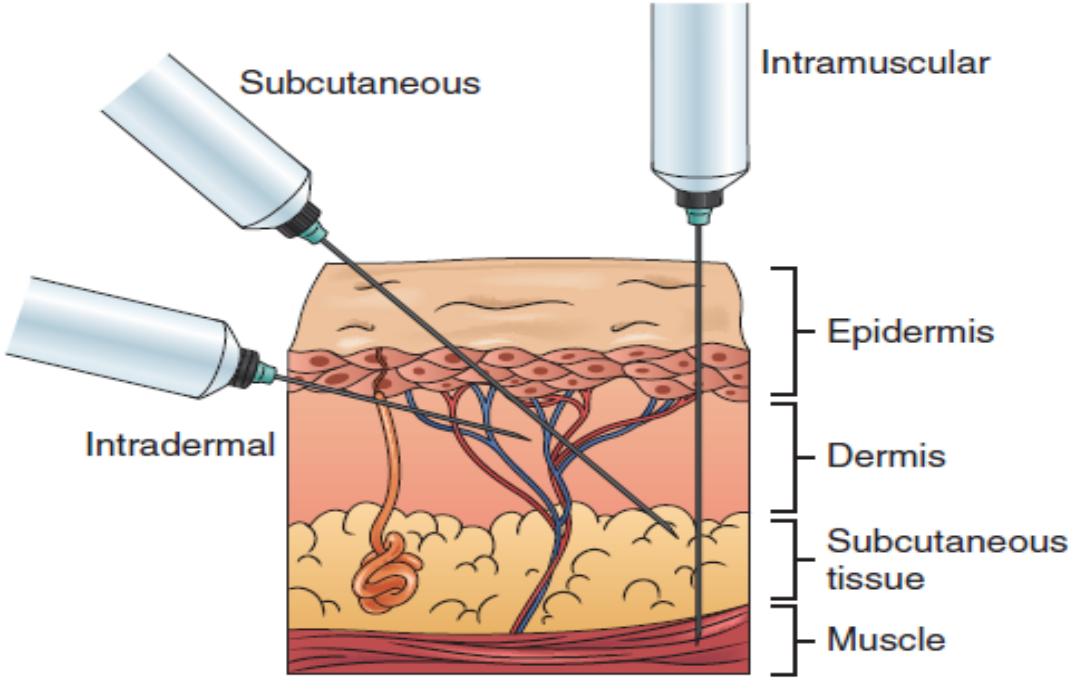


Figure 3-1 Hypodermic needles inserted within the skin (intradermal), under the skin (subcutaneous), or in between the muscular layer (intramuscular).

Table 3-2 Prefixes of Position—cont'd

Prefix	Meaning	Word Analysis
infra-	under, below	**infra**/cost/al (ĭn-fră-KŎS-tăl): below the ribs *cost:* ribs *-al:* pertaining to
sub-		**sub**/cutane/ous (sŭb-kū-TĀ-nē-ŭs): pertaining to under the skin *cutane:* skin *-ous:* pertaining to ***The subcutaneous tissue is the lowest layer of skin. It binds the dermis to underlying structures.***
inter-	between	**inter**/cost/al (ĭn-tĕr-KŎS-tăl): between the ribs *cost:* ribs *-al:* pertaining to
retro-	backward, behind	**retro**/version (rĕt-rō-VĔR-shŭn): turning backward *-version:* turning **Retroversion** ***refers to tipping backward of an organ (such as the uterus) from its normal position.***

Table 3-3 Prefixes of Number and Measurement

This table lists commonly used prefixes of number and measurement, along with their meanings and word analyses.

Prefix	Meaning	Word Analysis
bi-	two	**bi**/later/al (bī-LĂT-ĕr-ăl): pertaining to two sides *later:* side *-al:* pertaining to
dipl-	double	**dipl**/opia (dĭp-LŌ-pē-ă): double vision *-opia:* vision
diplo-		**diplo**/bacteri/al (dĭp-lō-băk-TĒR-ē-ăl): bacteria linked together in pairs *bacteri:* bacteria *-al:* pertaining to ***Diplobacteria reproduce in such a manner that they are joined together in pairs.***
hemi-	one-half	**hemi**/plegia (hĕm-ē-PLĒ-jē-ă): paralysis of one half *-plegia:* paralysis ***Hemiplegia is a paralysis of one-half of the body, either the right side or the left side.***
hyper-	excessive, above normal	**hyper**/calc/emia (hī-pĕr-kăl-SĒ-mē-ă): excessive calcium in the blood *calc:* calcium *-emia:* blood condition

Table 3-3 Prefixes of Number and Measurement

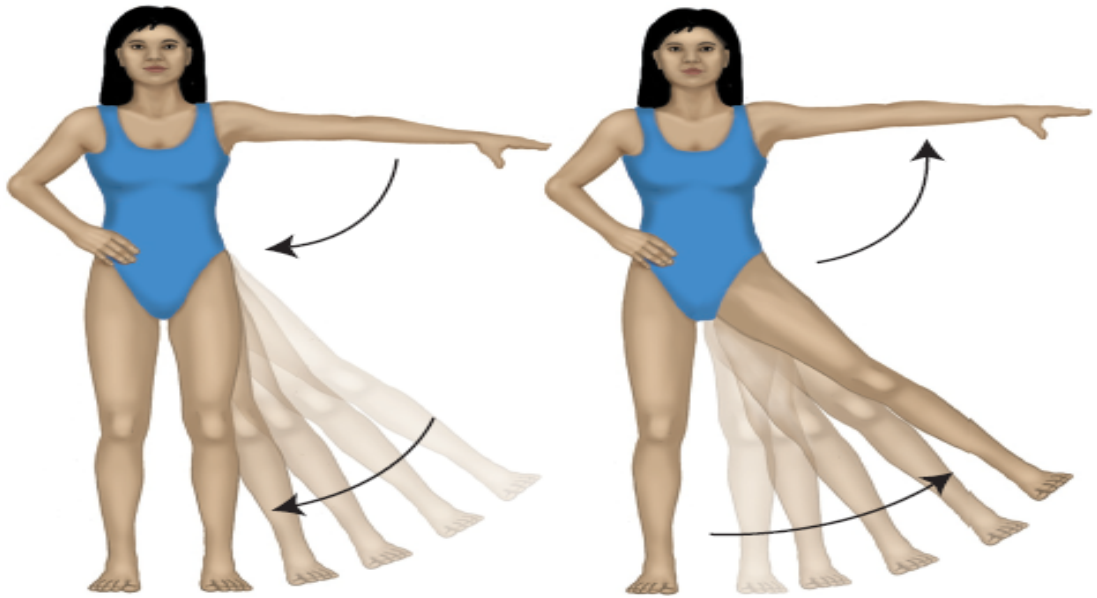
This table lists commonly used prefixes of number and measurement, along with their meanings and word analyses.

Prefix	Meaning	Word Analysis
macro-	large	**macro/**cyte (MĂK-rō-sīt): large cell *-cyte:* cell
micro-	small	**micro/**scope (MĪ-krō-skōp): instrument for examining small (objects) *-scope:* instrument for examining ***A microscope is an optical instrument that greatly magnifies minute objects.***
mono-	one	**mono/**therapy (MŎN-ō-thĕr-ă-pē): one treatment *-therapy:* treatment ***An example of monotherapy is treatment using only a single drug or a single treatment modality.***
uni-		**uni/**nucle/ar (ū-nĭ-NŪ-klē-ăr): pertaining to one nucleus *nucle:* nucleus *-ar:* pertaining to
multi-	many, much	**multi/**gravida (mŭl-tĭ-GRĂV-ĭ-dă): woman who has been pregnant more than once *-gravida:* pregnant woman
poly		**poly/**phobia (pŏl-ē-FŌ-bē-ă): fear of many things *-phobia:* fear
quadri-	four	**quadri/**plegia (kwŏd-rĭ-PLĒ-jē-ă): paralysis of four *-plegia:* paralysis ***Quadriplegia is a paralysis of all four extremities, usually caused by an injury to or disease of the cervical spinal cord.***
tri-	three	**tri/**ceps (TRĪ-cĕps): three heads *-ceps:* head ***Triceps describes a muscle having three heads or points of origin.***

Table 3-4 Prefixes of Direction

This table lists commonly used prefixes of direction as well as their meanings and word analyses.

Prefix	Meaning	Word Analysis
ab-	from, away from	**ab**/duction (ăb-DŬK-shŭn): movement of a limb away from (the body) *-duction:* act of leading, bringing, conducting ***Abduction is a body movement away from the midline or axis of the body. (See Fig. 3-2.)***
ad-	toward	**ad**/duction (ăb-DŬK-shŭn): movement of a limb toward (the body) *-duction:* act of leading, bringing, conducting ***Adduction is a body movement toward to the midline or axis of the body. (See Fig. 3-2.)***



A Adduction **B** Abduction

Figure 3-2 Abduction and adduction.

Table 3-4 Prefixes of Direction—cont'd

Prefix	Meaning	Word Analysis
circum-	around	**circum**/ren/al (sĕr-kŭm-RĒ-năl): pertaining to around the kidney *ren:* kidney *-al:* pertaining to
peri-		**peri**/odont/al (pĕr-ē-ō-DŎN-tăl): pertaining to around a tooth *odont:* teeth *-al:* pertaining to
dia-	through, across	**dia**/rrhea (dī-ă-RĒ-ă): flow through *-rrhea:* discharge, flow ***Diarrhea is a condition of abnormally frequent discharge or flow of fluid fecal matter through the bowel.***
trans-		**trans**/vagin/al (trăns-VĂJ-ĭn-ăl): pertaining to across or through the vagina *vagin:* vagina *-al:* pertaining to
ecto-	outside, outward	**ecto**/gen/ous (ĕk-TŎJ-ĕ-nŭs): forming outside (the body or structure) *gen:* forming, producing, origin *-ous:* pertaining to ***An ectogenous infection is one that originates outside of the body.***
exo-		**exo**/tropia (ĕks-ō-TRŌ-pē-ă): turning outward *-tropia:* turning ***Exotropia refers to the turning outward of the eyes.***
extra-		**extra**/crani/al (ĕks-tră-KRĀ-nē-ăl): pertaining to outside the skull *crani:* cranium (skull) *-al:* pertaining to

Table 3-4 Prefixes of Direction—cont'd

Prefix	Meaning	Word Analysis
para-*	near, beside; beyond	**para**/nas/al (păr-ă-NĀ-săl): beside the nose *nas:* nose *-al:* pertaining to
super-	upper, above	**super**/ior (soo-PĒ-rē-or): pertaining to above or the upper part of a structure *-ior:* pertaining to
supra-	above; excessive; superior	**supra**/ren/al (soo-pră-RĒ-năl): pertaining to above the kidney *ren:* kidney *-al:* pertaining to
ultra-	excess, beyond	**ultra**/son/ic (ŭl-tră-SŎN-ĭk): pertaining to sound beyond (that which can be heard by the human ear) *son:* sound *-ic:* pertaining to

**Para-* may also be used as a suffix meaning *to bear (offspring)*.

Other Common Prefixes

Many other common prefixes may also change the meaning of a word.

See Table 3-5 for a list of other common prefixes.

Table 3-5 Other Common Prefixes

This table lists other commonly used prefixes, along with their meanings and word analyses.

Prefix	Meaning	Word Analysis
a-*	without, not	**a**/mast/ia (ă-MĂS-tē-ă): without a breast *mast:* breast *-ia:* condition ***Amastia may be the result of a congenital defect, an endocrine disorder, or mastectomy.***
an-**		**an**/esthesia (ăn-ĕs-THĒ-zē-ă): without feeling *-esthesia:* feeling ***Anesthesia may be a partial or complete loss of sensation with or without loss of consciousness.***

Table 3-5 Other Common Prefixes—cont'd

Prefix	Meaning	Word Analysis
anti-	against	**anti**/bacteri/al (ăn-tĭ-băk-TĒR-ē-ăl): against bacteria *bacteri:* bacteria *-al:* pertaining to ***Antibacterials are substances that kill bacteria or inhibit their growth or replication.***
contra-		**contra**/ception (kŏn-tră-SĔP-shŭn): against conceiving *-ception:* conceiving ***Contraceptive techniques prevent pregnancy by means of medication, a device, or a method that blocks or alters one or more of the processes of reproduction.***

*The prefix *a-* is usually used before a consonant.

** The prefix *an-* is usually used before a vowel.

Table 3-5 Other Common Prefixes—cont'd

Prefix	Meaning	Word Analysis
auto-	self, own	**auto**/graft (AW-tō-grăft); transplantation to self *-graft:* transplantation ***An autograft is tissue transplanted from one site and grafted to another site of the same person. (See Fig. 3-3.)***

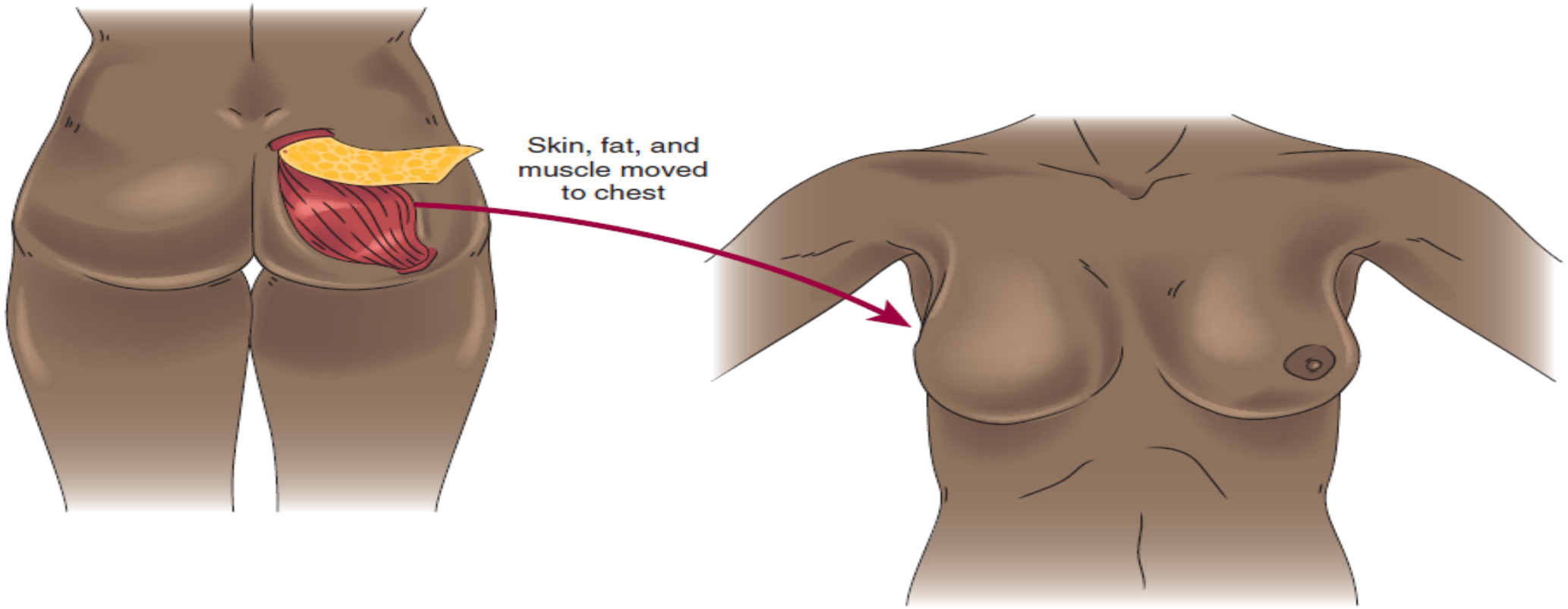


Figure 3-3 Autograft in which tissue from the patient's buttocks is transplanted to her breast.

Table 3-5 Other Common Prefixes—cont'd

Prefix	Meaning	Word Analysis
hetero-	different	**hetero/**graft (HĔT-ĕ-rō-grăft): different transplantation; also called *xenograft* *-graft:* transplantation ***A heterograft is a transplant from one species to another. (See Fig. 3-4.)***

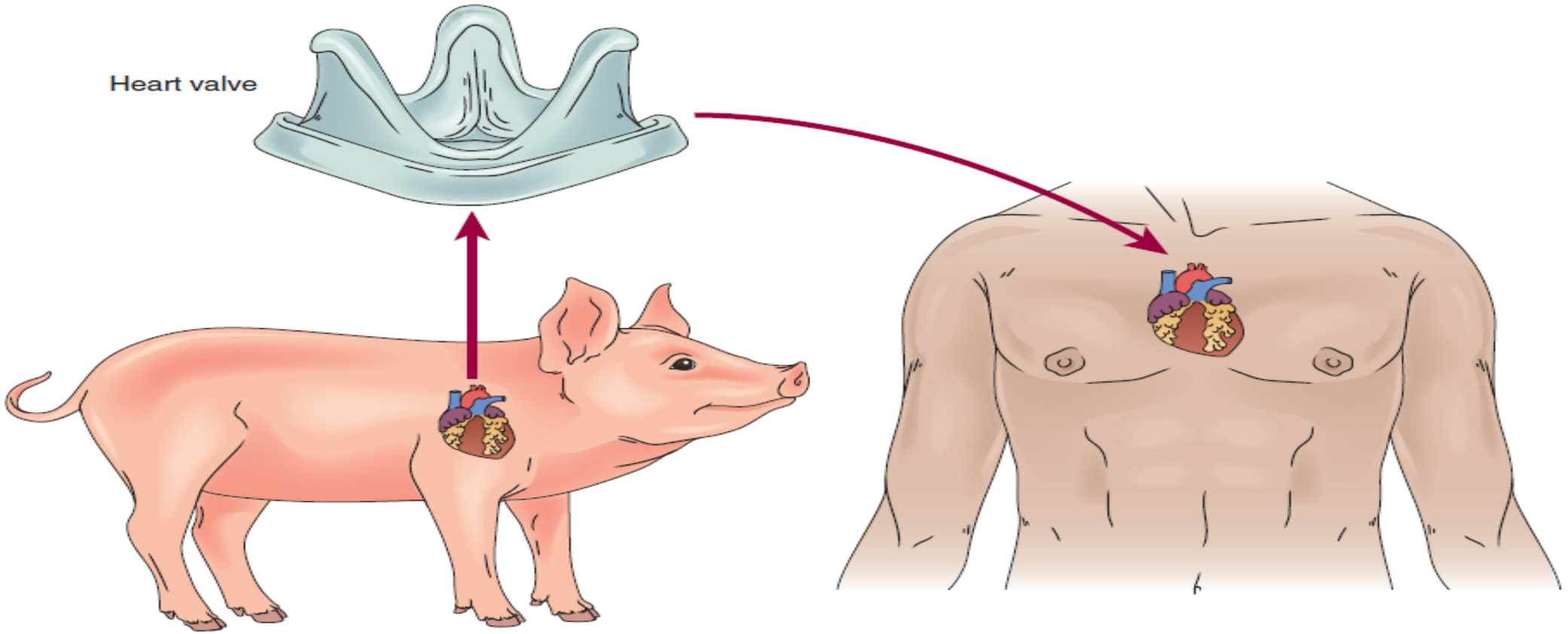


Figure 3-4 Heterograft in which tissue (heart valve) from one species (pig) is transplanted to another (human).

Table 3-5 Other Common Prefixes—cont'd

Prefix	Meaning	Word Analysis
homo-	same	**homo/**graft (HŌ-mō-grăft): same transplantation *-graft:* transplantation ***A homograft, also called an allograft, is a graft of tissue or an organ taken from a donor of the same species as the recipient. Commonly transplanted organs include the kidneys, lungs, and heart. (See Fig. 3-5.)***
homeo-		**homeo/**plasia (hō-mē-ō-PLĀ-zē-ă): formation or growth of same (or similar tissue) *-plasia:* formation, growth

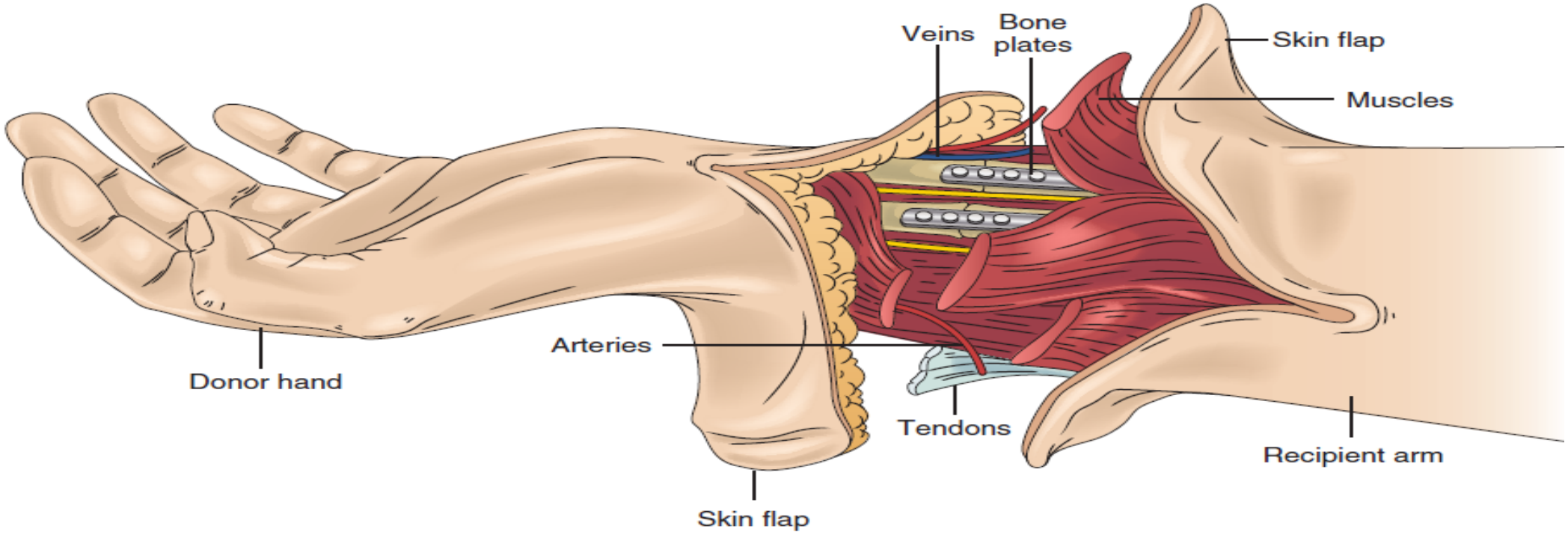


Figure 3-5 Homograft of an arm taken from a human donor and transplanted to another human.

Table 3-5 Other Common Prefixes—cont'd

Prefix	Meaning	Word Analysis
tachy-	rapid	**tachy**/pnea (tăk-ĭp-NĒ-ă): rapid breathing *-pnea:* breathing
brady-	slow	**brady**/cardia (brăd-ē-KĂR-dē-ă): slow heart rate *-cardia:* heart
dys-	bad; painful; difficult	**dys**/tocia (dĭs-TŌ-sē-ă): difficult childbirth *-tocia:* childbirth, labor
eu-	good, normal	**eu**/pnea (ūp-NĒ-ă): normal breathing *-pnea:* breathing

It is time to review prefixes by completing Learning Activities 3-1, 3-2, and 3-3.

Learning Activities

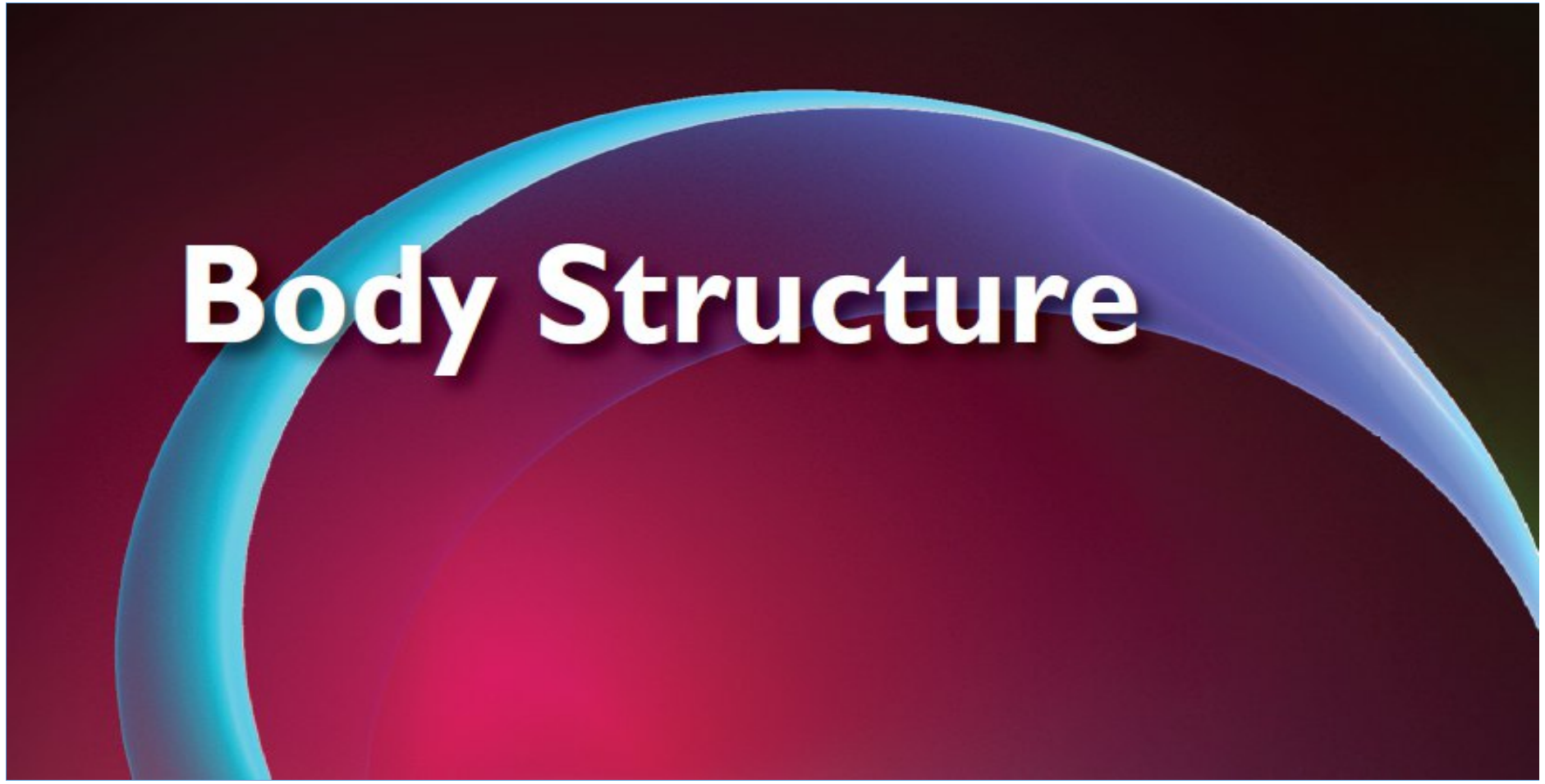
The activities that follow provide

a review of the prefixes introduced in this chapter.

Complete each activity from 3-1 to 3-3 and review your answers to evaluate your understanding of chapter 3.



CHAPTER 4



Chapter Outline

- Objectives
- Introduction
- Body Structure Key Terms
- Levels of Organization
  - Cells
  - Tissues
  - Organs
  - Systems
  -Organism

Chapter Outline (cont.)

- Anatomical Position
- Body Planes
- Directional Terms
- Spine
- Medical Word Elements

Objectives

Upon completion of this chapter, you will be able to:

- List the levels of organization of the body.
- Define and identify three planes of the body.
- Identify the cavities, quadrants, and regions of the body.
- List and identify terms related to direction, position, and planes of the body.
- Recognize, pronounce, spell, and build words related to body structure.
- Describe diseases, conditions, and procedures related to body structure.
- Demonstrate your knowledge of this chapter by completing the learning and documenting health-care activities.

Introduction

This chapter provides an orientation to the body as a whole and contains general terms that are relevant to all body systems.

Learning these terms and how they are used to locate and describe structures within the body helps to master the material presented in subsequent chapters of this course.

Levels of Organization

The human body contains several levels of structure and function. Each of these levels builds on the previous level and contributes to the structure and function of the entire organism.

Five levels of organization are relevant to understanding anatomy, physiology, and pathology: the cells, tissues, organs, systems, and organism. (See Fig. 4-1.)



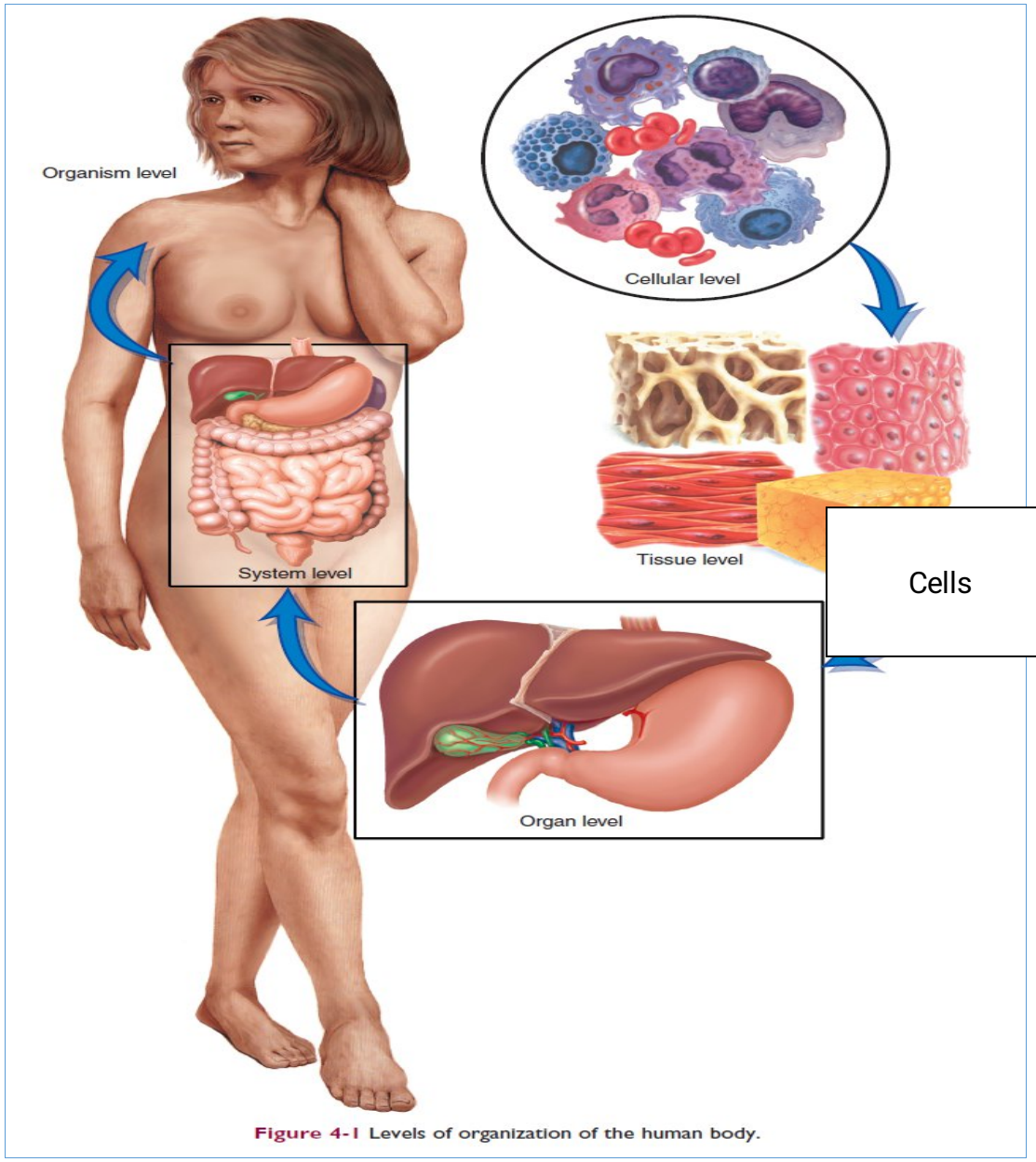


Figure 4-1 Levels of organization of the human body.

Cells Tissues Organs Systems C

Anatomical Position

Anatomical position is a body posture used among anatomists and clinicians as a position of reference to ensure uniformity and consistency in locating anatomical parts or divisions of the human body.

In the anatomical position, the person stands erect, facing forward, and the arms are at the sides of the body, with the palms of the hands turned forward and the feet parallel to each another.

No matter how the body is actually positioned—standing or lying down, facing forward or backward or how the limbs are actually placed, the positions and relationships of a structure are always described as if the body were in the anatomical position.

Body Planes

A plane is an imaginary flat surface that divides the body into two sections.

When the body is in anatomical position, the planes serve as points of reference to identify the different sections of the body.

The most commonly used planes are 3; coronal (frontal), transverse (horizontal), and midsagittal (median).

- A coronal (frontal) plane divides the body into an anterior and posterior sections.
- A transverse (horizontal) plane divides the body into superior and inferior sections.
- The midsagittal (median) plane runs through the center of the body, dividing the body into equal right and left halves. (See Fig. 4-2.)

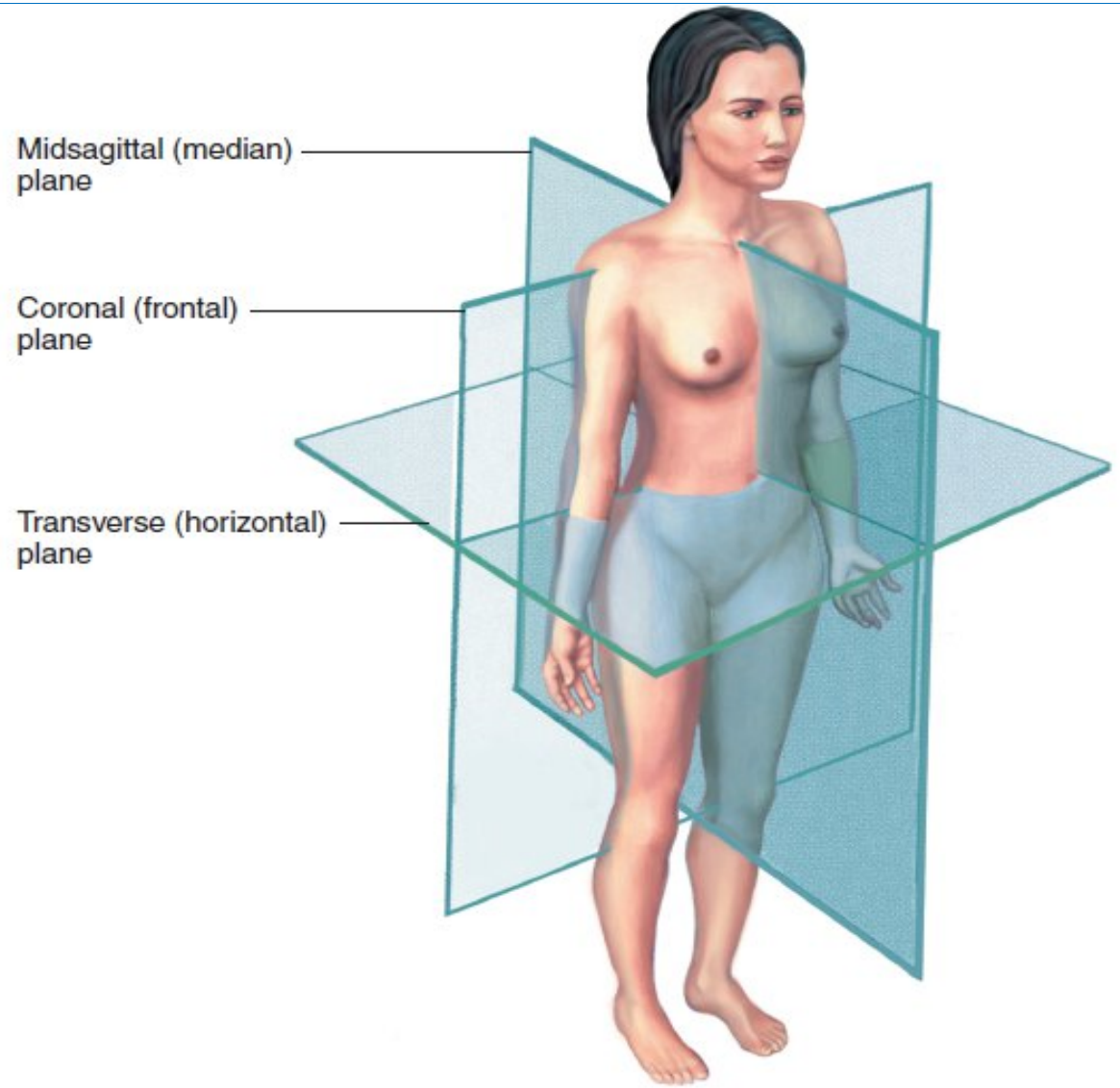


Figure 4-2 Body planes. (Note that the body is in anatomical position.)

Directional Terms

Directional terms help indicate the position of structures, surfaces, and regions of the body.

These terms are always identified relative to the anatomical position.

For example, the knee is superior to the ankle; the legs are inferior to the trunk.

(Refer to Figure 4-3 to locate the directional terms superior and inferior.)

Directional terms identify the location of diseases, injuries, and surgical sites.

In a clinical setting, health-care providers may describe the location of a heart attack to the patient as occurring in the front, or anterior, part of the heart.

A tumor on the back of the kidney may be described as being located on the posterior surface of the kidney.

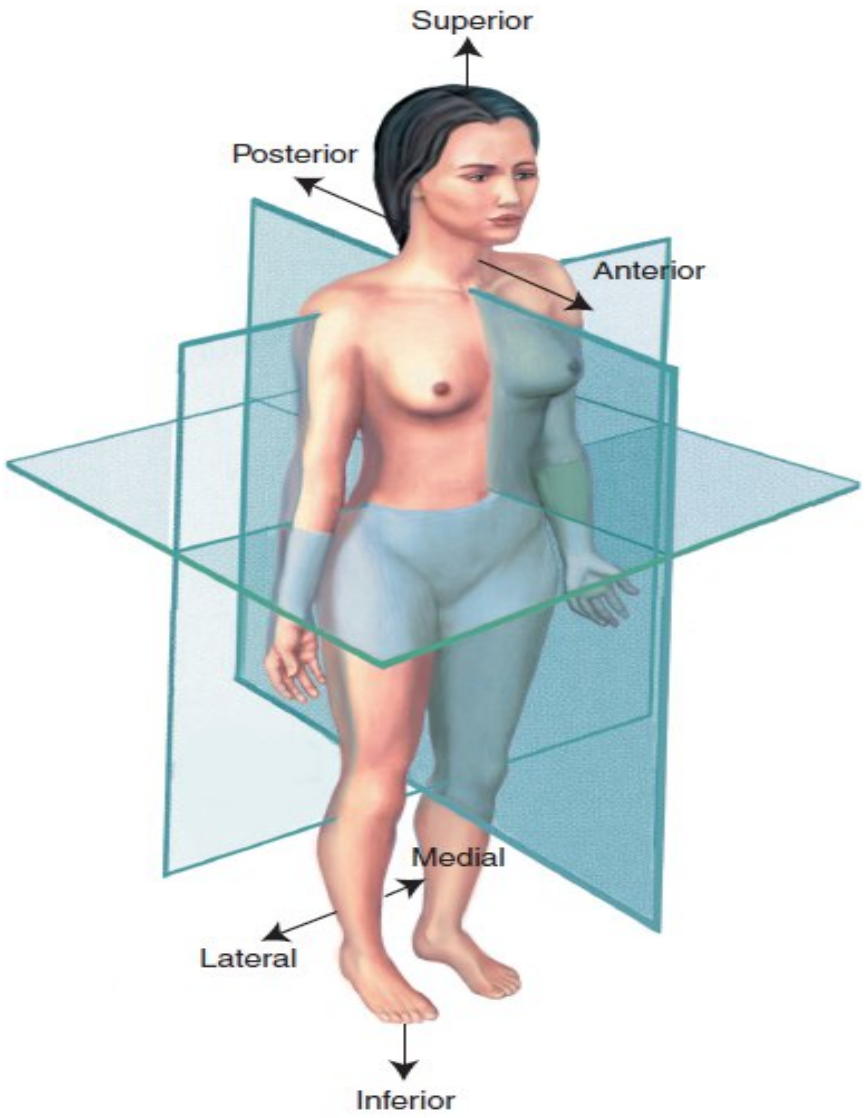


Figure 4-3 Directional terms.

Table 4-1 Directional Terms

This table provides a comprehensive summary of directional terms, along with their definitions. Opposing terms are presented consecutively to aid memorization.

Term	Definition
Abduction	Movement away from the midsagittal (median) plane of the body or one of its parts
Adduction	Movement toward the midsagittal (median) plane of the body
Medial	Pertaining to the midline of the body or structure
Lateral	Pertaining to a side
Superior (cephalad)	Toward the head or upper portion of a structure
Inferior (caudal)	Away from the head, or toward the tail or lower part of a structure
Proximal	Nearer to the center (trunk of the body) or to the point of attachment to the body
Distal	Further from the center (trunk of the body) or from the point of attachment to the body
Anterior (ventral)	Front of the body
Posterior (dorsal)	Back of the body

Table 4-1 Directional Terms

This table provides a comprehensive summary of directional terms, along with their definitions. Opposing terms are presented consecutively to aid memorization.

Term	Definition
Parietal	Pertaining to the outer wall of the body cavity
Visceral	Pertaining to the viscera, or internal organs, especially the abdominal organs
Prone	Lying on the abdomen, face down
Supine	Lying horizontally on the back, face up
Inversion	Turning inward or inside out
Eversion	Turning outward
Palmar	Pertaining to the palm of the hand
Plantar	Pertaining to the sole of the foot
Superficial	Toward the surface of the body (external)
Deep	Away from the surface of the body (internal)

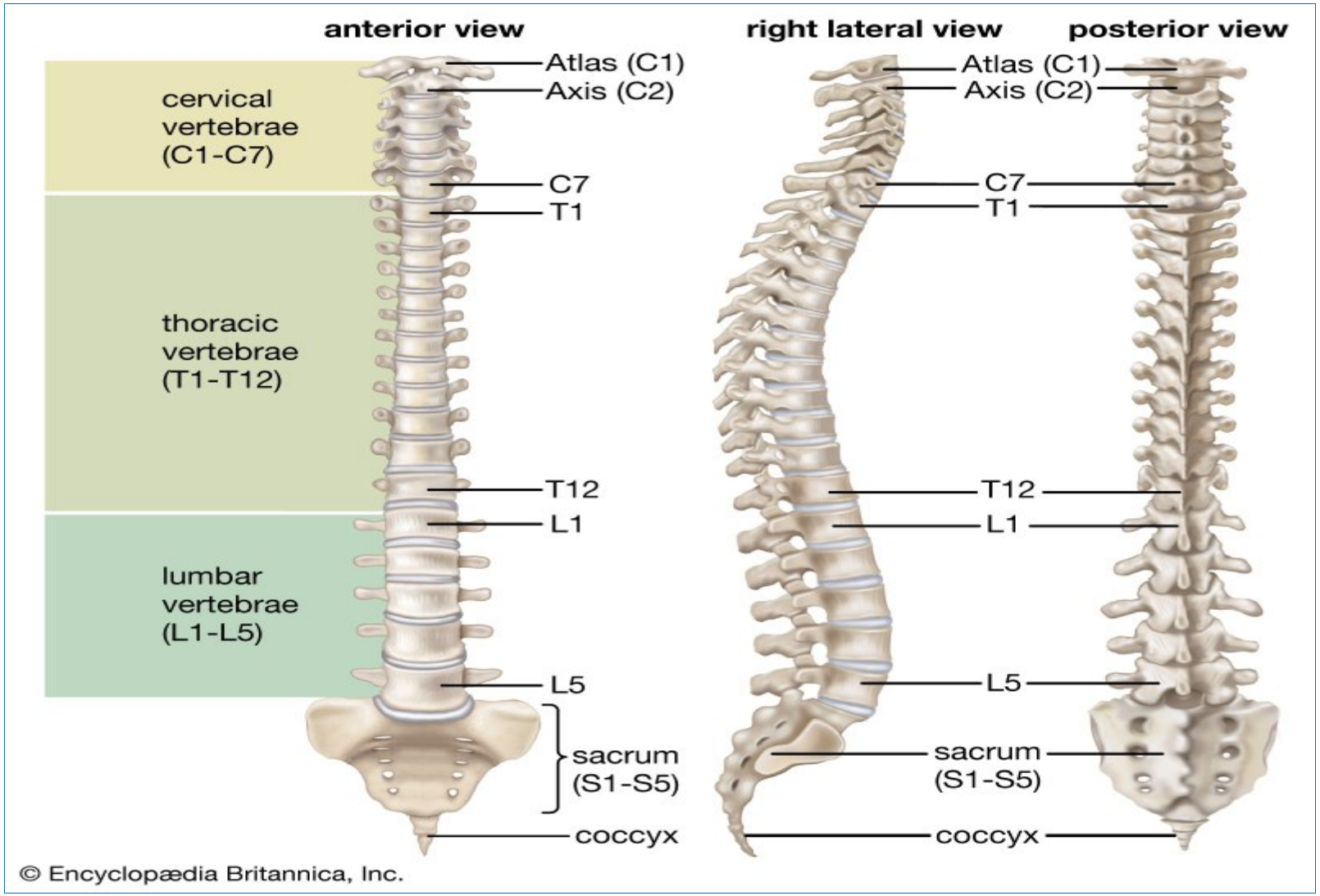
The Spine

The spine (vertebral column or backbone) is composed of a series of bones that extend from the base of the skull to the pelvis. It is formed from 26 irregular bones (vertebrae, singular: vertebra) and connective tissue in such a way that a flexible, curved structure results.

The spine is divided into sections corresponding to the vertebrae located in the spinal column. These divisions are as follows:

- Cervical (neck)
- Thoracic (chest)
- Lumbar (loin)
- Sacral (lower back)
- Coccyx (tailbone)

(see the following figure of the spine)



The spine / vertebral column or backbone

It is time to review body cavities, the spine, and directional terms by completing Learning Activity 4-1.

Medical Word Elements

This section introduces combining forms, suffixes, and prefixes related to body structure. Word analyses are also provided. From the information provided, complete the meaning of the medical words in the right-hand column. The first one is completed for you.

Element	Meaning	Word Analysis and Meaning
Combining Forms		
Cellular Structure		
cyt/o	cell	**cyt/o**/logist (sī-TŎL-ō-jĭst): *specialist in the study of cells* *-logist:* specialist in the study of *Cytologists study the formation, structure, and function of cells.*
hist/o	tissue	**hist/o**/logy (hĭs-TŎL-ō-jē): _____ *-logy:* study of *Histology is the branch of science that investigates the microscopic structures and functions of tissues.*
kary/o	nucleus	kary/o/lysis (kăr-ē-ŎL-ĭ-sĭs): _____ *-lysis:* separation; destruction; loosening *Karyolysis results in death of the cell.*
nucle/o		nucle/ar (NŪ-klē-ăr): _____ *-ar:* pertaining to

Medical Word Elements—cont'd

Element	Meaning	Word Analysis and Meaning
Combining Forms		
Position and Direction		
anter/o	anterior, front	**anter**/ior (ăn-TĒR-ē-or): _____ *-ior:* pertaining to
caud/o	tail	**caud**/ad (KAW-dăd): _____ *-ad:* toward
dors/o	back (of body)	**dors**/al (DOR-săl): _____ *-al:* pertaining to
infer/o	lower, below	**infer**/ior (ĭn-FĒR-rē-or): _____ *-ior:* pertaining to
later/o	side, to one	**later**/al (LĂT-ĕr-ăl): _____

Medical Word Elements—cont'd

Element	Meaning	Word Analysis and Meaning
Combining Forms		
Position and Direction		
medi/o	middle	**medi**/ad (MĒ-dē-ăd): _____ *-ad:* toward
poster/o	back (of body), behind, posterior	**poster**/ior (pŏs-TĒR-ē-or): _____ *-ior:* pertaining to
proxim/o	near, nearest	**proxim**/al (PRŎK-sĭm-ăl): _____
ventr/o	belly, belly side	**ventr**/al (VĔN-trăl): _____ *-al:* pertaining to

Medical Word Elements—cont'd

Element	Meaning	Word Analysis and Meaning
Combining Forms		
Color		
albin/o	white	**albin**/ism (ĂL-bĭn-ĭzm): _____ *–ism:* condition *Albinism is characterized by a partial or total lack of pigment in the skin, hair, and eyes.*
leuk/o		**leuk**/o/cyte (LOO-kō-sīt): _____ *–cyte:* cell *A leukocyte is a white blood cell.*
chrom/o	color	hetero/**chrom**/ic (hĕt-ĕr-ō-KRŌ-mĭk): _____ *hetero–:* different *–ic:* pertaining to *Heterochromia is associated with the iris or sections of the iris of the eyes. Thus, the individual with heterochromia may have one brown iris and one blue iris.*

Medical Word Elements—cont'd

Element	Meaning	Word Analysis and Meaning
Combining Forms		
Color		
cirrh/o	yellow	**cirrh**/osis (sĭr-RŌ-sĭs): _____ *-osis:* abnormal condition; increase (used primarily with blood cells) *In cirrhosis, the skin, sclera of the eyes, and mucous membranes take on a yellow color.*
jaund/o		**jaund**/ice (JAWN-dĭs): _____ *-ice:* noun ending *Jaundice is the yellowing of the skin, mucous membranes, and sclera caused by excessive bilirubin in the blood.*

Medical Word Elements—cont'd

Element	Meaning	Word Analysis and Meaning
Combining Forms		
Color		
xanth/o		**xanth**/oma (ZĂN-thō-sĭs): _____ *-oma:* tumor *A xanthoma is a nodule or patch composed of lipoid material commonly associated with disturbances in lipid metabolism.*
cyan/o	blue	**cyan/o**/tic (sī-ăn-ŎT-ĭk): _____ *-tic:* pertaining to *Cyanosis is associated with lack of oxygen in the blood and a bluish discoloration of the skin.*
erythr/o	red	**erythr/o**/cyte (ĕ-RĬTH-rō-sīt): _____ *-cyte:* cell *An erythrocyte is a red blood cell.*

Medical Word Elements—cont'd

Element	Meaning	Word Analysis and Meaning
Combining Forms		
Color		
melan/o	black	**melan**/oma (mĕl-ă-NŌ-mă): _____ *-oma:* tumor *Melanoma is a malignancy that arises from melanocytes.*
poli/o	gray; gray matter (of the brain or spinal cord)	**poli/o**/myel/itis (pō-lē-ō-mī-ĕ-LĪ-tĭs): _____ *myel:* bone marrow; spinal cord *-itis:* inflammation

Medical Word Elements—cont'd

Element	Meaning	Word Analysis and Meaning
Combining Forms		
Other		
radi/o	radiation, x-ray; radius (lower arm bone on thumb side)	**radi/o**/logist (rā-dē-ŎL-ŏ-jĭst): _____ *-logist:* specialist in the study of *Radiologists are physicians who employ imaging techniques to help medical doctors diagnose and treat diseases.*
tom/o	to cut	**tom/o**/graphy (tō-MŎG-ră-fē): _____ *-graphy:* process of recording *Tomography is an imaging procedure that employs a computer to produce images that appear as cuts (or slices) of an organ or structure.*
viscer/o	internal organs	**viscer**/al (VĬS-ĕr-ăl): _____ *-al:* pertaining to

Medical Word Elements—cont'd

Element	Meaning	Word Analysis and Meaning
Suffixes		
-ar	pertaining to	lumb/ar (LŬM-băr): _____ *lumb/o:* loins (lower back)
-ic	pertaining to	gastr/ic (ĕp-ĭ-GĂS-trĭk): _____ *epi:* above, upon *-ic:* pertaining to
-graphy	process of recording	son/o/graphy (sō-NŎG-ră-fĕ): _____ *son/o:* sound *Sonography is a diagnostic imaging procedure using sound in the inaudible range (ultrasound) to visualize internal structures.*

Medical Word Elements—cont'd

Element	Meaning	Word Analysis and Meaning
Prefixes		
infra-	below, under	**infra**/cost/al (ĭn-fră-KŎS-tăl): _____ *cost:* ribs *–al:* pertaining to
peri-	around	**peri**/umbilic/al (pĕr-ē-ŭm-BĬL-ĭ-kăl): _____ *umbilic:* umbilicus, navel *–al:* pertaining to
super-	upper, above	**super**/ior (soo-PĒ-rē-or): _____ *–ior:* pertaining to *Superior is a directional term meaning toward the head or upper portion of a structure.*
ultra-	excess, beyond	**ultra**/son/ic (ŭl-tră-SŎN-ĭk): _____ *son:* sound *–ic:* pertaining to *Ultrasound includes sound frequencies too high to be perceived by the human ear.*

It is time to review medical word elements by completing Learning Activities 4-2 and 4-3.

Learning Activities

The activities that follow provide a review of the body structure terms introduced in this chapter.

Complete each activity from 4-1 to 4-3 and review your answers to evaluate your understanding of the chapter 4.



CHAPTER 5



Chapter Outline

- Objectives
- Anatomy and Physiology
  - Skin
    - Epidermis
    - Dermis
  - Accessory Organs (appendages) of the Skin
    - Glands
    - Hair
    - Nails

Medical Word Elements

Learning Activities

Objectives

Upon completion of this chapter, you will be able to:

- Locate the major organs of the integumentary system and describe their structure and function.
- Describe the functional relationship between the integumentary system and other body systems.
- Pronounce, spell, and build words related to the integumentary system.
- Explain pharmacology associated with the treatment of skin disorders.
- Demonstrate your knowledge of this chapter by completing the learning and documenting health-care activities.

Anatomy and Physiology

** The Skin

Also called the integument, is the largest organ in the body.

Together with its accessory Organs/appendages (hair, nails, and glands), the skin makes up the integumentary system.

This elaborate system of distinct tissues includes glands that produce several types of secretions, nerves that transmit impulses, and blood vessels that help regulate body temperature. The skin covers and protects all outer surfaces of the body and performs many vital functions, including the sense of touch.

The skin consists of 2 layers; the epidermis, and the dermis (corium) (See Fig. 5-1.)

1. The epidermis:
- It's the outermost layer of the skin.
- It is in a state of continuous shedding.
- Cells that are formed in the innermost part of the epidermis mature and are pushed to the outermost layer. These cells are finally sloughed and replaced by new cells.

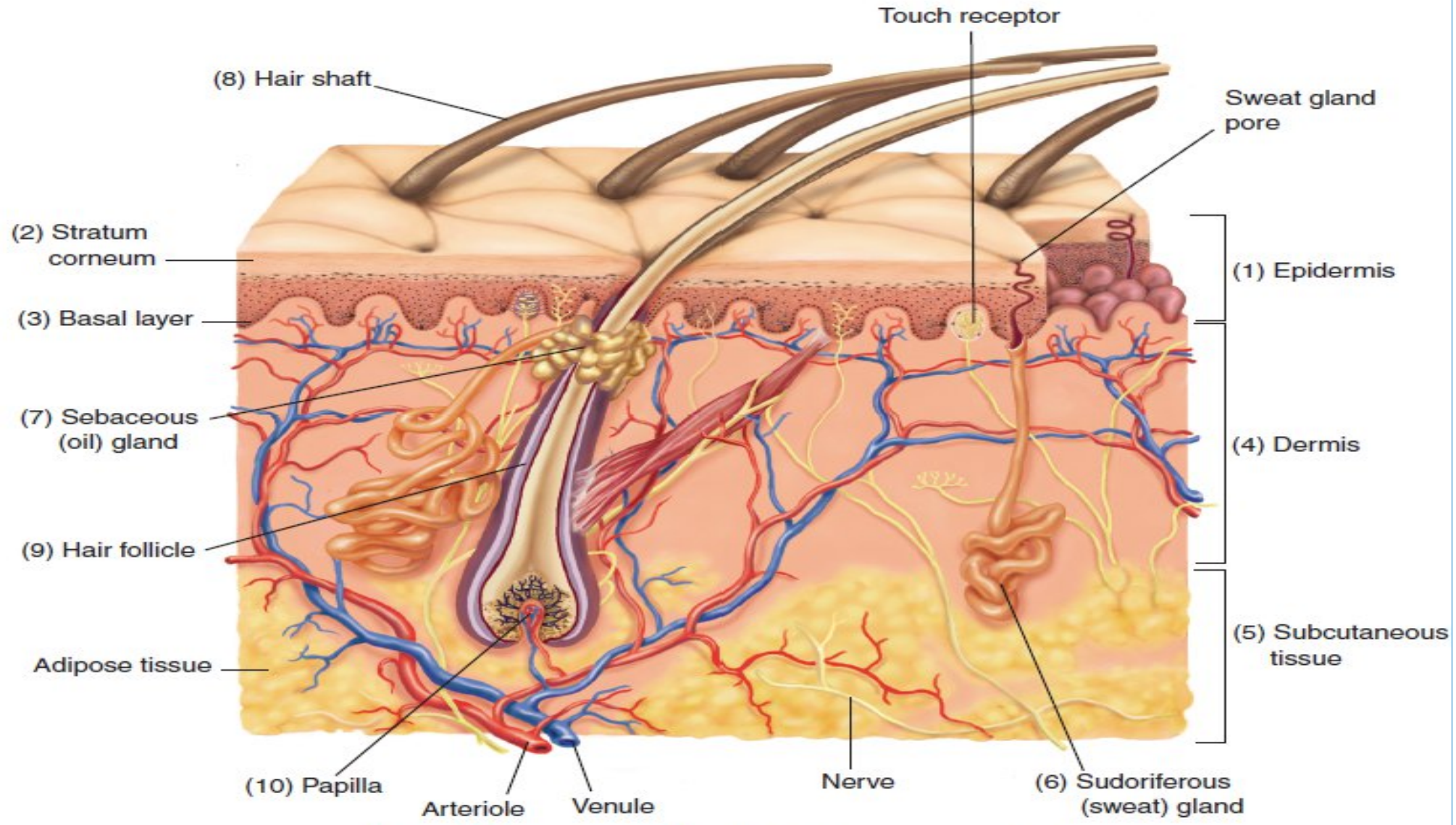


Figure 5-1 Structure of the skin and subcutaneous tissue.

Anatomy and Physiology (Cont.)

2. Dermis (Corium):

- It's the layer of skin lying immediately under the epidermis.
- It is composed of living tissue that consists of numerous capillaries, lymphatics, and nerve endings.
- Hair follicles, sebaceous glands, and sweat glands are also located in the dermis.

** Accessory Organs of the Skin

1. Hair:

Hair is distributed over the entire body, except the bottom (plantar surface) of the feet and the palms (ventral surface) of the hands.

The visible part of the hair is referred to as the shaft, while that which is embedded in the dermis is called the root, together with its coverings, forms the hair follicle.

At the bottom of the follicle is a loop of capillaries enclosed in a covering called the papilla (Fig. 5.1).

Anatomy and Physiology (Cont.)

The cluster of epithelial cells lying over the papilla reproduced and is responsible for the eventual formation of the hair shaft. As long as these cells remain alive, hair will regenerate even though it is cut or plucked or otherwise removed.

Hair is kept soft and flexible by sebaceous glands, which secrete varying amounts of oily sebum into the upper part of the hair follicle located near the surface or the skin.

2. Nails:

Nails protect the tips of the fingers and toes from bruises and injuries.

(See Fig. 5-2.) Each nail is formed in the (1) nail root . As the nail grows, it stays attached and slides forward over the layer of epithelium called the (2) nailbed. This epithelial layer is continuous with the epithelium of the skin.

Most of the (3) nail body appears pink because of the underlying vascular tissue

The half-moon–shaped area at the base of the nail, the (4) lunula, is the region where new growth occurs. The lunula has a whitish appearance because the vascular tissue underneath does not show through.

of the skin. Most of the (3) nail body

appears pink because of the underlyi

vascular tissue. The half-moon–shap

area at the base of the nail,

the (4) lunula, is the region where ne

growth occurs.

The lunula has a whitish appearance

because the vascular tissue underne

does not show through.

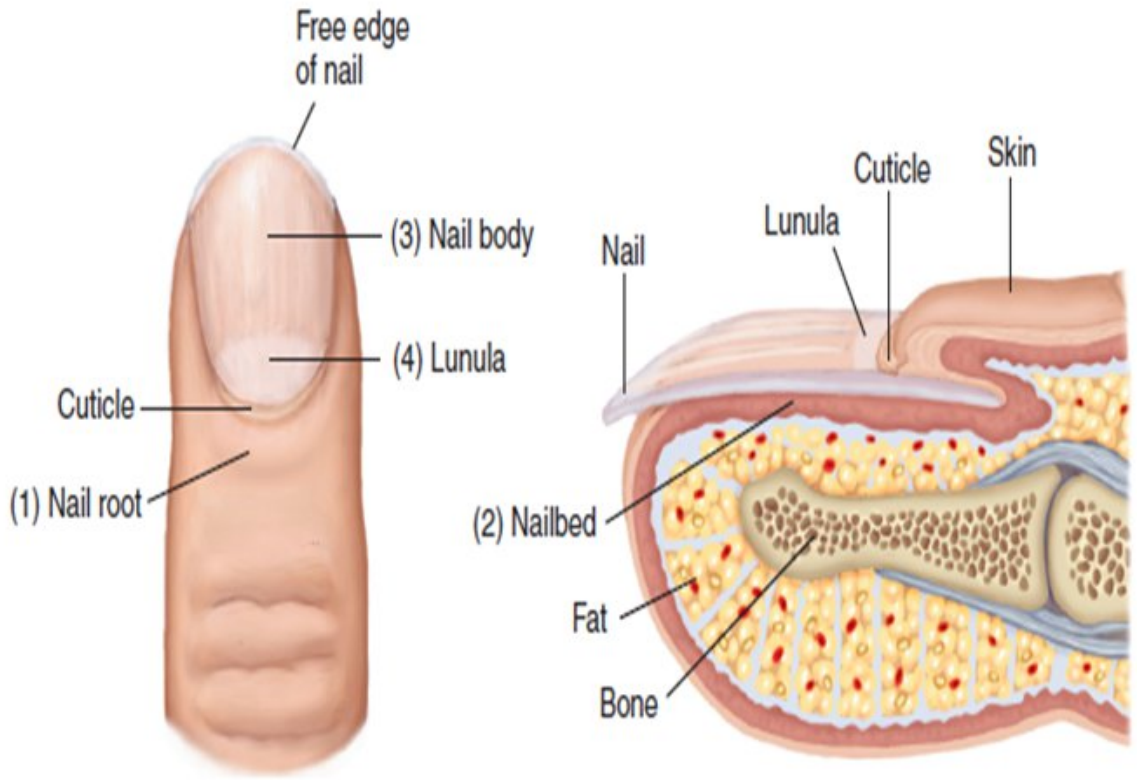


Figure 5-2 Structure of a fingernail.

Anatomy and Physiology (Cont.)

3. Glands

Three important glands located in the dermis produce secretions: The (6) sudoriferous (sweat) glands produce sweat, the (7) sebaceous (oil) glands produce oil, and ceraminous glands.

These glands are exocrine glands because they secrete substances through ducts to an outer surface of the body rather than directly into the bloodstream.

The sudoriferous glands secrete perspiration, or sweat, onto the surface of the skin through pores. Pores are most plentiful on the palms, soles, forehead, and axillae (armpits). The main functions of the sudoriferous glands are to cool the body by evaporation, excrete waste products, and moisten surface cells.

The sebaceous glands contain fatty droplets. As these cells disintegrate, they secrete an oily material called sebum. The acidic nature of sebum helps destroy harmful organisms on the skin, thus preventing infection.

Sebaceous glands are present over the entire body except on the soles of the feet and the palms of the hands. They are especially prevalent on the scalp and face; around such openings as the nose, mouth, external ear, and anus; and on the upper back.

Ceruminous glands: They are modified sweat glands located in the skin that lines the external auditory canal.

Medical Word Elements

This section introduces combining forms, suffixes, and prefixes related to the integumentary system. Word analyses are also provided. From the information provided, complete the meaning of each medical word in the right-hand column. The first one is completed for you.

Element	Meaning	Word Analysis
Combining Forms		
adip/o	fat	**adip**/osis (ăd-ĭ-PŌ-sĭs): *abnormal condition of fat* *-osis:* abnormal condition; increase (used primarily with blood cells) *Adiposis is an abnormal accumulation of fatty tissue in the body.*
lip/o		**lip**/oma (lĭ-PŌ-mă): _____ *-oma:* tumor
cutane/o	skin	sub/**cutane**/ous (sŭb-kū-TĀ-nē-ŭs): _____ *sub-:* under, below *-ous:* pertaining to
dermat/o		**dermat/o**/plasty (DĚR-mă-tō-plăs-tē): _____ *-plasty:* surgical repair
derm/o		hypo/**derm**/ic (hī-pō-DĚR-mĭk): _____ *hypo-:* under, below, deficient *-ic:* pertaining to *Hypodermic needles are used for subcutaneous injections.*

Medical Word Elements—cont'd

Element	Meaning	Word Analysis
Combining Forms		
hidr/o	sweat	**hidr**/aden/itis (hī-drăd-ĕ-NĪ-tĭs): _____ *aden:* gland *–itis:* inflammation *Do not confuse* hidr/o *(sweat) with* hydr/o *(water).*
sudor/o		**sudor**/esis (soo-dō-RĒ-sĭs): _____ *–esis:* condition *Sudoresis is the body's physiological means to regulate body temperature.*
ichthy/o	dry, scaly	**ichthy**/osis (ĭk-thē-Ō-sĭs): _____ *–osis:* abnormal condition; increase (used primarily with blood cells) *Ichthyosis can be any of several dermatological conditions in which the skin is dry and hardened (hyperkeratotic), resembling fish scales. A mild form of ichthyosis, called* winter itch, *is commonly seen on the legs of older patients, especially during the winter months.*
kerat/o	horny tissue; hard; cornea	**kerat**/osis (kĕr-ă-TŌ-sĭs): _____ *–osis:* abnormal condition; increase (used primarily with blood cells) *Keratosis is a thickened area of the epidermis or any horny growth on the skin, such as a callus or wart.*

Medical Word Elements—cont'd

Element	Meaning	Word Analysis
Combining Forms		
melan/o	black	**melan**/oma (mĕl-ă-NŌ-mă): _____ *-oma:* tumor *Melanoma is a malignant tumor of melanocytes that commonly begins in a darkly pigmented mole and can metastasize widely.*
myc/o	fungus (plural, fungi)	dermat/o/**myc**/osis (dĕr-mă-tō-mī-KŌ-sĭs): _____ *dermat/o:* skin *-osis:* abnormal condition; increase (used primarily with blood cells)
onych/o	nail	**onych/o**/malacia (ŏn-ĭ-kō-mă-LĀ-shē-ă): _____ *-malacia:* softening
ungu/o		**ungu**/al (ŬNG-gwăl): _____ *-al:* pertaining to
pil/o	hair	**pil/o**/nid/al (pī-lō-NĪ-dăl): _____ *nid:* nest *-al:* pertaining to *A pilonidal cyst commonly develops in the skin at the base of the spine. It develops as a growth of hair in a dermoid cyst.*
trich/o		**trich/o**/pathy (trĭk-ŎP-ă-thē): _____ *-pathy:* disease

Medical Word Elements—cont'd

Element	Meaning	Word Analysis
Combining Forms		
scler/o	hardening; sclera (white of eye)	**scler/o**/derma (sklĕ-rō-DĔR-mă): _____ *-derma:* skin *Scleroderma is an autoimmune disorder that causes the skin and internal organs to become progressively hardened as a result of deposits of collagen. It may occur as a localized form or as a systemic disease.*
seb/o	sebum, sebaceous	**seb/o**/rrhea (sĕb-ō-RĒ-ă): _____ *-rrhea:* discharge, flow *Seborrhea is an overactivity of the sebaceous glands.*
squam/o	scale	**squam**/ous (SKWĀ-mŭs): _____ *-ous:* pertaining to
xen/o	foreign, strange	**xen/o**/graft (ZĔN-ō-grăft): _____ *-graft:* transplantation *Xenografts are used as a temporary graft to protect the patient against infection and fluid loss.*
xer/o	dry	**xer/o**/derma (zē-rō-DĔR-mă): _____ *-derma:* skin *Xeroderma is a chronic skin condition characterized by dryness and roughness and is a mild form of ichthyosis.*

Medical Word Elements—cont'd

Element	Meaning	Word Analysis and Meaning
Suffixes		
-cyte	cell	lip/o/**cyte** (LĬP-ō-sīt): _____ *lip/o:* fat
-derma	skin	py/o/**derma** (pī-ō-DĚR-mă): _____ *py/o:* pus *Pyoderma is an acute, inflammatory, purulent bacterial dermatitis. It may be primary, such as impetigo, or secondary to a previous skin condition.*
-logist	specialist in the study of	dermat/o/**logist** (dĕr-mă-TŎL-ō-jĭst): _____ *dermat/o:* skin
-logy	study of	dermat/o/**logy** (dĕr-mă-TŎL-ō-jē): _____ *dermat/o:* skin
-therapy	treatment	cry/o/**therapy** (krī-ō-THĚR-ă-pē): _____ *cry/o:* cold *Cryotherapy is used to destroy tissue by freezing with liquid nitrogen. Cutaneous warts and actinic keratosis are common skin disorders that respond well to cryotherapy treatment.*

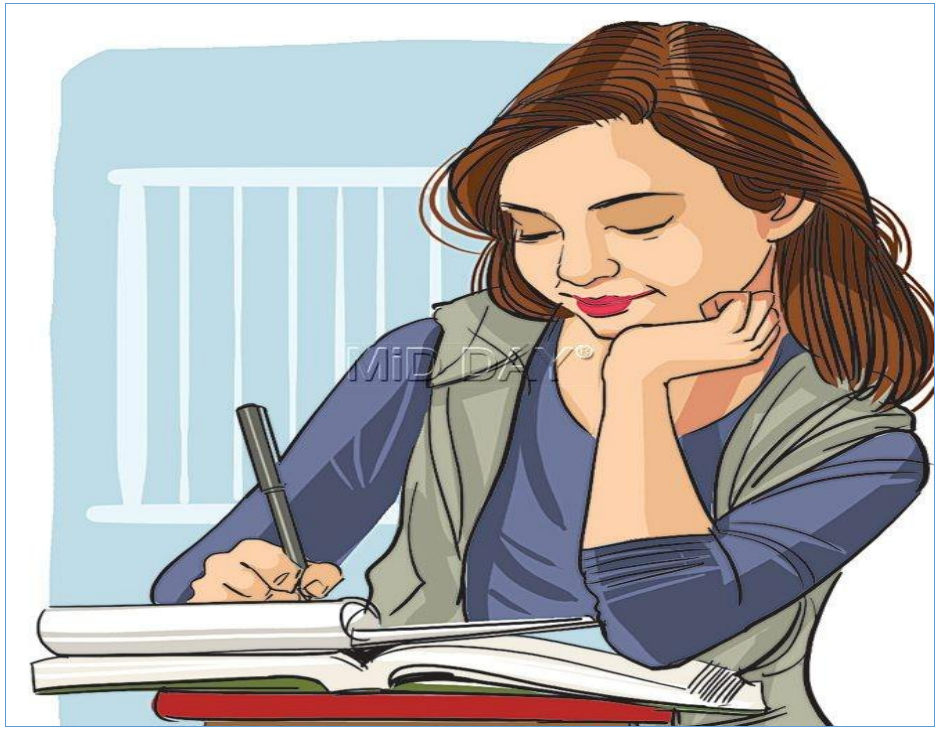
Medical Word Elements—cont'd

Element	Meaning	Word Analysis
Prefixes		
an-	without, not	**an**/hidr/osis (ăn-hī-DRŌ-sĭs): _____ *hidr:* sweat *-osis:* abnormal condition; increase (used primarily with blood cells)
epi-	above, upon	**epi**/derm/is (ĕp-ĭ-DĔR-mĭs): _____ *derm:* skin *-is:* noun ending *The epidermis is the outermost layer of the skin.*
homo-	same	**homo**/graft (HŌ-mō-grăft): _____ *-graft:* transplantation

It is time to review medical word elements by completing Learning Activities 5-1 and 5-2.

Learning Activities

The activities that follow provide a review of the a review of the integumentary system terms introduced in this chapter.



Complete each activity from 5-1 and 5-2 and review your answers to evaluate your understanding of chapter 5.

CHAPTER 6



Chapter Outline

- Objectives
- Anatomy and Physiology
  - Muscles
    - Muscle Types
  - Bones
    - Bone Types
- Medical Word Elements
- Learning Activities

Objectives

Upon completion of this chapter, you will be able to:

- Locate and describe the structures of the musculoskeletal system.
- Pronounce, spell, and build words related to the musculoskeletal system.
- Demonstrate your knowledge of this chapter by completing the learning and documenting health-care activities.

Anatomy and Physiology

The musculoskeletal system includes muscles, bones, joints, and related structures, such as the tendons and connective tissue that function in the support and movement of body parts and organs. (See Fig. 10-1.)

** Muscles

Muscle tissue is composed of contractile cells, or fibers, that provide movement of an organ or body part. Muscles contribute to posture, produce body heat, and act as a protective covering for internal organs.

Muscles have the ability to contract, relax, and return to their original size and shape. Whether muscles are attached to bones or found in internal organs and blood vessels, their primary responsibility is movement.

Apparent motions provided by muscles include walking and talking.

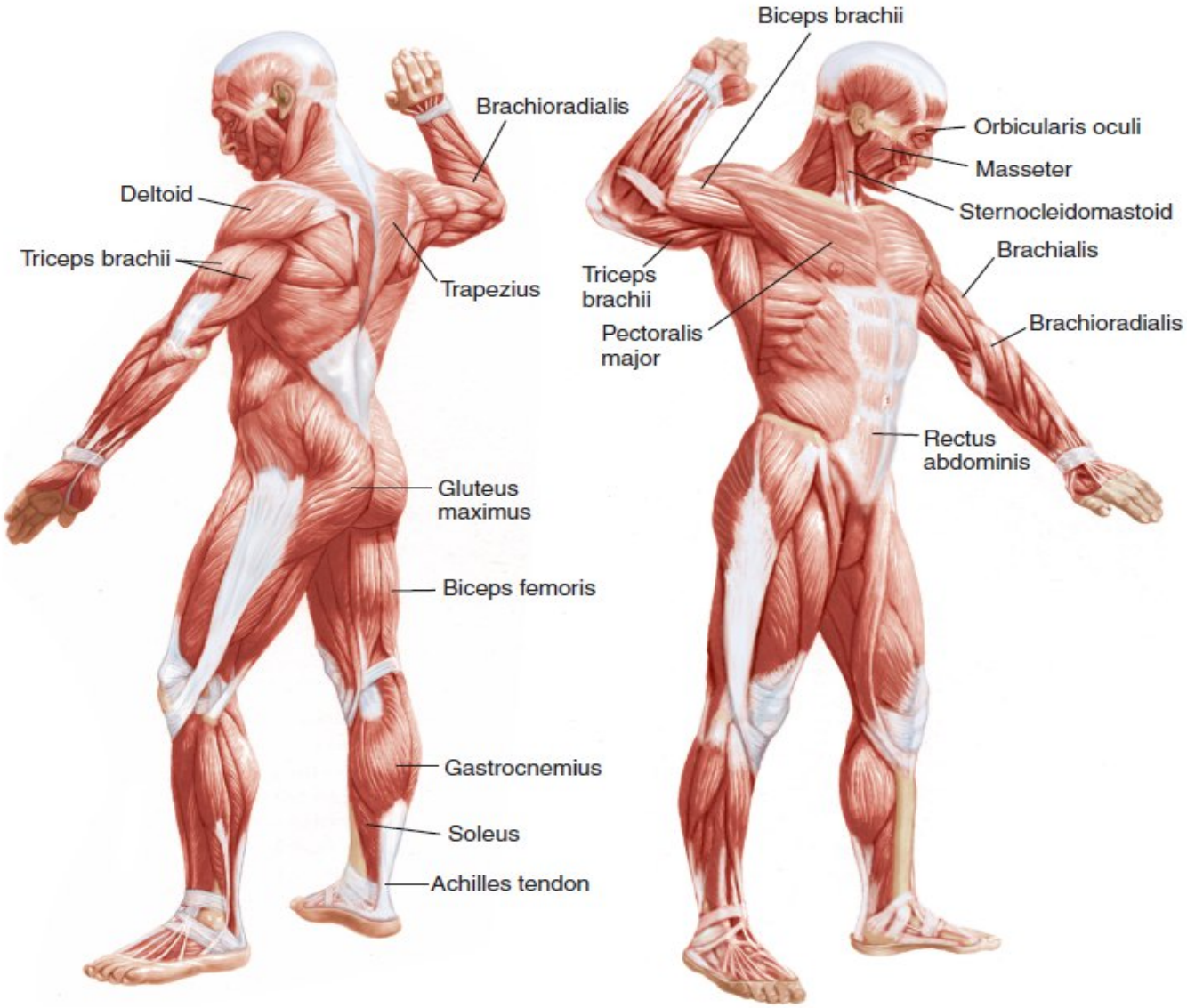


Figure 10-1 Selected muscles of the body.

(Cont.)

* Types of Muscles:

There are three types of muscle tissue in the body:

- Skeletal muscles:

They are attached to bones and provide the means for movement. Skeletal muscles are voluntary muscles that contract and relax in response to conscious thought. Some examples of voluntary muscles are muscles that move the eyeballs, tongue, and bones.

- Smooth muscles:

They are mainly responsible for assisting internal processes, such as digestion, circulation, and urination. Thus, they are called involuntary or visceral muscles. Because their movement is not under conscious control but functions under the control of the autonomic (involuntary) nervous system. Some examples of involuntary muscles are those in the digestive tract that propel food through the alimentary canal and those in the urinary system that control urination.

- Cardiac muscle:

(Cont.)

** Bones

Bones provide the framework of the body, protect internal organs, allow for movement, store calcium and other minerals, and produce red blood cells within bone marrow (hematopoiesis). The bones of the skull protect the brain; the rib cage protects the heart and lungs; the pelvic bones protect the developing fetus and reproductive organs. Movement is possible because bones provide points of attachment for muscles, tendons, and ligaments. Bones serve as a storehouse for minerals, particularly phosphorus and calcium.

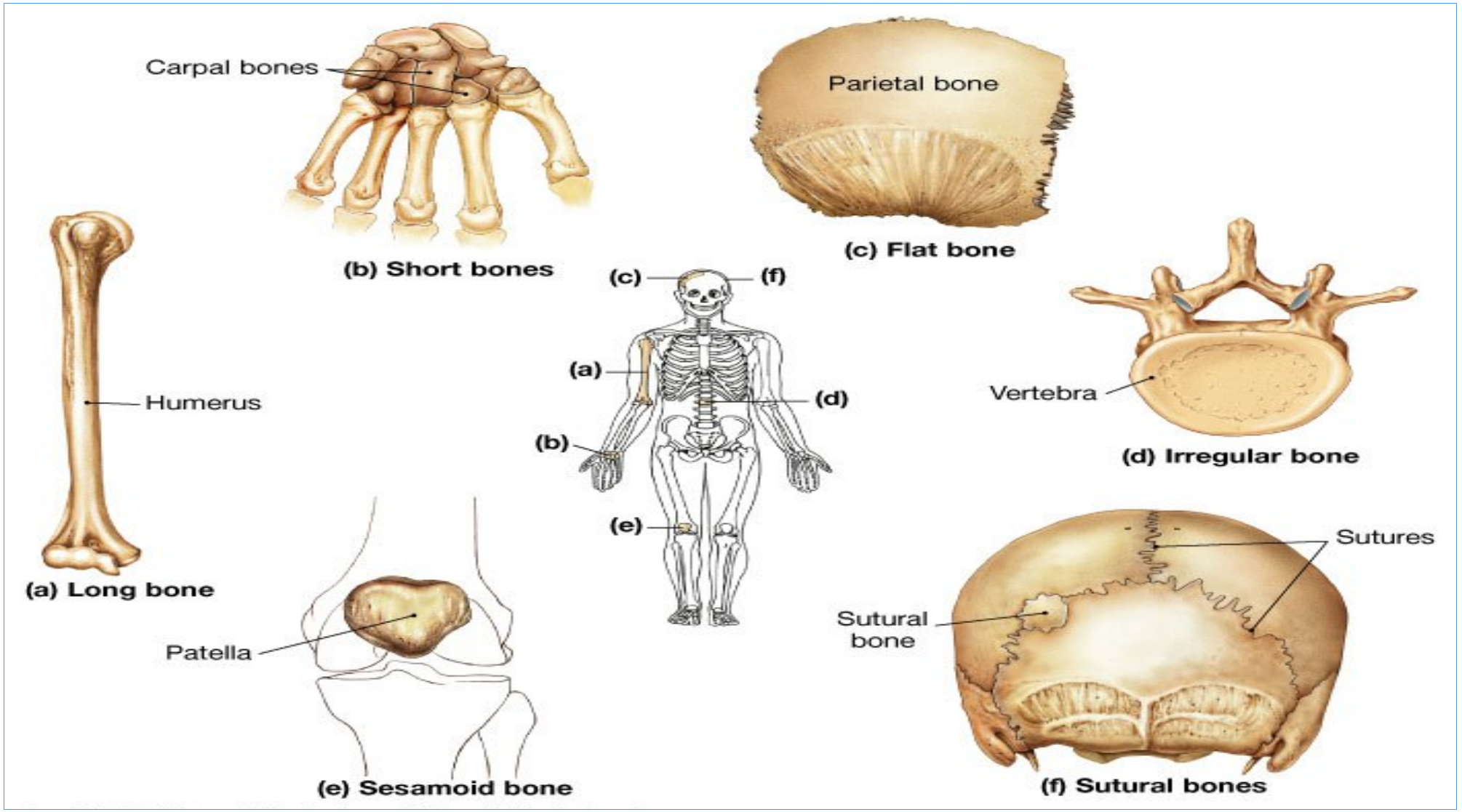
* Bone Types

There are four principal types of bone: (See Fig. 10-1)

1- Short bones:

They are somewhat cube-shaped. Examples of short bones include the bones of the wrist (carpals) and ankles (tarsals).

2- Irregular bones include the bones that cannot be classified as short or long because of their complex shapes. Examples of irregular bones include



(Fig. 10-1) Types of bones by shape

Anatomy and Physiology (Cont.)

3- Flat bones:

They are exactly what their name suggests. They provide broad surfaces for muscular attachment or protection for internal organs. Examples of flat bones include bones of the skull, shoulder blades, and sternum.

4- Sesamoid Bone:

It has a sesamoid shape, and only one bone like of this type "the patella or the knee cap".

5- Long bones:

They are found in the extremities of the body, such as the legs, arms, and fingers. The long bones have regular, well-defined shapes. (See Fig. 10-2) Each long bone has three main parts:

• The (1) diaphysis is the shaft, or long, main portion of a bone. It consists of compact bone that forms a cylinder and surrounds a central canal called the (2) medullary cavity.

The medullary cavity is filled with "yellow bone marrow."

• The (3) distal epiphysis and (4) proximal epiphysis (plural, epiphyses) are the

Anatomy and Physiology (Cont.)

- (5) Articular cartilage, which is a thin outer layer of cartilage where bones meet to form joints,

- Thick, dense layer of hard (7) compact bone

- Inner layer of porous tissue called (8) spongy or cancellous bone that is less dense than

compact bone and is filled with "red bone marrow ".

• The (9) periosteum, a dense, white, fibrous membrane, covers the remaining surface of the bone except the articular cartilage. It contains numerous blood and lymph vessels and nerves.

In growing bones, the inner layer of periosteum contains the bone-forming cells known as osteoblasts. The periosteum provides a means for bone repair and general bone nutrition. The periosteum also serves as a point of attachment for muscles, ligaments, and tendons.

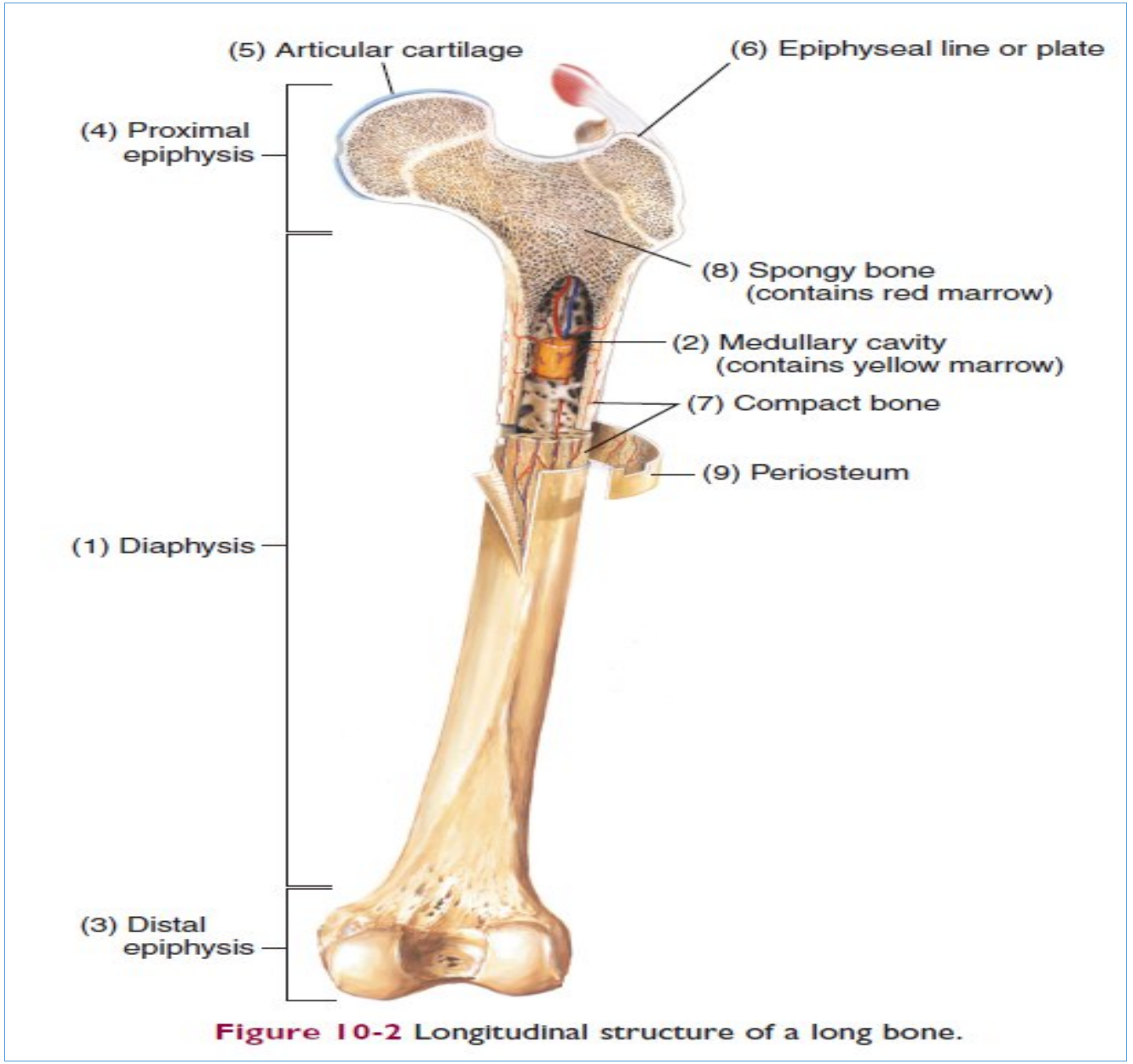


Figure 10-2 Longitudinal structure of a long bone.

Medical Word Elements

This section introduces combining forms, suffixes, and prefixes related to the musculoskeletal system. Word analyses are also provided. From the information provided, complete the meaning of the medical words in the right-hand column. The first one is completed for you.

Element	Meaning	Word Analysis and Meaning
Combining Forms		
Muscular System		
muscul/o	muscle	**muscul**/ar (MŬS-kū-lăr): _____ *-ar:* pertaining to
my/o		**my**/oma (mī-Ō-mă): _____ *-oma:* tumor
Skeletal System		
Bones of the Upper Body		
brachi/o	arm	**brachi**/algia (brā-kē-ĂL-jē-ă): _____ *-algia:* pain
carp/o	carpus (wrist bone)	**carp/o**/ptosis (kăr-pŏp-TŌ-sĭs): _____ *-ptosis:* prolapse, downward displacement *Carpoptosis is commonly called* wrist drop.

Medical Word Elements—cont'd

Element	Meaning	Word Analysis and Meaning
Combining Forms		
cephal/o	head	**cephal**/ad (SĚF-ă-lăd): _____ *–ad:* toward
cervic/o	neck; cervix uteri (neck of the uterus)	**cervic/o**/dynia (sĕr-vĭ-kō-DĬN-ē-ă): _____ *–dynia:* pain *Cervicodynia is also called* cervical neuralgia.
clavicul/o	clavicle (collar bone)	**clavicul**/ar (klă-VĬK-ū-lăr): _____ *–ar:* pertaining to
cost/o	ribs	**cost**/ectomy (kŏs-TĚK-tō-mē): _____ *–ectomy:* excision, removal
crani/o	cranium (skull)	**crani/o**/tomy (krā-nē-ŎT-ō-mē): _____ *–tomy:* incision
dactyl/o	fingers; toes	**dactyl**/itis (dăk-tĭl-Ī-tĭs): _____ *–itis:* inflammation
humer/o	humerus (upper arm bone)	**humer/o**/scapul/ar (hū-mĕr-ō-SKĂP-ū-lăr): _____ *scapul:* scapula (shoulder blade) *–ar:* pertaining to
metacarp/o	metacarpus (hand bones)	**metacarp**/ectomy (mĕt-ă-kăr-PĚK-tō-mē): _____ *–ectomy:* excision, removal

Medical Word Elements—cont'd

Element	Meaning	Word Analysis and Meaning
Combining Forms		
phalang/o	phalanges (bones of the fingers and toes)	**phalang**/ectomy (făl-ăn-JĔK-tō-mē): _____ *-ectomy:* excision, removal
radi/o	radiation, x-ray; radius (lower arm bone on the thumb side)	**radi**/al (RĀ-dē-ăl): _____ *-al:* pertaining to
spondyl/o	vertebrae (backbone)	**spondyl**/itis (spŏn-dĭl-Ī-tĭs): _____ *-itis:* inflammation *The combining form* spondyl/o *describes diseases and conditions.*
vertebr/o		inter/**vertebr**/al (ĭn-tĕr-VĔRT-ĕ-brĕl): _____ *inter-:* between *-al:* pertaining to *The combining form* vertebr/o *indicates anatomical terms.*
stern/o	sternum (breastbone)	**stern**/ad (STĔR-năd): _____ *-ad:* toward
thorac/o	chest	**thorac/o**/dynia (thō-răk-ō-DĬN-ē-ă): _____ *-dynia:* pain

Medical Word Elements—cont'd

Element	Meaning	Word Analysis and Meaning
Combining Forms		
Bones of the Lower Body		
calcane/o	calcaneum (heel bone)	**calcane/o**/dynia (kăl-kā-nē-ō-DĬN-ē-ă): _____ *-dynia:* pain
femor/o	femur (thigh bone)	**femor**/al (FĔM-or-ăl): _____ *-al:* pertaining to
fibul/o	fibula (smaller bone of the lower leg)	**fibul/o**/calcane/al (fĭb-ū-lō-kăl-KĀ-nē-ăl): _____ *calcane:* calcaneum (heel bone) *-al:* pertaining to
ili/o	ilium (lateral, flaring portion of the hip bone)	**ili/o**/pelv/ic (ĭl-ē-ō-PĔL-vĭk): _____ *pelv:* pelvis *-ic:* pertaining to
ischi/o	ischium (lower portion of the hip bone)	**ischi/o**/dynia (ĭs-kē-ō-DĬN-ē-ă): _____ *-dynia:* pain

Medical Word Elements—cont'd

Element	Meaning	Word Analysis and Meaning
Combining Forms		
lumb/o	loins (lower back)	**lumb/o**/dynia (lŭm-bō-DĬN-ē-ă): _____ *-dynia:* pain
metatars/o	metatarsus (foot bones)	**metatars**/algia (mĕt-ă-tăr-SĂL-jē-ă): _____ *-algia:* pain *Metatarsalgia radiates from the head of the metatarsus and worsens with weight-bearing activity or palpation.*
patell/o	patella (kneecap)	**patell**/ectomy (păt-ĕ-LĔK-tō-mē): _____ *-ectomy:* excision, removal
pelv/i	pelvis	**pelv/i**/metry* (pĕl-VĬM-ĕt-rē): _____ *-metry:* act of measuring *Pelvimetry is routinely performed in obstetrical management.*
pelv/o		**pelv**/ic (PĔL-vĭc): _____ *-ic:* pertaining to *A woman's pelvis is usually less massive but wider and more circular than a man's pelvis.*

Medical Word Elements—cont'd

Element	Meaning	Word Analysis and Meaning
Combining Forms		
pod/o	foot	**pod**/iatry (pō-DĪ-ă-trē): _____ *-iatry:* medicine, treatment
pub/o	pubis (anterior part of the pelvic bone)	**pub/o**/coccyg/eal (pū-bō-k ŏk-SĬJ-ē-ăl): _____ *coccyg:* coccyx (tailbone) *-eal:* pertaining to
tibi/o	tibia (larger bone of the lower leg)	**tibi/o**/femor/al (tĭb-ē-ō-FĔM-or-ăl): _____ *femor:* femur *-al:* pertaining to
Other		
ankyl/o	stiffness; bent, crooked	**ankyl**/osis (ăng-k ĭ-LŌ-sĭs): _____ *-osis:* abnormal condition; increase (used primarily with blood cells) *Ankylosis results in immobility and stiffness of a joint. It may be the result of trauma, surgery, or disease and most commonly occurs in rheumatoid arthritis.*
arthr/o	joint	**arthr**/itis (ăr-THRĪ-tĭs): _____ *-itis:* inflammation

Medical Word Elements—cont'd

Element	Meaning	Word Analysis and Meaning
Combining Forms		
chondr/o	cartilage	**chondr**/itis (kŏn-DRĪ-tĭs): _____ –*itis:* inflammation
fasci/o	band, fascia (fibrous membrane supporting and separating muscles)	**fasci/o**/plasty (FĂSH-ē-ō-plăs-tē): _____ –*plasty:* surgical repair
fibr/o	fiber, fibrous tissue	**fibr**/oma (fī-BRŌ-mă): _____ –*oma:* tumor
kyph/o	humpback	**kyph**/osis (kī-FŌ-sĭs): _____ –*osis:* abnormal condition; increase (used primarily with blood cells)
lamin/o	lamina (part of vertebral arch)	**lamin**/ectomy (lăm-ĭ-NĔK-tō-mē): _____ –*ectomy:* excision, removal *Laminectomy is usually performed to relieve compression of the spinal cord or remove a lesion or herniated disk.*
lord/o	curve, swayback	**lord**/osis (lor-DŌ-sĭs): _____ –*osis:* abnormal condition; increase (used primarily with blood cells)

Medical Word Elements—cont'd

Element	Meaning	Word Analysis and Meaning
Combining Forms		
myel/o	bone marrow; spinal cord	**myel/o**/cyte (MĪ-ĕl-ō-sīt): _____ *-cyte:* cell
orth/o	straight	**orth/o**/ped/ist (or-thō-PĒ-dĭst): _____ *ped:* foot; child *-ist:* specialist *Historically, an orthopedist corrected deformities and straightened children's bones. In today's medical practice, however, the orthopedist treats musculoskeletal disorders and associated structures in persons of all ages.*
oste/o	bone	**oste**/oma (ŏs-tē-Ō-mă): _____ *-oma:* tumor *Osteomas are benign bony tumors.*
ped/o	foot; child	**ped/o**/graph (PĔD-ō-grăf): _____ *-graph:* instrument for recording *A pedograph produces an imprint of the foot and studies the gait (manner of walking).*
ped/*i*		**ped/i**/cure** (PĔD-ĭ-kūr): _____
scoli/o	crooked, bent	**scoli**/osis (skō-lē-Ō-sĭs): _____ *-osis:* abnormal condition; increase (used primarily with blood cells) *Scoliosis is characterized by a lateral spinal curvature.*

Medical Word Elements—cont'd

Element	Meaning	Word Analysis and Meaning
Combining Forms		
	membrane, synovial fluid	*-ectomy:* excision, removal
ten/o	tendon	**ten/o**/desis (tĕn-ŌD-ĕ-sĭs): _____ *-desis:* binding, fixation (of a bone or joint)
tend/o		**tend/o**/plasty (TĚN-dō-plăs-tē): _____ *-plasty:* surgical repair
tendin/o		**tendin**/itis (tĕn-dĭn-Ī-tĭs): _____ *-itis:* inflammation
Suffixes		
-asthenia	weakness, debility	my/**asthenia** (mī-ăs-THĒ-nē-ă): _____ *my:* muscle
-clasia	to break; surgical fracture	oste/o/**clasia** (ŏs-tē-ō-KLĀ-zē-ă): _____ *oste/o:* bone *Osteoclasia is the intentional fracture of a bone to correct a deformity and is also called* osteoclasis.

Medical Word Elements—cont'd

Element	Meaning	Word Analysis and Meaning
Suffixes		
-clasia	to break; surgical fracture	oste/o/**clasia** (ŏs-tē-ō-KLĀ-zē-ă): _____ *oste/o:* bone *Osteoclasia is the intentional fracture of a bone to correct a deformity and is also called* osteoclasis.
-clast	to break; surgical fracture	oste/o/**clast** (ŎS-tē-ō-klăst): _____ *oste/o:* bone *An osteoclast is a cell that breaks down the matrix of bone. Osteoblasts and osteoclasts work together to maintain a constant bone size in adults. An osteoclast also refers to an instrument used to surgically fracture a bone (osteoclasis).*
-desis	binding, fixation (of a bone or joint)	arthr/o/**desis** (ăr-thrō-DĒ-sĭs): _____ *arthr/o:* joint *This procedure immobilizes a joint to relieve intractable pain.*
-malacia	softening	chondr/o/**malacia** (kŏn-drō-măl-Ā-shē-ă): _____ *chondr/o:* cartilage *Chondromalacia is a deterioration of the articular cartilage, usually involving the patella.*

Medical Word Elements—cont'd

Element	Meaning	Word Analysis and Meaning
Suffixes		
-porosis	porous	oste/o/**porosis** (ŏs-tē-ō-pŏ-RŌ-sĭs): _____ *oste/o:* bone *Osteoporosis is a disorder characterized by loss of bone density. It may cause pain, especially in the lower back; pathological fractures; loss of stature; and hairline fractures.*
-sarcoma	malignant tumor of connective tissue	chondr/o/**sarcoma** (kŏn-drō-săr-KŌ-mă): _____ *chondr/o:* cartilage *Connective tissue is found in cartilage, fat, blood vessels, bone, nerves, and muscles.*
a-	without, not	**a**/trophy (ĂT-rō-fē): _____ *-trophy:* development, nourishment *Atrophy causes a wasting or decrease in size or physiological activity of a part of the body because of disease or other influences.*
dys-	bad; painful; difficult	**dys**/trophy (DĬS-trō-fē): _____ *-trophy:* development, nourishment
syn-	union, together, joined	**syn**/dactyl/ism (sĭn-DĂK-tĭl-ĭzm): _____ *dactyl:* fingers, toes *-ism:* condition *Syndactylism results in a fusion of two or more fingers or toes.*

It is time to review medical word elements by completing Learning Activities 10-1 and 10-2.

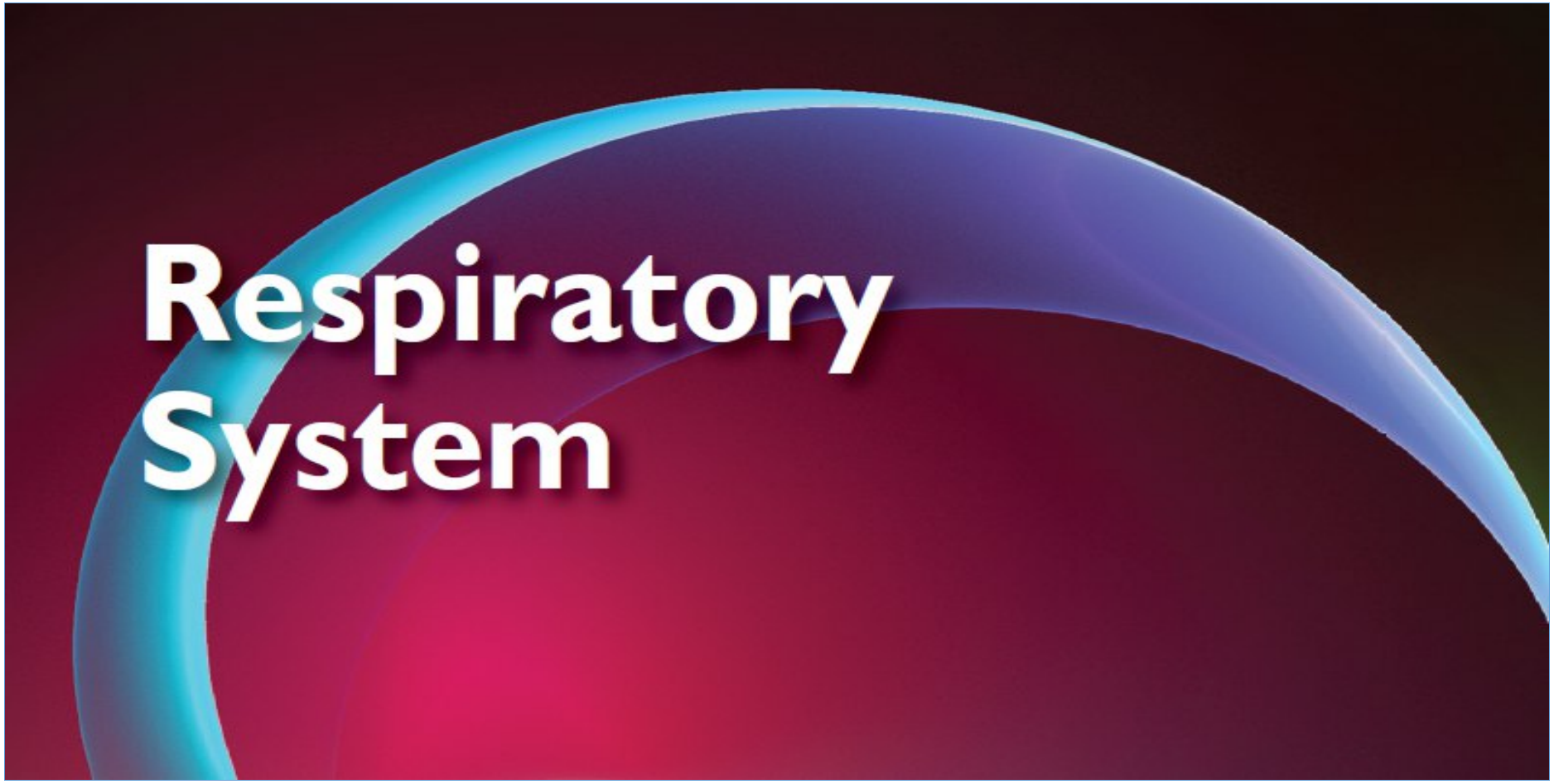
Learning Activities

The activities that follow provide a review of the a review of the musculoskeletal system terms introduced in this chapter.

Complete each activity from 10-1 and 10-2 and review your answers to evaluate your understandir of chapter 6.



CHAPTER7



Chapter Outline

- Objectives
- Anatomy and Physiology
  - Anatomy and Physiology Key Terms
  - Upper Respiratory Tract
  - Lower Respiratory Tract
  - Pulmonary Respiration
- Medical Word Elements
- Learning Activities

Objectives

Upon completion of this chapter, you will be able to:

- Locate and describe the structures of the respiratory system.
- Pronounce, spell, and build words related to the respiratory system.
- Demonstrate your knowledge of this chapter by completing the learning

  and documenting health-care activities.

Anatomy and Physiology

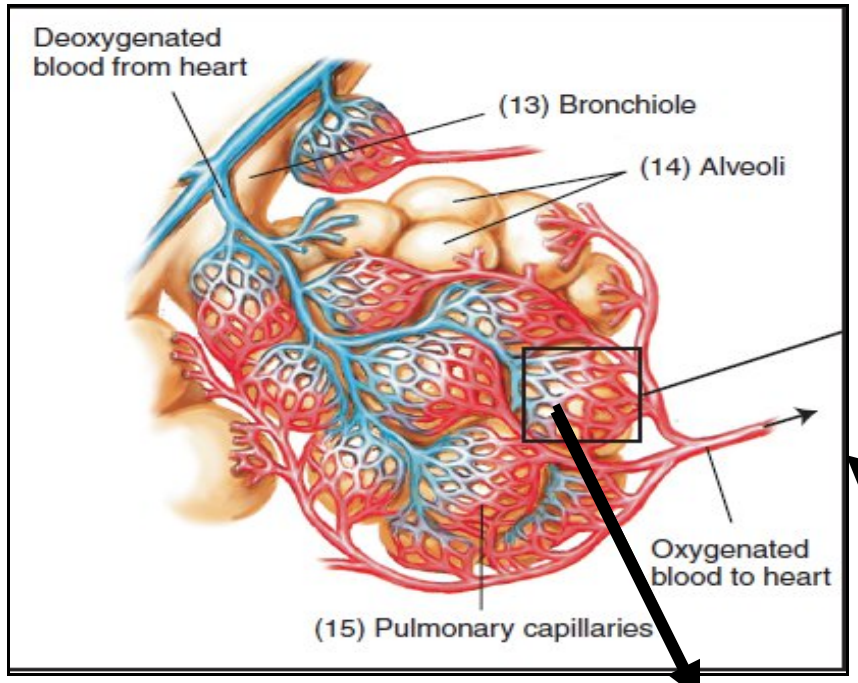
The respiratory system is responsible for the exchange of oxygen (O2) and carbon dioxide (CO2).

Oxygen is essential for life. It is carried to all cells of the body in exchange for CO2, a waste product.

The lungs and airways transport oxygen-enriched air from the atmosphere to the lungs and carry waste CO2 from the lungs to the atmosphere by a process called breathing (ventilation).

** Upper Respiratory Tract (outside the lungs)

The breathing process begins with inhalation. (See Fig. 7-1.) Air is drawn into the (1) nasal cavity, a chamber lined with mucous membranes and tiny hairs called cilia. Here, air is filtered, heated, and moistened to prepare it for its journey to the lungs. The nasal cavity is divided into a right and left side by a vertical partition of cartilage called the nasal septum.



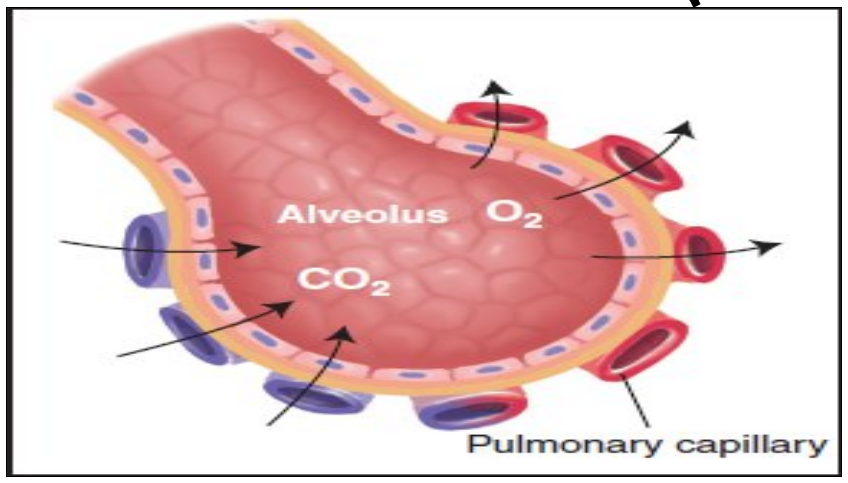
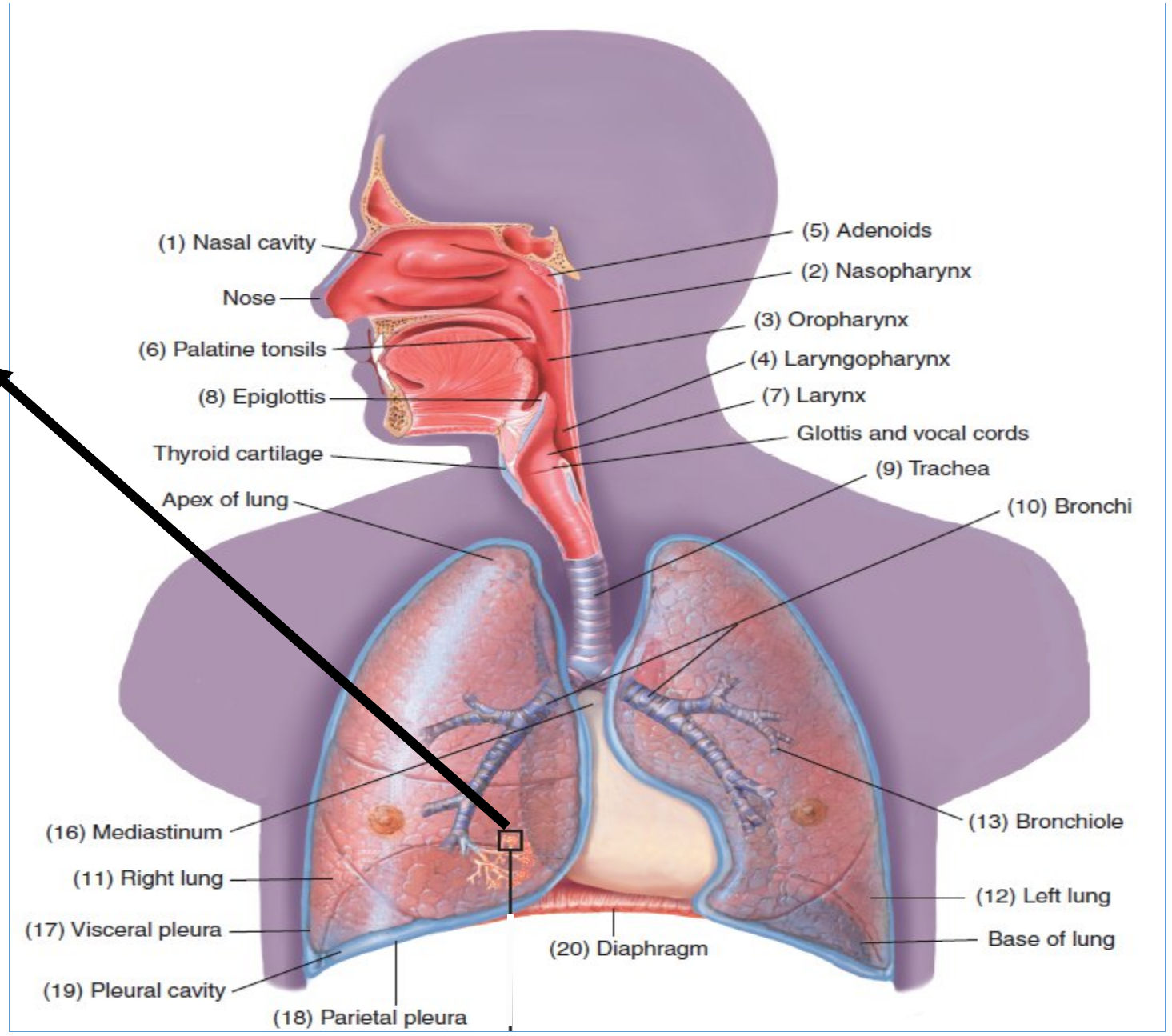


Figure 7-1 Anterior view of the upper and lower respiratory tracts.



Anatomy and Physiology (Cont.)

Air passes from the nasal cavity to the throat (pharynx), a muscular tube that serves as a passageway for food and air.

The pharynx consists of three sections:

- The (2) nasopharynx, posterior to the nose.
- The (3) oropharynx, posterior to the mouth.
- and the (4) laryngopharynx, superior to the larynx.

Within the nasopharynx is a collection of lymphoid tissue known as (5) adenoids (pharyngeal tonsils).

The (6) palatine tonsils, more commonly known as tonsils, are located in the oropharynx. They protect the opening to the respiratory tract from microscopic organisms that may enter this route.

The (7) larynx (voice box) contains the structures that make vocal sounds possible.

Anatomy and Physiology (Cont.)

A leaf-shaped structure on top of the larynx, the (8) epiglottis, seals off the air passage to the lungs during swallowing. This function ensures that food or liquids do not obstruct the flow of air to the lungs.

The larynx is a short passage that joins the pharynx with the (9) trachea (windpipe).

The trachea is composed of smooth muscle embedded with C-shaped rings of cartilage, which provide rigidity to keep the air passage open.

** Lower Respiratory Tract (within the lungs)

The trachea divides into two branches called (10) bronchi (singular, bronchus). One branch leads to the (11) right lung and the other to the (12) left lung. The inner walls of the trachea and bronchi are composed of mucous membrane (mucosa) embedded with cilia. This membrane traps incoming particles, and the cilia move the entrapped material upward into the pharynx, where it is expelled by coughing, sneezing, or swallowing.

Like the trachea, bronchi contain C-shaped rings of cartilage. Each bronchus divides

Anatomy and Physiology (Cont.)

At the end of the bronchioles are tiny air sacs called (14) alveoli (singular, alveolus). An alveolus resembles a small balloon because it expands and contracts with inflow and outflow of air.

The (15) pulmonary capillaries lie next to the thin tissue membranes of the alveoli. Carbon dioxide diffuses from the blood within the pulmonary capillaries and enters the alveolar spaces, and O2 from the alveoli diffuses into the blood. After the exchange of gases, freshly oxygenated blood returns to the heart. Oxygen is now ready for delivery to all body tissues.

The lungs are divided into lobes: three lobes in the right lung and two lobes in the left lung. The space between the right and left lungs is called the (16) mediastinum. It contains the heart, aorta, esophagus, and bronchi.

A serous membrane, the pleura, covers the lobes of the lungs and folds over to line the walls of the thoracic cavity. The membrane lying closest to the lung is the (17) visceral pleura; the membrane that lines the thoracic cavity is the (18) parietal pleura.

The space between these two membranes is the (19) pleural cavity. It contains a small amount of lubricating fluid, which permits the visceral pleura to glide

Anatomy and Physiology (Cont.)

Ventilation depends on a pressure differential between the atmosphere and chest cavity. A large muscular partition, the (20) diaphragm, lies between the chest and abdominal cavities. The diaphragm assists in changing the volume of the thoracic cavity to produce the needed pressure differential for ventilation. When the diaphragm contracts, it partially descends into the abdominal cavity, thus decreasing the pressure within the chest and drawing air into the lungs (inspiration). When the diaphragm relaxes, it slowly reenters the thoracic cavity, thus increasing the pressure within the chest. As pressure increases, air leaves the lungs (expiration). The intercostal muscles assist the diaphragm in changing the volume of the thoracic cavity by elevating and lowering the rib cage.

(See Fig. 7-2.)

** Pulmonary Respiration

Pulmonary respiration is the process by which O2 is taken from air and carried to body cells for their use, and CO2 and waste products generated by these cells, are

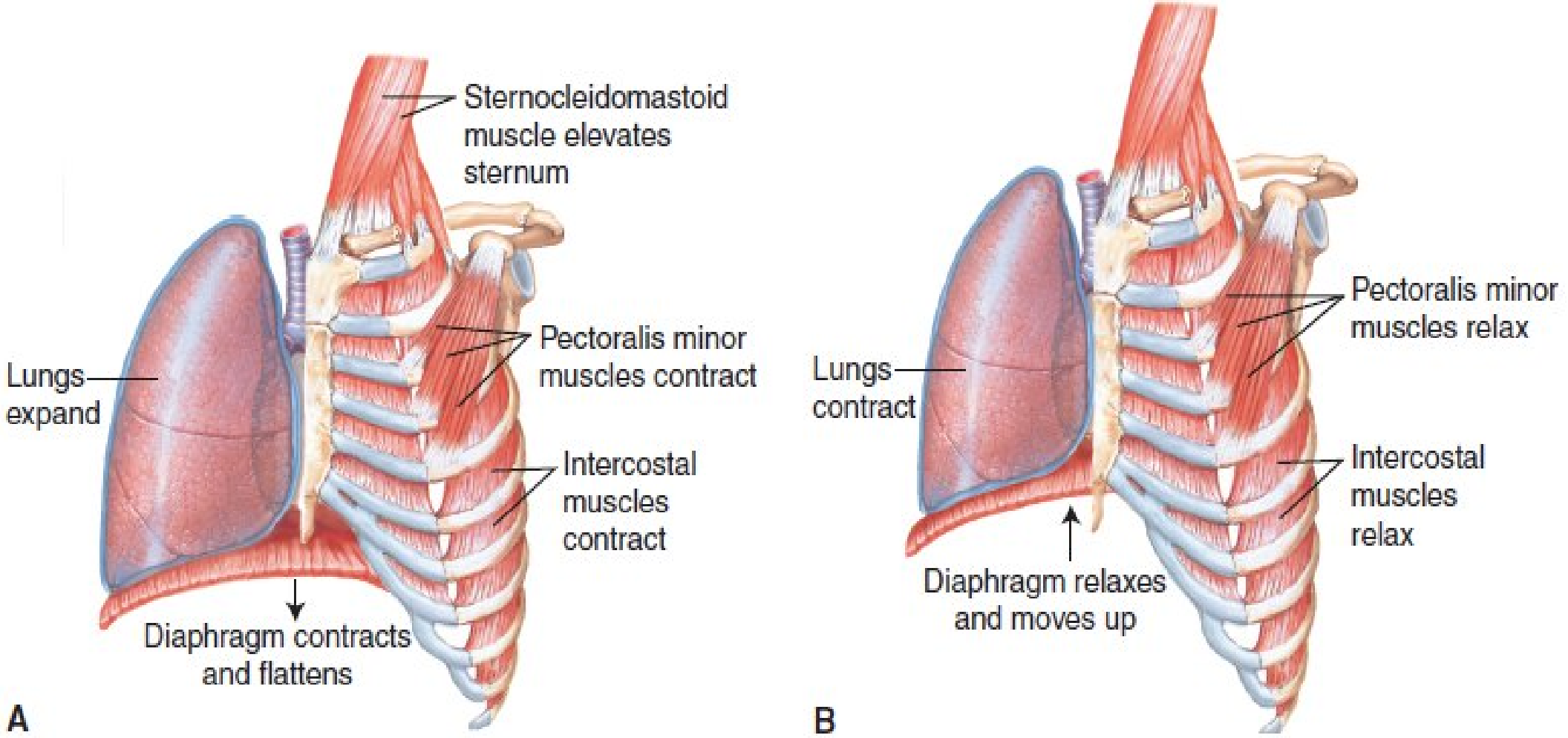


Figure 7-2 Breathing muscles. (A) Inspiration. (B) Expiration.

Anatomy and Physiology (Cont.)

Respiration includes four separate processes:

- Pulmonary ventilation (breathing), a largely involuntary action that moves air into (inspiration) and out of (expiration) the lungs in response to changes in blood O2 and CO2 levels and nervous stimulation of the diaphragm and intercostal muscles
- External respiration, the exchange of O2 and CO2 between the alveoli and the blood in the pulmonary capillaries
- Transport of respiratory gases, the movement of O2 to body cells and CO2 to the lungs by means of the cardiovascular system
- Internal respiration, the exchange of O2 and CO2 between body cells and the blood in systemic capillaries.

Medical Word Elements—cont'd

Element	Meaning	Word Analysis
Combining Forms		
Upper Respiratory Tract		
nas/o	nose	**nas**/al (NĀ-zl): *pertaining to the nose* *–al:* pertaining to
rhin/o		**rhin/o**/plasty (RĪ-nō-plăs-tē): _____ *–plasty:* surgical repair *Rhinoplasty is performed to correct birth defects or for cosmetic purposes.*
sept/o	septum	**sept/o**/plasty (SĔP-tō-plăs-tē): _____ *–plasty:* surgical repair *Septoplasty is commonly performed to correct a deviated septum.*
sinus/o	sinus, cavity	**sinus/o**/tomy (sī-nŭs-ŎT-ō-mē): _____ *–tomy:* incision *Sinusotomy is performed to improve breathing or for drainage in unresponsive sinusitis.*
pharyng/o	pharynx (throat)	**pharyng/o**/scope (făr-ĬN-gō-skōp): _____ *–scope:* instrument for examining

Element	Meaning	Word Analysis
Combining Forms		
Upper Respiratory Tract		
adenoid/o	adenoids	**adenoid**/ectomy (ăd-ĕ-noyd-ĔK-tō-mē): _____ *-ectomy:* excision, removal
tonsill/o	tonsils	peri/**tonsill**/ar (pĕr-ĭ-TŎN-sĭ-lăr): _____ *peri-:* around *-ar:* pertaining to
epiglott/o	epiglottis	**epiglott**/itis (ĕp-ĭ-glŏt-Ī-tĭs): _____ *-itis:* inflammation *Because the epiglottis seals the opening to the lungs, inflammation can lead to severe airway obstruction and death. Epiglottitis is treated as a medical emergency.*
laryng/o	larynx (voice box)	**laryng/o**/plegia (lă-rĭn-gō-PLĒ-jē-ă): _____ *-plegia:* paralysis
trache/o	trachea (windpipe)	**trache/o**/plasty (TRĀ-kē-ō-plăs-tē): _____ *-plasty:* surgical repair *Tracheoplasty is performed to correct a narrow or stenotic trachea.*

Medical Word Elements—cont'd

Element	Meaning	Word Analysis
Lower Respiratory Tract		
bronchi/o	bronchus (plural, bronchi)	**bronchi**/ectasis (brŏng-kē-ĔK-tă-sĭs): _____ *-ectasis:* dilation, expansion *Bronchiectasis is associated with various lung conditions and is commonly accompanied by chronic infection.*
bronch/o		**bronch/o**/scope (BRŎNG-kō-skōp): _____ *-scope:* instrument for examining *A bronchoscope is a flexible tube that is passed through the nose or mouth to enable inspection of the lungs and collection of tissue biopsies and secretions for analysis.*
bronchiol/o	bronchiole	**bronchiol**/itis (brŏng-kē-ō-LĪ-tĭs): _____ *-itis:* inflammation
alveol/o	alveolus;	**alveol**/ar (ăl-VĒ-ō-lăr): _____

Medical Word Elements—cont'd

Element	Meaning	Word Analysis
Lower Respiratory Tract		
pleur/o	pleura	**pleur/o**/scopy (ploo-RŎS-kō-pē): _____ *-scopy:* visual examination *Pleuroscopy involves insertion of a specialized endoscope through a small incision in the chest to allow an inspection of the pleural space, obtain lung tissue for analysis, inject medications, and perform other diagnostic or therapeutic procedures.*
pneum/o	air; lung	**pneum/o**/lith (NŪ-mō-lĭth): _____ *-lith:* stone, calculus
pneumon/o		**pneumon**/ia (nū-MŌ-nē-ă): _____ *-ia:* condition
pulmon/o	lung	**pulmon/o**/logist (pŭl-mŏ-NŎL-ŏ-jĭst): _____ *-logist:* specialist in the study of

Medical Word Elements—cont'd

Element	Meaning	Word Analysis
Other		
anthrac/o	coal, coal dust	**anthrac**/osis (ăn-thră-KŌ-sĭs): _____ *–osis:* abnormal condition; increase (used primarily with blood cells) *Anthracosis (black lung disease) is a chronic occupational disease found in coal miners and those associated with the coal industry.*
atel/o	incomplete; imperfect	**atel**/ectasis (ăt-ĕ-LĔK-tă-sĭs): _____ *–ectasis:* dilation, expansion *Atelectasis (airless or collapsed lung) is usually caused by a blocked airway, trauma to the chest cavity, or infection.*
coni/o	dust	pneum/o/**coni**/osis (nū-mō-kō-nē-Ō-sĭs): _____ *pneum/o:* air; lung *–osis:* abnormal condition; increase (used primarily with blood cells) *Pneumoconiosis is caused by mineral dusts of occupational or environmental origin and includes silicosis, asbestosis, and anthracosis.*
cyan/o	blue	**cyan**/osis (sī-ă-NŌ-sĭs): _____ *–osis:* abnormal condition; increase (used primarily with blood cells) *Cyanosis results from poor circulation or inadequate oxygenation of the blood.*

Medical Word Elements—cont'd

Element	Meaning	Word Analysis
Other		
lob/o	lobe	**lob**/ectomy (lō-BĔK-tō-mē): _____ *-ectomy:* excision *Lobectomies are performed when a malignancy is confined to a single lobe of any lobed organ, such as the lungs, liver, and thyroid gland.*
orth/o	straight	**orth/o**/pnea (or-THŎP-nē-ă): _____ *-pnea:* breathing *Discomfort in breathing that is relieved by sitting or standing in an erect position.*
ox/o	oxygen (O_2)	hyp/**ox**/emia (hī-pŏks-Ē-mē-ă): _____ *hyp-:* under, below, deficient *-emia:* blood condition *Hypoxemia is an abnormal decrease of oxygen in arterial blood.*

Medical Word Elements—cont'd

Element	Meaning	Word Analysis
Other		
pector/o	chest	**pector**/algia (pĕk-tō-RĂL-jē-ă): _____ *-algia:* pain *Pectoralgia is also called* thoracalgia *or* thoracodynia.
steth/o		**steth/o**/scope (STĔTH-ō-skōp): _____ *-scope:* instrument for examining *A stethoscope enables evaluation of sounds in the chest and the abdomen, an assessment technique known as auscultation.*
thorac/o		**thorac/o**/pathy (thō-răk-ŎP-ă-thē): _____ *-pathy:* disease
phren/o	diaphragm; mind	**phren/o**/spasm (FRĔN-ō-spăzm): _____ *-spasm:* involuntary contraction, twitching
spir/o	breathe	**spir/o**/meter (spī-RŎM-ĕt-ĕr): _____ *-meter:* instrument for measuring *A spirometer evaluates the movement of air into and out of the lungs (ventilation).*

Medical Word Elements—cont'd

Element	Meaning	Word Analysis
Suffixes		
-capnia	carbon dioxide (CO_2)	hyper/**capnia** (hī-pĕr-KĂP-nē-ă): _____ *hyper–:* excessive, above normal
-osmia	smell	an/**osmia** (ăn-ŎZ-mē-ă): _____ *an–:* without, not *Anosmia is a loss, usually partial, of the sense of smell. It can be temporary or permanent, depending on the cause.*
-phonia	voice	dys/**phonia** (dĭs-FŌ-nē-ă): _____ *dys–:* bad; painful; difficult *Dysphonia usually signifies dysfunction in the muscles needed to produce sound.*
-pnea	breathing	a/**pnea** (ĂP-nē-ă): _____ *a–:* without, not
-ptysis	spitting	hem/o/**ptysis** (hē-MŎP-tĭ-sĭs): _____ *hem/o:* blood *Hemoptysis is usually a sign of a serious condition of the lungs.*
-thorax	chest	hem/o/**thorax** (hē-mō-THŌ-răks): _____ *hem/o:* blood *Hemothorax is a type of pleural effusion containing blood and commonly associated with severe trauma to the chest.*

Medical Word Elements—cont'd

Element	Meaning	Word Analysis
Prefixes		
brady-	slow	**brady**/pnea (brăd-ĭp-NĒ-ă): _____ -*pnea:* breathing
dys-	bad; painful; difficult	**dys**/pnea (DĬSP-nē-ă): _____ -*pnea:* breathing
eu-	good, normal	**eu**/pnea (ŪP-nē-ă): _____ -*pnea:* breathing
tachy-	rapid	**tachy**/pnea (tăk-ĭp-NĒ-ă): _____ -*pnea:* breathing

It is time to review medical word elements by completing Learning Activities 7-1 and 2-2.

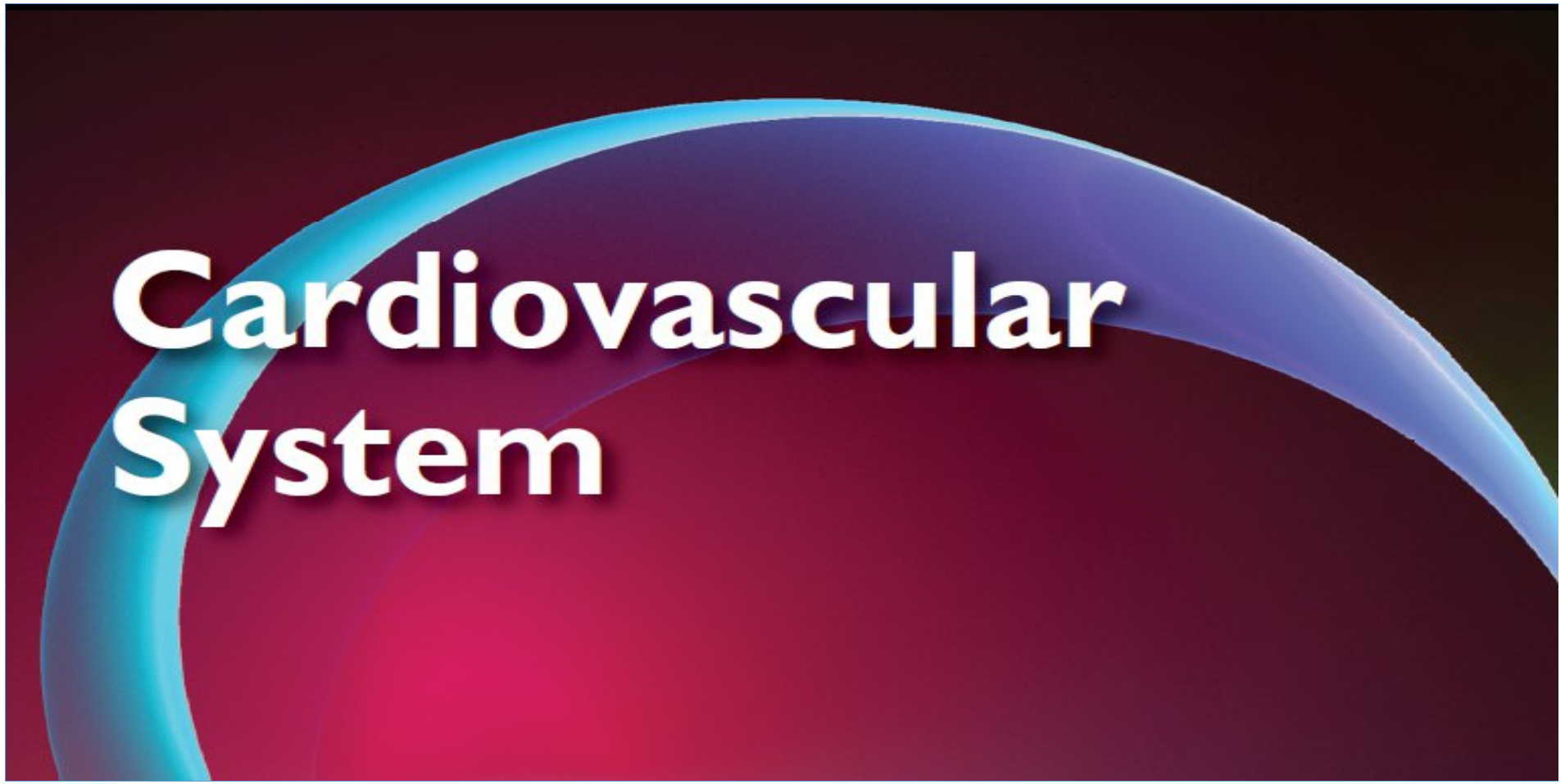
Learning Activities

The activities that follow provide a review of the a review of the respiratory system terms introduced in this chapter.

Complete each activity from 7-1 and 7-2 and review your answers to evaluate your understandin of chapter 6.



CHAPTER 8



Chapter Outline

- Objectives
- Anatomy and Physiology
  - -Vascular System
    - Arteries
    - Capillaries
    - Veins
  - Heart
- Medical Word Elements
- Learning Activities

Objectives

Upon completion of this chapter, you will be able to:

- Locate and describe the structures of the cardiovascular system.
- Pronounce, spell, and build words related to the cardiovascular system.
- Demonstrate your knowledge of this chapter by completing the learning and documenting health-care activities.

Anatomy and Physiology

The cardiovascular (CV) system is composed of the heart and blood vessels.

The heart is a hollow, muscular organ lying in the mediastinum (the center of the thoracic cavity between the lungs).

The pumping action of the heart propels blood that contains oxygen (O2), nutrients, and other vital products from the heart to body cells through a vast network of blood vessels called arteries.

Arteries branch into smaller vessels until they become microscopic vessels called capillaries.

At the capillary level, an exchange of products occurs between body cells and blood. Capillaries merge to form larger blood vessels called venules, which then combine to form veins, the vessels that return blood to the heart to begin the cycle again.

Millions of body cells rely on the CV system for their survival. When it fails, life at the cellular level is not possible and, ultimately, death occurs.

(Cont.)

** Vascular System

Three major types of vessels—(1) artery, (2) capillary, and (3) vein—carry blood throughout the body. (See Fig. 8-1.) Each type of vessel differs in structure, depending on its function.

1- Arteries

Arteries carry blood from the heart to all cells of the body. Because the pumping action of the heart propels blood through the arteries, the walls of the arteries must be strong and flexible enough to withstand the surge of blood that results from each contraction.

The walls of large arteries have three layers to provide toughness and elasticity.

- The (4) tunica externa is the outer layer, composed of connective tissue that provides strength and flexibility.
- The(5) tunica media is the middle layer, composed of smooth muscle. Depending on the needs of the body, this muscle can alter the size of the (7) lumen of the vessel. When it contracts, the tunica media causes vasoconstriction, resulting in decreased blood flow. When it relaxes, it causes vasodilatation or vasodilation, resulting in increased blood flow.
- The (6) tunica intima is the thin, inner lining of the lumen of the vessel, composed of endothelial cells that provide a smooth surface on the inside of the vessel.

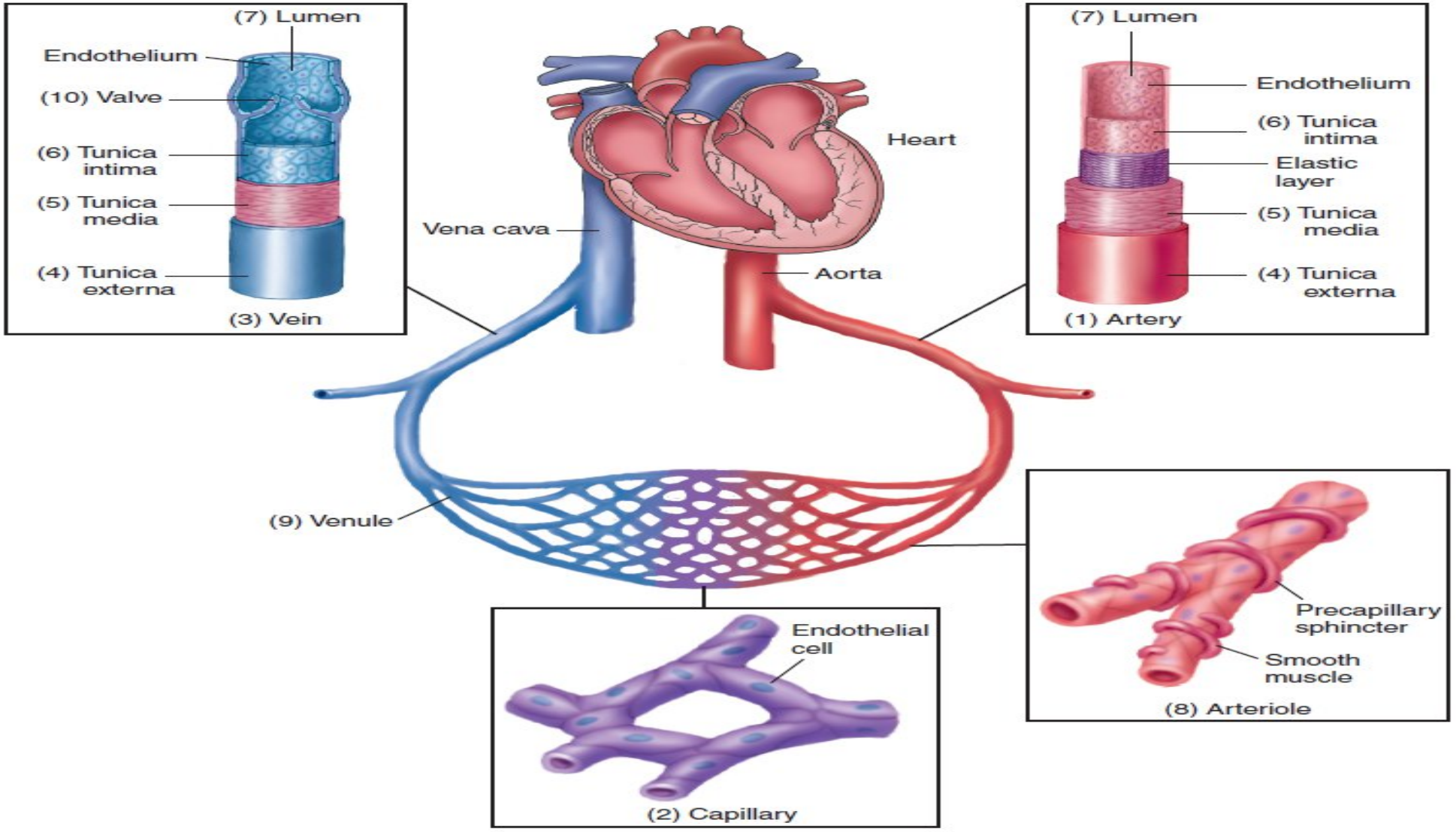


Figure 8-1 Vascular structures.

Anatomy and Physiology (Cont.)

2- Capillaries

Capillaries are microscopic vessels that join the arterial system with the venous system. Although they might seem like the most insignificant of the three vessel types because of their microscopic size, they are actually the most important because of their function. Because capillary walls are composed of only a single layer of endothelial cells, they are very thin. This thinness enables the exchange of water, gases, macromolecules, metabolites, and wastes between the blood and the cells adjacent to the capillary bed. The vast number of capillaries branching from arterioles causes blood to flow very slowly, providing sufficient time for exchange of essential substances.

3- Veins

Veins return blood to the heart. They are formed from smaller vessels called (9) venules that develop from the union of capillaries. veins use many methods to return blood to the heart, including the following:

Anatomy and Physiology (Cont.)

- skeletal muscle contraction
- gravity
- respiratory activity
- valves

The (10) valves are small structures within veins that prevent the backflow of blood. Valves are found mainly in the extremities and are especially important for returning blood from the legs to the heart because blood must travel a long distance against the force of gravity to reach the heart from the legs.

Blood carried in veins (except for the blood in the pulmonary veins) contains a low concentration of O2 and a correspondingly high concentration of carbon dioxide (CO2). This blood takes is said to be deoxygenated.

Anatomy and Physiology (Cont.)

** Heart

The heart is a muscular pump that propels blood to the entire body through a closed vascular network. It allows a dual circulatory system as follows

- Pulmonary circulation which is provided by the right side of the heart. It delivers blood to the lungs, where CO2 is exchanged for O2.
- Systemic circulation which is provided by the left side of the heart. It delivers blood to body tissues, where O2 is exchanged for CO2, a waste product that will be expelled by the lungs.

Both systemic and pulmonary circulatory activities occur simultaneously. (See Fig. 8-2.)

The heart is enclosed in a sac called the pericardium and is composed of

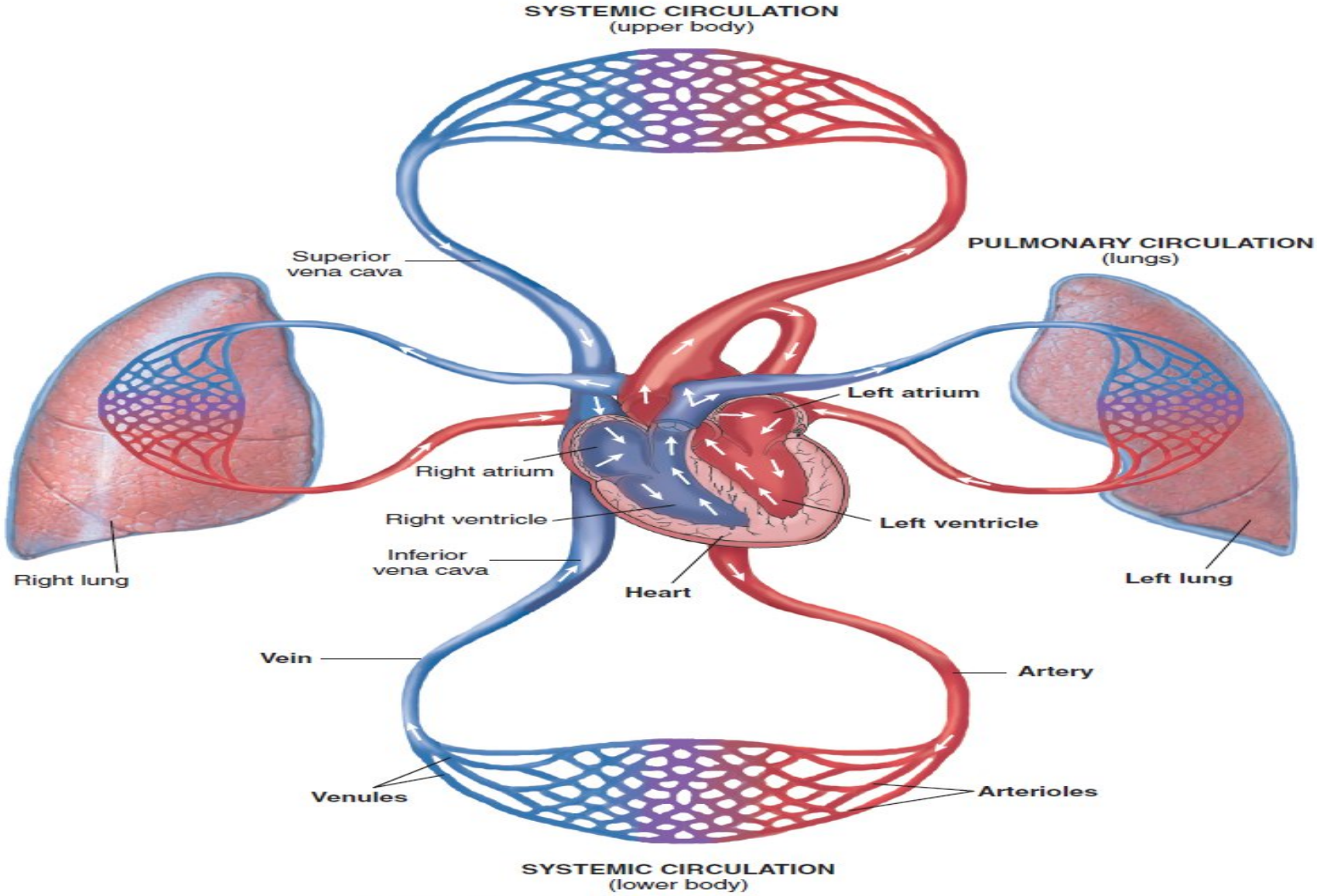


Figure 8-2 Systemic and pulmonary circulation.

Anatomy and Physiology (Cont.)

three distinct layers:

- Endocardium, a serous membrane that lines the four chambers of the heart and its valves and is continuous with the endothelium of the arteries and veins
- Myocardium, the muscular layer of the heart
- Epicardium, the outermost layer of the heart.

The heart is divided into four chambers. (See Fig. 8-3, page 226.)

- The two upper chambers, the (1) right atrium (RA) and (2) left atrium (LA), collect the deoxygenated blood coming from the body.
- The two lower chambers, the (3) right ventricle (RV) and (4) left ventricle (LV), pump the oxygenated blood from the heart to the body. The right ventricle pumps blood to the lungs (pulmonary circulation) for oxygenation, and the left ventricle pumps oxygenated blood to the entire body (systemic circulation).

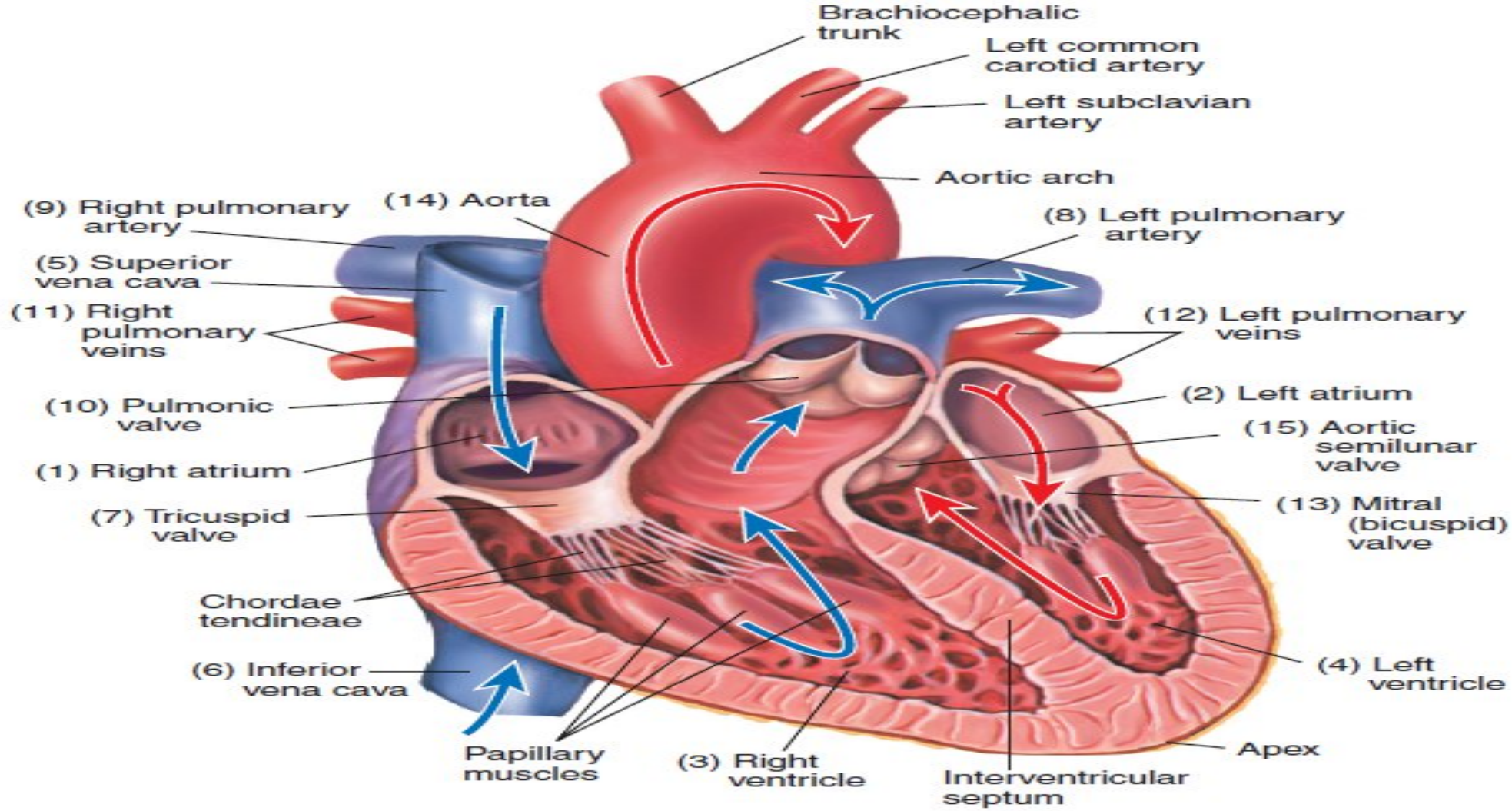


Figure 8-3 Internal structures of the heart, with red arrows designating oxygen-rich blood flow and blue arrows designating oxygen-poor blood flow.

Anatomy and Physiology (Cont.)

Deoxygenated blood from the body returns to the right atrium by way of two large veins: the (5) superior vena cava, which collects and carries blood from the upper body, and the (6) inferior vena cava, which collects and carries blood from the lower body.

From the right atrium, blood passes through the (7) tricuspid valve, consisting of three leaflets, to the right ventricle. When the heart contracts, blood leaves the right ventricle by way of the (8) left pulmonary artery and (9) right pulmonary artery and travels to the lungs.

During contraction of the right ventricle, the tricuspid valve closes to prevent a backflow of blood to the right atrium. The (10) pulmonic valve (or pulmonary semilunar valve) prevents regurgitation of blood into the right ventricle from the pulmonary artery.

Anatomy and Physiology (Cont.)

In the lungs, the pulmonary artery branches into millions of capillaries, each lying close to an alveolus. Here, carbon dioxide in the blood is exchanged for oxygen. Pulmonary capillaries unite to form four pulmonary veins—two (11) right pulmonary veins and two (12) left pulmonary veins. These vessels carry oxygenated blood back to the heart. They deposit blood in the left atrium. From there, blood passes to the left ventricle through the (13) mitral (bicuspid) valve, a structure consisting of two leaflets.

Upon contraction of the ventricles, the oxygenated blood leaves the heart through the largest artery of the body, the (14) aorta.

The aorta contains the (15) aortic semilunar valve (aortic valve) that permits blood to flow in only one direction—from the left ventricle to the aorta. The aorta branches into many smaller arteries that carry blood to all parts of the body.

It is important to understand that the myocardium cannot use the blood that passes through the chambers of the heart as a source of oxygen and nutrients. Instead, an arterial system composed of the coronary arteries branches from the aorta and provides

Anatomy and Physiology (Cont.)

the myocardium with its own blood supply. (See Fig. 8-4.) The artery vascularizing the right side of the heart is the (1) right coronary artery. The artery vascularizing the left side of the heart is the (2) left coronary artery.

The left coronary artery divides into two branches, the (3) left anterior descending artery and the (4) circumflex artery. If blood flow in the coronary arteries is diminished, damage to the heart muscle may result. When severe damage occurs, part of the heart muscle may die.

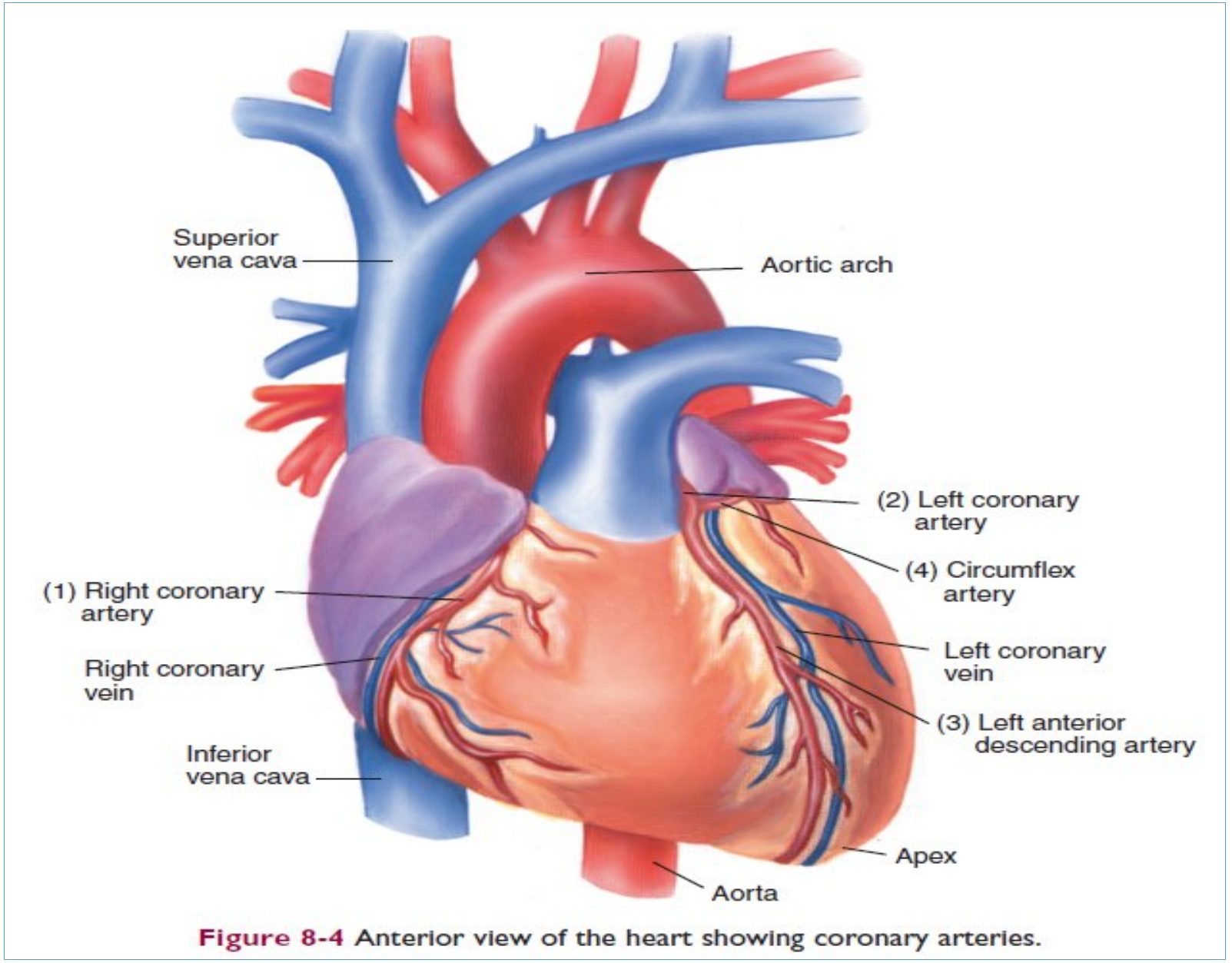


Figure 8-4 Anterior view of the heart showing coronary arteries.

Medical Word Elements

This section introduces combining forms, suffixes, and prefixes related to the cardiovascular system. Word analyses are also provided. From the information provided, complete the meaning of the medical words in the right-hand column. The first one is completed for you.

Element	Meaning	Word Analysis
Combining Forms		
aneurysm/o	aneurysm (widened blood vessel)	**aneurysm/o**/rrhaphy (ăn-ū-rĭz-MOR-ă-fē): *suture of an aneurysm* *-rrhaphy:* suture
angi/o	vessel (usually blood or lymph)	**angi/o**/plasty (ĂN-jē-ō-plăs-tē): _____ *-plasty:* surgical repair *Angioplasty is a procedure that reopens narrowed blood vessels and restores blood flow using a balloon-tipped catheter.*
vascul/o		**vascul**/itis (văs-kū-LĪ-tĭs): _____ *-itis:* inflammation
aort/o	aorta	**aort/o**/stenosis (ā-or-tō-stĕ-NŌ-sĭs): _____ *-stenosis:* narrowing, stricture
arteri/o	artery	**arteri/o**/rrhexis (ăr-tē-rē-ō-RĔK-sĭs): _____ *-rrhexis:* rupture

Medical Word Elements—cont'd

Element	Meaning	Word Analysis
arteriol/o	arteriole	**arteriol**/itis (ăr-tēr-ē-ō-LĪ-tĭs): _____ *–itis:* inflammation
atri/o	atrium	**atri/o**/megaly (ā-trē-ō-MĔG-ă-lē): _____ *–megaly:* enlargement
ather/o	fatty plaque	**ather**/oma (ăth-ĕr-Ō-mă): _____ *–oma:* tumor *Atheromas are formed when fatty plaque builds up on the inner lining of arterial walls.*
cardi/o	heart	**cardi/o**/megaly (kăr-dē-ō-MĔG-ă-lē): _____ *–megaly:* enlargement
coron/o		**coron**/ary (KOR-ō-nă-rē): _____ *–ary:* pertaining to *Coronary artery disease (CAD) is the most common type of heart disease and the leading cause of death in the United States in men and women.*
electr/o	electricity	**electr/o**/cardi/o/gram (ē-lĕk-trō-KĂR-dē-ō-grăm): _____ *cardi/o:* heart *–gram:* record, recording *An electrocardiogram helps detect many heart problems including heart attacks, arrhythmias, and heart failure.*

Medical Word Elements—cont'd

Element	Meaning	Word Analysis
embol/o	embolus (plug)	**embol**/ectomy (ĕm-bō-LĔK-tō-mē): _____ *–ectomy:* excision, removal *An embolectomy is performed in emergency situations to open blood vessels and reestablish blood flow.*
hemangi/o	blood vessel	**hemangi**/oma (hē-măn-jē-Ō-mă): _____ *–oma:* tumor *Hemangiomas, also called strawberry marks and found mostly in neonates, are benign tumors of cells that line blood vessels and usually disappear over time.*
my/o	muscle	**my/o**/cardi/al (mī-ō-KĂR-dē-ăl): _____ *cardi:* heart *–al:* pertaining to
phleb/o	vein	**phleb**/ectasis (flĕ-BĔK-tă-sĭs): _____ *–ectasis:* dilation, expansion
ven/o		**ven/o**/stasis (vē-nō-STĀ-sĭs): _____ *–stasis:* standing still *Venostasis, also called phlebostasis, is an abnormally slow blood flow in the veins and is a major risk factor for clot formation.*

Medical Word Elements—cont'd

Element	Meaning	Word Analysis
scler/o	hardening; sclera (white of eye)	**arteri/o**/scler/osis (ăr-tē-rē-ō-sklĕ-RŌ-sĭs): _____ *arteri/o:* artery *-osis:* abnormal condition; increase (used primarily with blood cells) *The most common cause of arteriosclerosis is the presence of an atheroma in the vessel. Other causes include smoking, diabetes, high blood pressure, obesity, and familial tendency.*
sept/o	septum	**sept/o**/stomy (sĕp-TŎS-tō-mē): _____ *-stomy:* forming an opening (mouth) *Septostomy is a temporary procedure performed to increase systemic oxygenation in infants with congenital heart defects until corrective surgery can be performed.*
sphygm/o	pulse	**sphygm**/oid (SFĬG-moyd): _____ *-oid:* resembling
sten/o	narrowing, stricture	**sten/o**/tic (stĕ-NŎT-ĭk): _____ *-tic:* pertaining to
thromb/o	blood clot	**thromb/o**/lysis (thrŏm-BŎL-ĭ-sĭs): _____ *-lysis:* separation; destruction; loosening *In thrombolysis, enzymes that destroy blood clots are infused into the occluded vessel.*

Medical Word Elements—cont'd

Element	Meaning	Word Analysis
valv/o	valve	**valv/o**/tomy (văl-VŎT-ō-mē): _____ *-tomy:* incision *Valvotomy commonly involves use of a balloon catheter passed through a blood vessel in the groin to gain access to a stenosed valve of the heart.*
valvul/o		**valvul/o**/plasty (VĂL-vū-lō-plăs-tē): _____ *-plasty:* surgical repair
ventricul/o	ventricle (of the heart or brain)	**ventricul**/ar (v ĕn-TRĬK-ū-lăr): _____ *-ar:* pertaining to
Suffixes		
-cardia	heart condition	tachy/**cardia** (tăk-ē-KĂR-dē-ă): _____ *tachy-:* rapid
-stenosis	narrowing, stricture	aort/o/**stenosis** (ā-or-tō-st ĕ-NŌ-s ĭs): _____ *aort/o:* aorta

Medical Word Elements—cont'd

Element	Meaning	Word Analysis
Prefixes		
brady-	slow	**brady**/cardia (brăd-ē-KĂR-dē-ă): _____ *–cardia:* heart condition
endo-	in, within	**endo**/vascul/ar (ĕn-dō-VĂS-kū-lăr): _____ *vascul:* vessel (usually blood or lymph) *–ar:* pertaining to *Endovascular procedures are those that occur within the lumen of a vessel.*
extra-	outside	**extra**/vascul/ar (ĕks-tră-VĂS-kū-lăr): _____ *vascul:* vessel (usually blood or lymph) *–ar:* pertaining to
peri-	around	**peri**/cardi/al (pĕr-ĭ-KĂR-dē-ăl): _____ *cardi:* heart *–al:* pertaining to *Pericardial refers to the membrane that surrounds the heart, the pericardium.*
trans-	across	**trans**/sept/al (trăns-SĔP-tăl): _____ *sept:* septum *–al:* pertaining to



It is time to review medical word elements by completing Learning Activities 8-1 and 8-2.

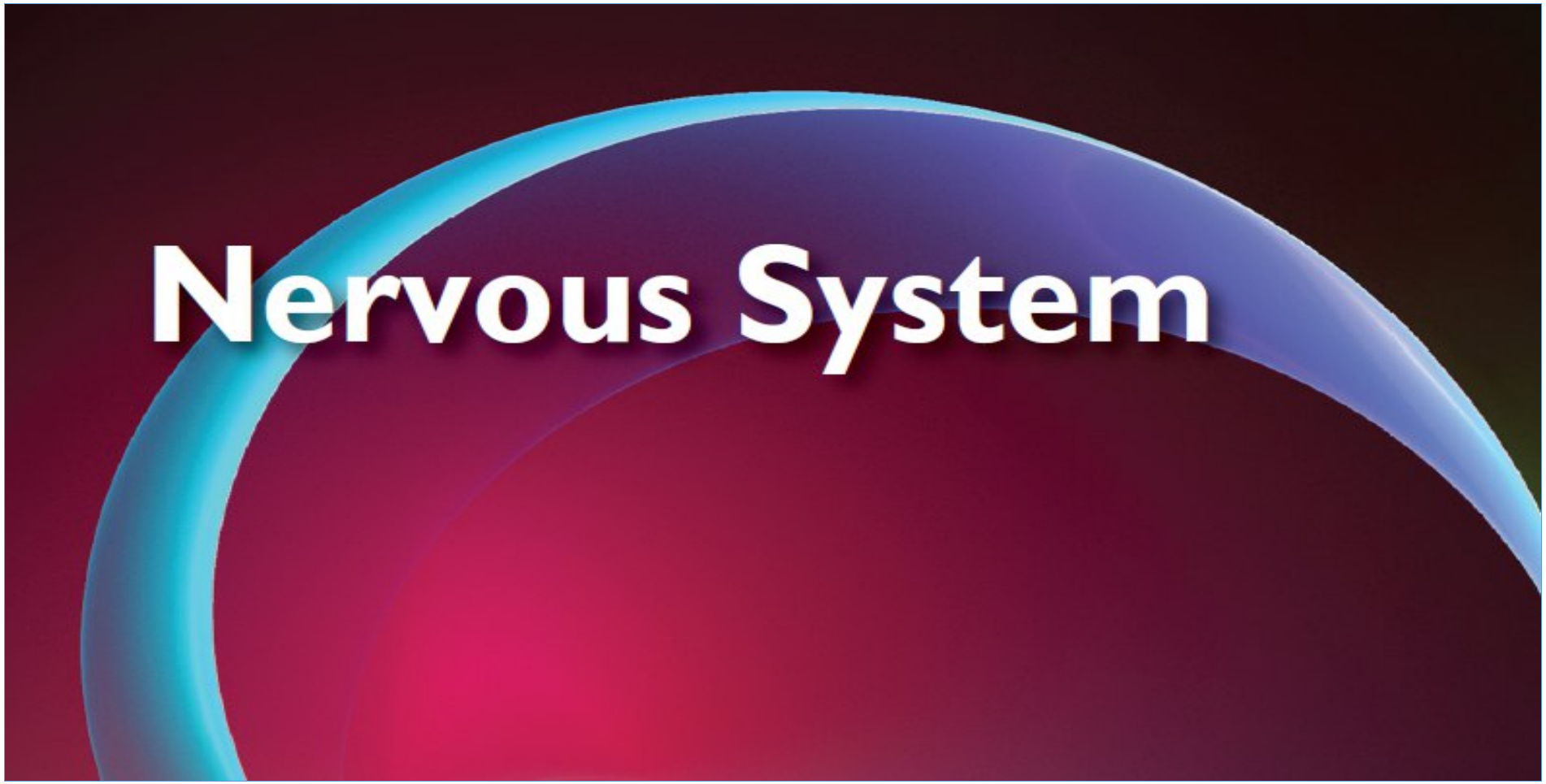
Learning Activities

The activities that follow provide a review of the a review of the cardiovascular system terms introduced in this chapter.

Complete each activity from 8-1 and 8-2 and review your answers to evaluate your understandin of chapter 8.



CHAPTER 9



Chapter Outline

- Objectives
- Anatomy and Physiology
  - Cellular Structure of the Nervous System
    - Neurons
    - Neuroglia
  - Nervous System Divisions
    - Central Nervous System
    - Peripheral Nervous System
- Medical Word Elements

Objectives

Upon completion of this chapter, you will be able to:

- Locate and describe the structures of the nervous system.
- Pronounce, spell, and build words related to the nervous system.
- Demonstrate your knowledge of this chapter by completing the learning and documenting health-care activities.

Anatomy and Physiology

The nervous system is one of the most complicated systems of the body in structure and function. It senses physical and chemical changes in the internal and external environments, processes them, and then responds to maintain homeostasis.

The nervous system coordinates, regulates, and integrates voluntary activities, such as walking and talking, and involuntary activities, such as digestion and circulation. The entire neural network of the body relies on

the transmission of electrochemical impulses that travel from one area of the body to another. The speed at which this transmission occurs is almost instantaneous, thus providing an immediate response to change.

Anatomy and Physiology (Cont.)

** Cellular Structure of the Nervous System

Despite its complexity, the nervous system is composed of only two principal cell types: neurons and neuroglia. Together, neurons and neuroglia constitute the nervous tissue of the body.

1- Neurons

Neurons transmit impulses. Depending on the direction in which they transmit information, neurons and nerves are classified as afferent or sensory (when the impulse moves toward the brain or spinal cord) or efferent or motor (when the impulse moves away from the brain or spinal cord). The three major structures of the neuron are the cell body, axon, and dendrites.

(See Fig. 15-1.)

Anatomy and Physiology (Cont.)

The (1) cell body is the enlarged structure of the neuron that contains the (2) nucleus of the cell and various organelles. Its branching cytoplasmic projections are (3) dendrites that carry impulses to the cell body and (4) axons that carry impulses from the cell body. Dendrites resemble tiny branches on a tree, providing additional surface area for receiving impulses from other neurons. Axons are threadlike extensions of nerve cells that transmit impulses to dendrites of other neurons and to muscles and glands.

Axons possess a white, lipoid covering called a (5) myelin sheath. This covering acts as an electrical insulator that reduces the possibility of an impulse stimulating adjacent nerves. It also accelerates impulse transmission through the axon.

On nerves in the peripheral nervous system, the myelin sheath is formed by a neuroglial cell called a (6) Schwann cell that wraps tightly around the axon. The exterior

surface of the Schwann cell forms a thin tube called the (7) neurilemma, or

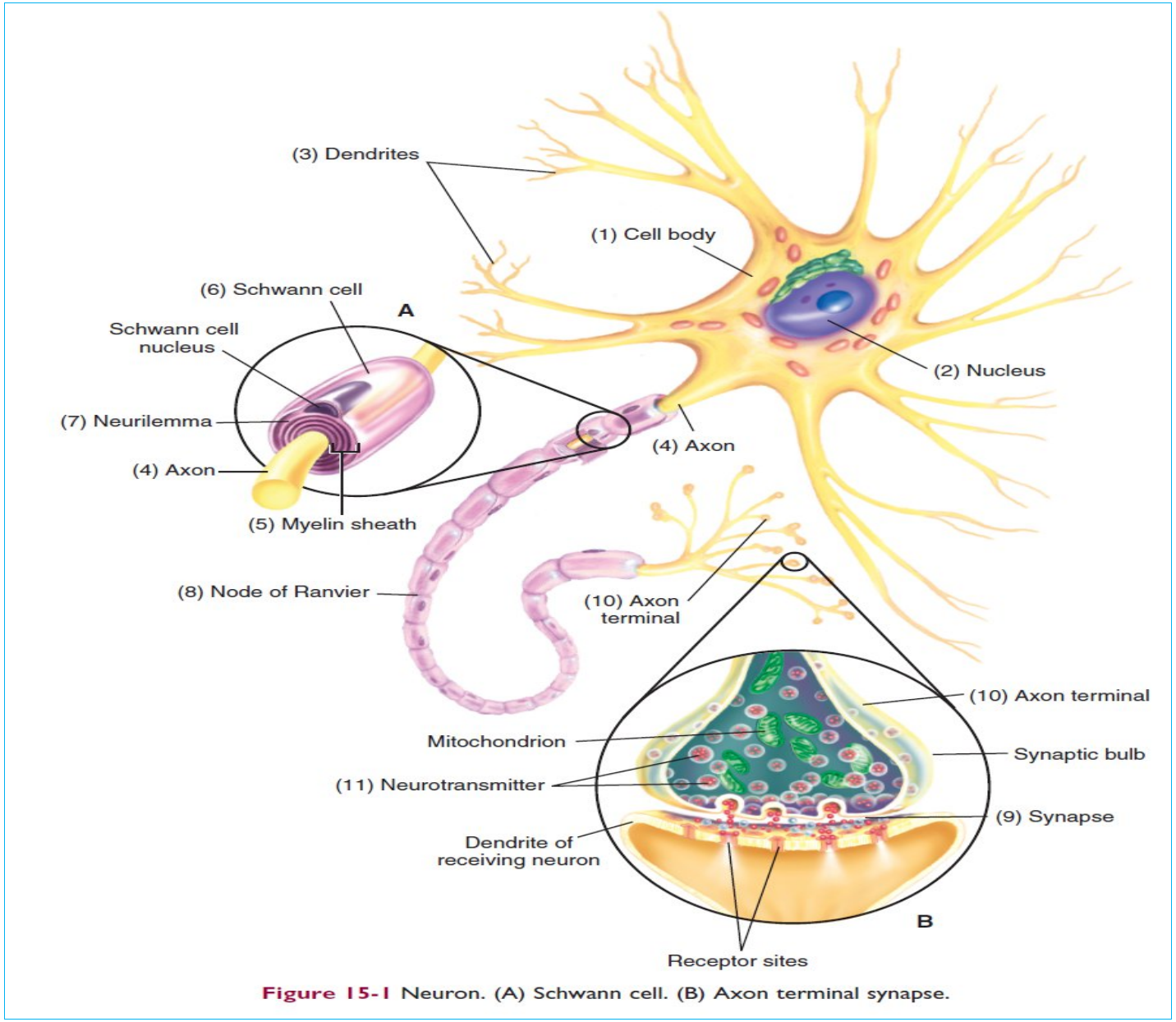


Figure 15-1 Neuron. (A) Schwann cell. (B) Axon terminal synapse.

Anatomy and Physiology (Cont.)

provides a pathway for possible neuron regeneration after injury.

Oligodendrocytes, rather than Schwann cells, form the myelin sheath that covers the axons in the central nervous system. Oligodendrocytes do not produce neurilemma, and thus injury or damage to neurons located in the central nervous system is irreparable. The short, unmyelinated spaces between adjacent segments of the myelin sheath are called (8) nodes of Ranvier. These nodes help speed the transmission of impulses down the axon.

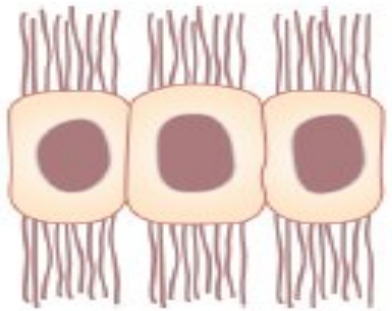
The functional connection between two neurons or between a neuron and its effector organ (muscle or gland) is a gap or space called a (9) synapse. Impulses must travel from the (10) axon terminal of one neuron to the dendrite of the next neuron or to its effector organ by crossing this synapse. The impulse within the transmitting axon produces a chemical substance called a (11) neurotransmitter to be released at the end of its axon. The neurotransmitter diffuses across the synapse and attaches to the receiving neuron.

Anatomy and Physiology (Cont.)

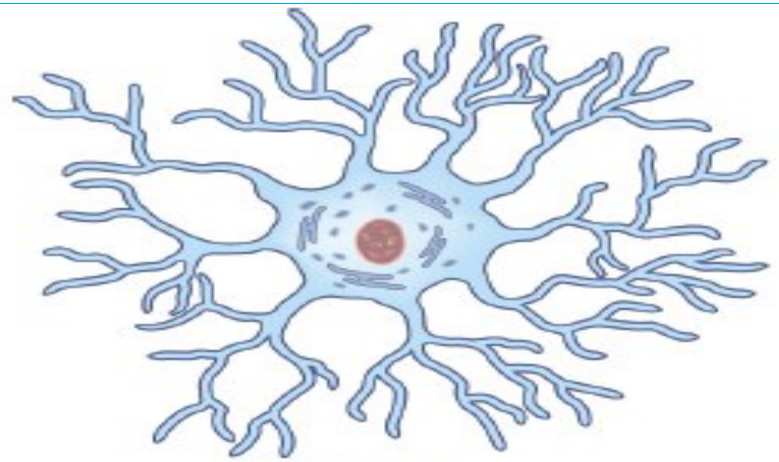
2. Neuroglia

Neuroglia are cells that support neurons and bind them to other neurons or other tissues of the body. Although they do not transmit impulses, they provide a variety of activities essential to the proper functioning of neurons. The term neuroglia literally means “nerve glue” because these cells were originally believed to serve only to bind neurons to each other and to other structures. They are now known to supply nutrients and oxygen to neurons and assist in other metabolic activities.

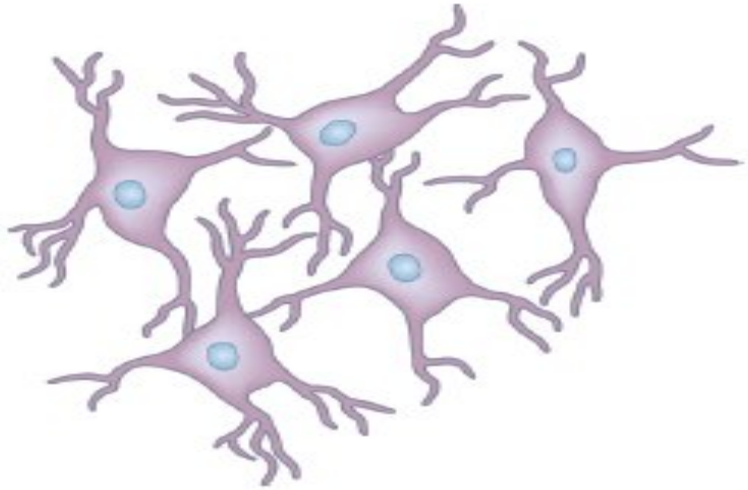
They also play an important role when the nervous system suffers injury or infection. The four major types of neuroglia are astrocytes, oligodendrocytes, microglia, and ependymocytes. (See Fig. 15-2.)



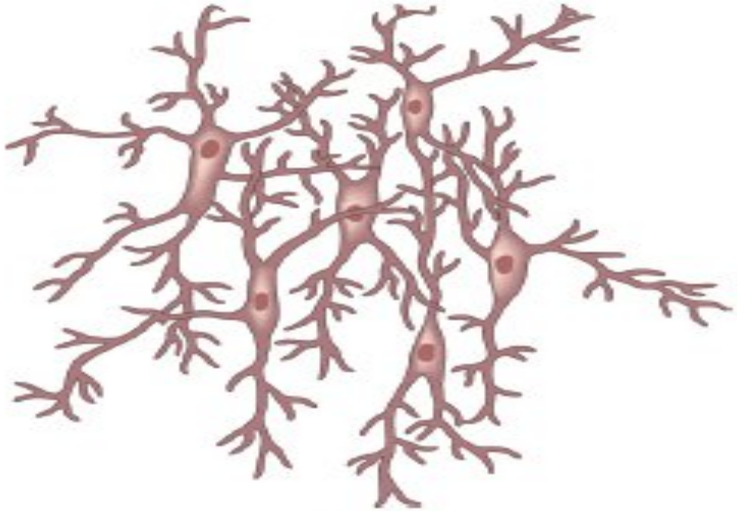
Ependymocytes



Astrocyte



Oligodendrocytes



Microglia

Figure 15-2 Four types of neuroglia.

Anatomy and Physiology (Cont.)

** Nervous System Divisions

The two major divisions of the nervous system are the central nervous system and the peripheral nervous system. The central nervous system consists of all nervous tissue located in the brain and spinal cord. The peripheral nervous system includes all nervous tissue located outside the central nervous system and consists of cranial and spinal nerves. (See Table 15-1.)

1. Central Nervous System

The central nervous system (CNS) consists of the brain and spinal cord. Its nervous tissue consists of white matter and gray matter. Bundles of axons and their white lipoid myelin sheaths constitute white matter. Unmyelinated fibers, dendrites, and nerve cell bodies make up the gray matter of the brain and spinal cord.

Table 15-1 Nervous System Structures and Functions

This table lists the structures of the nervous system, along with their functions.

Structures	Function
Central	
Brain	Center for thought and emotion, interpretation of sensory stimuli, and coordination of body functions
Spinal cord	Main pathway for transmission of information between the brain and body
Peripheral	
Cranial nerves	12 pairs of nerves that emerge from the base of the skull and may act in a motor capacity, sensory capacity, or both
Spinal nerves	31 pairs of nerves that emerge from the spine and act in motor and sensory capacities

Anatomy and Physiology (Cont.)

* Brain

In addition to being one of the largest organs of the body, the brain is highly complex in structure and function. (See Fig. 15-3.) It integrates almost every physical and mental activity of the body and is the center for memory, emotion, thought, judgment, reasoning, and consciousness. The four major structures of the brain are the following:

- Cerebrum
- Cerebellum
- Diencephalon
- Brainstem (consists of three parts; midbrain, pos, and medulla)

* Spinal Cord

The spinal cord transmits sensory impulses from the body to the brain and motor impulses from the brain to the muscles and organs of the body. The sensory nerve tracts

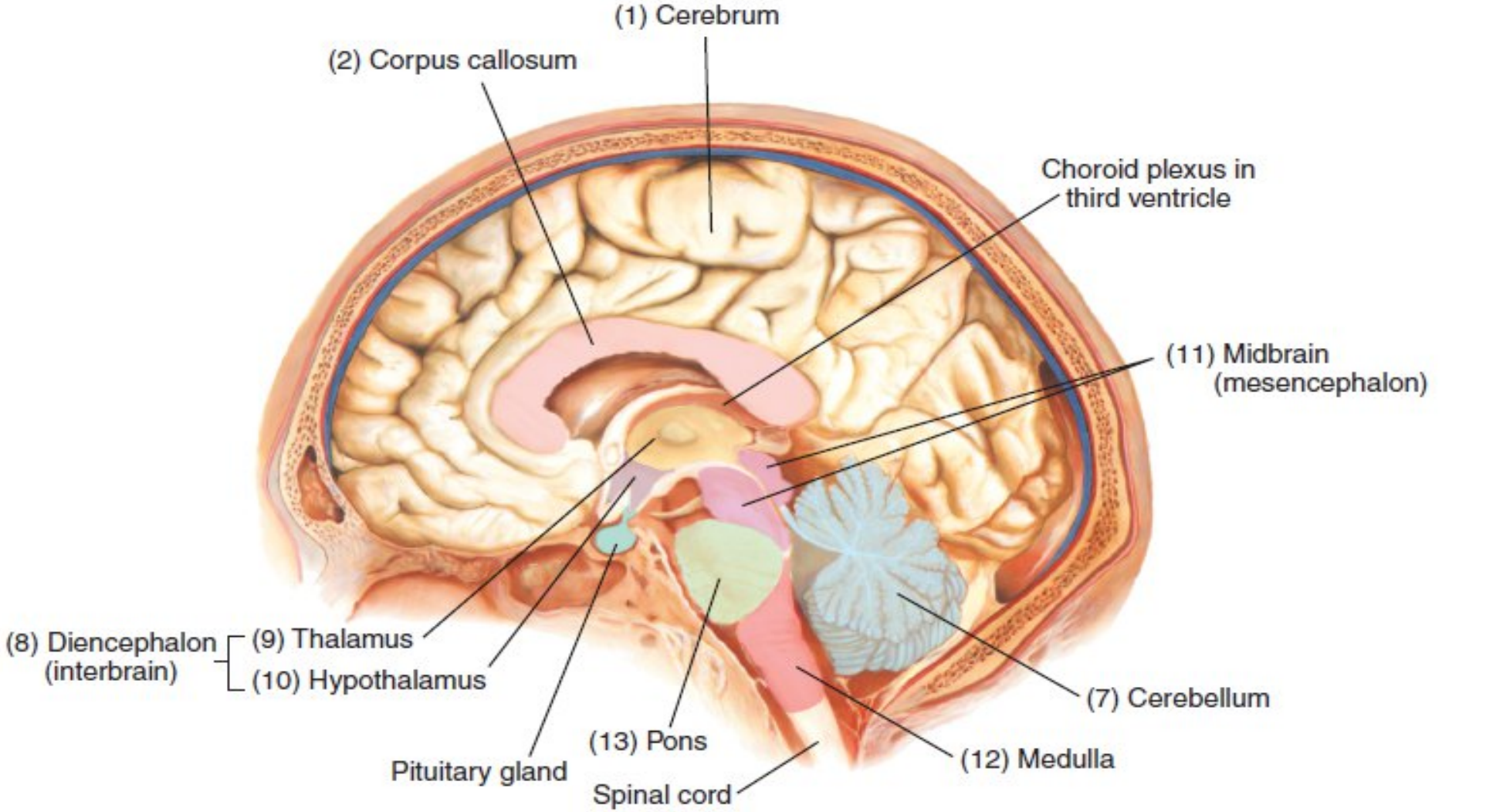


Figure 15-3 Brain structures.

Anatomy and Physiology (Cont.)

are called ascending tracts because the direction of the impulse is upward. Conversely, motor nerve tracts are called descending tracts because they carry impulses in a downward direction to muscles and organs. A

cross-sectional view of the spinal cord reveals an inner area of gray matter composed of cell bodies and dendrites and an outer area of white matter composed of myelinated tissue of the ascending and descending tracts.

The entire spinal cord is located within the spinal cavity of the vertebral column, with spinal nerves exiting between the intervertebral spaces throughout almost the entire length of the spinal column. Unlike the cranial nerves, which have specific names, the spinal nerves are identified by the region of the vertebral column from which they exit.

Note:

The brain and spinal cord receive limited protection from three coverings called meninges These coverings (from the outer to inner covering) are the dura mater, arachnoid, and pia mater.

(Cont.)

2. Peripheral Nervous System

The peripheral nervous system (PNS) is composed of all nervous tissue located outside the spinal column and skull. It consists of sensory neurons, which carry impulses from the body to the CNS (afferent), and motor neurons, which carry impulses from the brain and spinal cord to muscles and glands (efferent). The PNS is divided into the somatic nervous system and the autonomic nervous system. Some motor nerves in the peripheral nervous system innervate muscles under conscious control of the individual. They regulate such actions as walking and talking. The motor nerves that influence voluntary muscles comprise the somatic nervous system. Other motor nerves innervate involuntary muscles (muscles of the digestive or respiratory organs), glands, and cardiac muscles. The motor nerves that influence involuntary muscles, glands, and heart muscle make up the autonomic nervous system. (See Fig. 15-4.) which is divided into 2 parts; sympathetic and parasympathetic divisions. the sympathetic division and the parasympathetic division bring about opposite effects on the activity of the same organs. In other words, the sympathetic and parasympathetic divisions act as "increase" and "decrease" switches for controlling the actions

Anatomy and Physiology (Cont.)

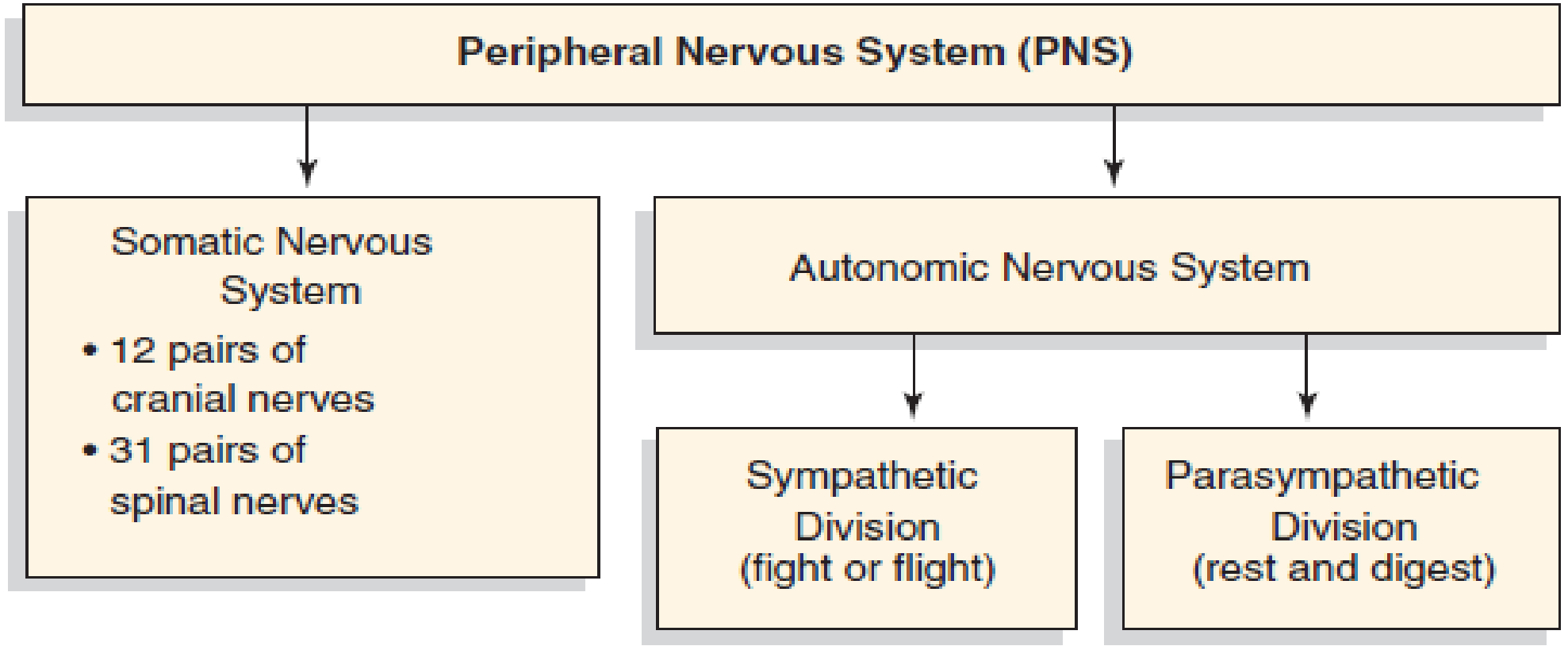


Figure 15-4 Divisions of the peripheral nervous system.

Medical Word Elements

This section introduces combining forms, suffixes, and prefixes related to the nervous system. Word analyses are also provided. From the information provided, complete the meaning of the medical words in the right-hand column. The first one is completed for you.

Element	Meaning	Word Analysis
Combining Forms		
cerebr/o	cerebrum	**cerebr/o**/tomy (sĕr-ĕ-BRŎT-ō-mē): *incision of the cerebrum* *-tomy:* incision
crani/o	cranium (skull)	**crani/o**/malacia (krā-nē-ō-mă-LĀ-shē-ă): _____ *-malacia:* softening
encephal/o	brain	**encephal/o**/cele (ĕn-SĔF-ă-lō-sēl): _____ *-cele:* hernia, swelling *Encephalocele is a condition in which portions of the brain and meninges protrude through a bony midline defect in the skull.*
gangli/o	ganglion (knot or knotlike mass)	**gangli**/ectomy (găng-glē-ĔK-tō-mē): _____ *-ectomy:* excision, removal *A ganglion is a mass of nerve cell bodies (gray matter) in the peripheral nervous system.*

Medical Word Elements—cont'd

Element	Meaning	Word Analysis
gli/o	glue; neuroglial tissue	**gli**/oma (glī-Ō-mă): _____ *-oma:* tumor
kinesi/o	movement	brady/**kines**/ia (brăd-ē-kĭ-NĒ-sē-ă): _____ *brady-:* slow *-ia:* condition
lept/o	thin, slender	**lept/o**/mening/o/pathy (lĕp-tō-mĕn-ĭn-GŎP-ă-thē): _____ *-mening/o:* meninges (membranes covering the brain and spinal cord) *-pathy:* disease *The leptomeninges include the pia mater and arachnoid, both of which are thin and delicate in structure, as opposed to the dura mater.*
lex/o	word, phrase	dys/**lex**/ia (dĭs-LĔK-sē-ă): _____ *dys-:* bad; painful; difficult *-ia:* condition *Dyslexia is a difficulty with reading or an inability to read, including the tendency to reverse letters or words when reading or writing.*
mening/o	meninges (membranes covering the brain and spinal cord)	**mening/o**/cele (mĕn-ĬN-gō-sēl): _____ *-cele:* hernia, swelling
meningi/o		**meningi**/oma (mĕn-ĭn-jē-Ō-mă): _____ *-oma:* tumor

Medical Word Elements—cont'd

Element	Meaning	Word Analysis
myel/o	bone marrow; spinal cord	**poli/o**/myel/itis (pōl-ē-ō-mī-ĕl-Ī-tĭs): _____ *poli/o:* gray; gray matter (of the brain or spinal cord) *-itis:* inflammation
narc/o	stupor; numbness; sleep	**narc/o**/tic (năr-KŎT-ĭk): _____ *-tic:* pertaining to *Narcotics depress the central nervous system, thus relieving pain and producing sleep.*
neur/o	nerve	**neur/o**/lysis (nū-RŎL-ĭs-ĭs): _____ *-lysis:* separation; destruction; loosening *Neurolysis is sometimes performed using cryoablation or radiofrequency techniques to relieve intractable pain as a temporary or permanent measure.*
radicul/o	nerve root	**radicul**/algia (ră-dĭk-ū-LĂL-jē-ă): _____ *-algia:* pain

Medical Word Elements—cont'd

Element	Meaning	Word Analysis
sthen/o	strength	hyper/**sthen**/ia (hī-pĕr-STHĒ-nē-ă): _____ *hyper–:* excessive, above normal *–ia:* condition *Hypersthenia is characterized by rigid muscles and muscle tension when resting; also called* hypertonia
thalam/o	thalamus	**thalam/o**/tomy (thăl-ă-MŎT-ō-mē): _____ *–tomy:* incision *Thalamotomy is performed to treat intractable pain or psychoses.*
thec/o	sheath (usually referring to the meninges)	intra/**thec**/al (ĭn-tră-THĒ-kăl): _____ *intra–:* in, within *–al:* pertaining to
ton/o	tension	dys/**ton**/ia (dĭs-TŌ-nē-ă): _____ *dys–:* bad; painful; difficult *–ia:* condition *Dystonia usually refers to a movement disorder characterized by sustained muscle contractions, resulting in a persistently abnormal posture.*
ventricul/o	ventricle (of the heart or brain)	**ventricul**/itis (vĕn-trĭk-ū-LĪ-tĭs): _____ *–itis:* inflammation

Medical Word Elements—cont'd

Element	Meaning	Word Analysis
Suffixes		
-algesia	pain	an/**algesia** (ăn-ăl-JĒ-zē-ă): _____ *an–:* without, not
-algia		syn/**algia** (sĭn-ĂL-jē-ă): _____ *syn–:* union, together, joined *Synalgia, commonly called referred pain, is pain experienced in a part of the body other than the place of pathology. For example, right shoulder pain is commonly associated with gallstones.*
-asthenia	weakness, debility	my/**asthenia** (mī-ăs-THĒ-nē-ă): _____ *my:* muscle
-esthesia	feeling	hyper/**esthesia** (hī-pĕr-ĕs-THĒ-zē-ă): _____ *hyper–:* excessive, above normal *Hyperesthesia involves a marked sensitivity to touch, pain, or other sensory stimuli.*
-kinesia	movement	hyper/**kinesia** (hī-pĕr-kĭ-NĒ-zē-ă): _____ *hyper–:* excessive, above normal

Medical Word Elements—cont'd

Element	Meaning	Word Analysis
Suffixes		
-lepsy	seizure	narc/o/**lepsy** (NĂR-kō-lĕp-sē): _____ *narc/o:* sleep *In narcolepsy, the individual has a sudden and uncontrollable urge to sleep at an inappropriate time, such as when driving.*
-paresis	partial paralysis	hemi/**paresis** (hĕm-ē-pă-RĒ-sĭs): _____ *hemi–:* one-half *When used alone, the term paresis refers to partial paralysis or motor weakness.*
-phasia	speech	a/**phasia** (ă-FĀ-zē-ă): _____ *a–:* without, not
-plegia	paralysis	quadri/**plegia** (kwŏd-rĭ-PLĒ-jē-ă): _____ *quadri–:* four
-taxia	order, coordination	a/**taxia** (ă-TĂK-sē-ă): _____ *a–:* without, not *Ataxia refers to poor muscle coordination, especially when voluntary movements are attempted.*

Medical Word Elements—cont'd

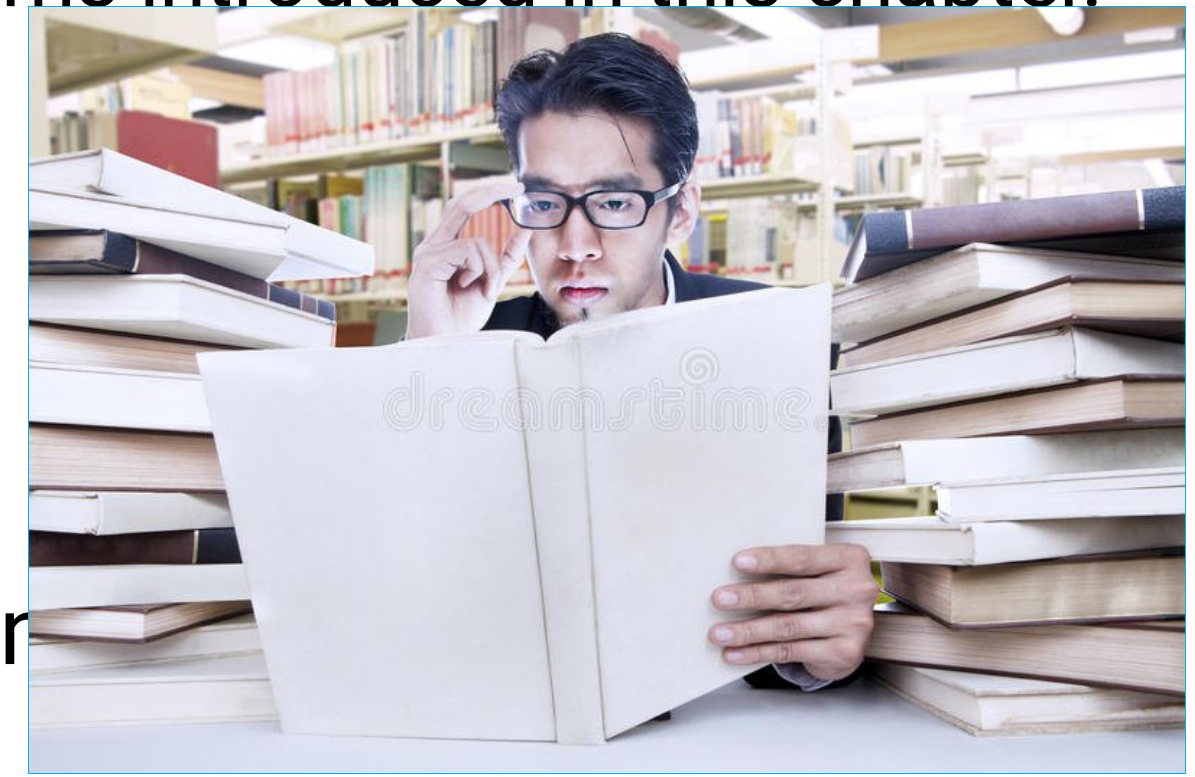
Element	Meaning	Word Analysis
Prefixes		
pachy-	thick	**pachy**/mening/itis (păk-ē-mĕn-ĭn-JĪ-tĭs): _____ *mening:* meninges (membranes covering the brain and spinal cord) *-itis:* inflammation *The dura mater (pachymeninx) is a thick membrane that provides protection for the brain and spinal cord.*
para-	near, beside; beyond	**para**/plegia (păr-ă-PLĒ-jē-ă): _____ *-plegia:* paralysis
syn-	union, together, joined	**syn**/algia (sĭn-ĂL-jē-ă): _____ *-algia:* pain *Pain in a deteriorated hip commonly causes referred pain in a healthy knee.*

It is time to review medical word elements by completing Learning Activities 15-1 and 15-2.

Learning Activities

The activities that follow provide a review of the a review of the cardiovascular system terms introduced in this chapter.

Complete each activity from 15-1 and 15-2 and review your answers to evaluate your understandir of chapter 9.



CHAPTER10



Chapter Outline

- Objectives
- Anatomy and Physiology
  - Mouth (Teeth, Tongue, Hard and Soft Palates)
  - Pharynx, Esophagus, and Stomach
  - Small Intestine
  - Large Intestine
  - Accessory Organs of Digestion
    - Liver
    - Pancreas
    - Gallbladder
- Medical Word Elements

Objectives

Upon completion of this chapter, you will be able to:

- Locate the major organs of the digestive system, an describe their structure and function.
- Describe the functional relationship between the digestive system and other body systems.
- Pronounce, spell, and build words related to the digestive system.
- Demonstrate your knowledge of this chapter by completing the learning and documenting health-care activities.

Anatomy and Physiology

The digestive system, also called the gastrointestinal (GI) system, consists of a digestive tube called the GI tract (or alimentary canal) and several accessory organs whose primary function is to break down food, prepare it for absorption, and eliminate waste. The GI tract, extending from the mouth to the anus, varies in size and structure in several distinct regions.

Food passing along the GI tract is mixed with digestive enzymes and broken down into nutrient molecules, which are absorbed in the bloodstream. Undigested waste materials not absorbed by the blood are then eliminated from the body through defecation. Included in the digestive system are the accessory organs of digestion: the liver, gallbladder, and pancreas. The process of digestion breaks down food into nutrients to nourish the body.

(See Fig. 6-1.)

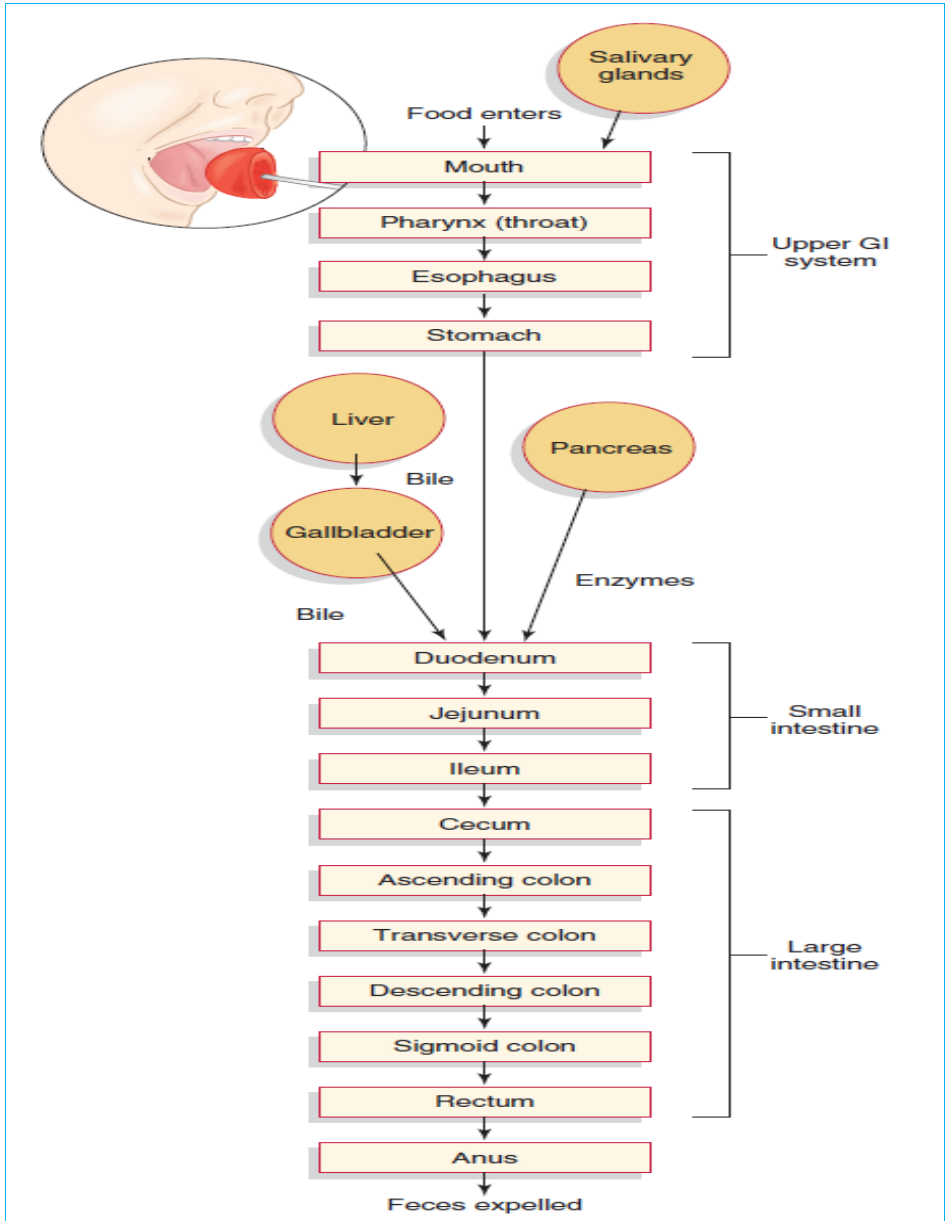


Figure 6-1 Pathway of food through the digestive system.

Anatomy and Physiology (Cont.)

** Gasrtointestinal (Alimentary) tract

1.Mouth

The process of digestion begins in the mouth. (See Fig. 6-2) The mouth, also known as the (1) oral cavity, is a receptacle for food. It is formed by the cheeks (bucca), lips, teeth, tongue, and hard and soft palates. Located around the oral cavity are three pairs of salivary glands that secrete saliva. Saliva contains important digestive enzymes that help begin the chemical breakdown of food. In the mouth, food is broken down mechanically (by the teeth) and chemically (by saliva) and then formed into a bolus.

2. Teeth

The (2) teeth play an important role in the initial stages of digestion by mechanically breaking down food (mastication) into smaller pieces as they mix it with saliva.

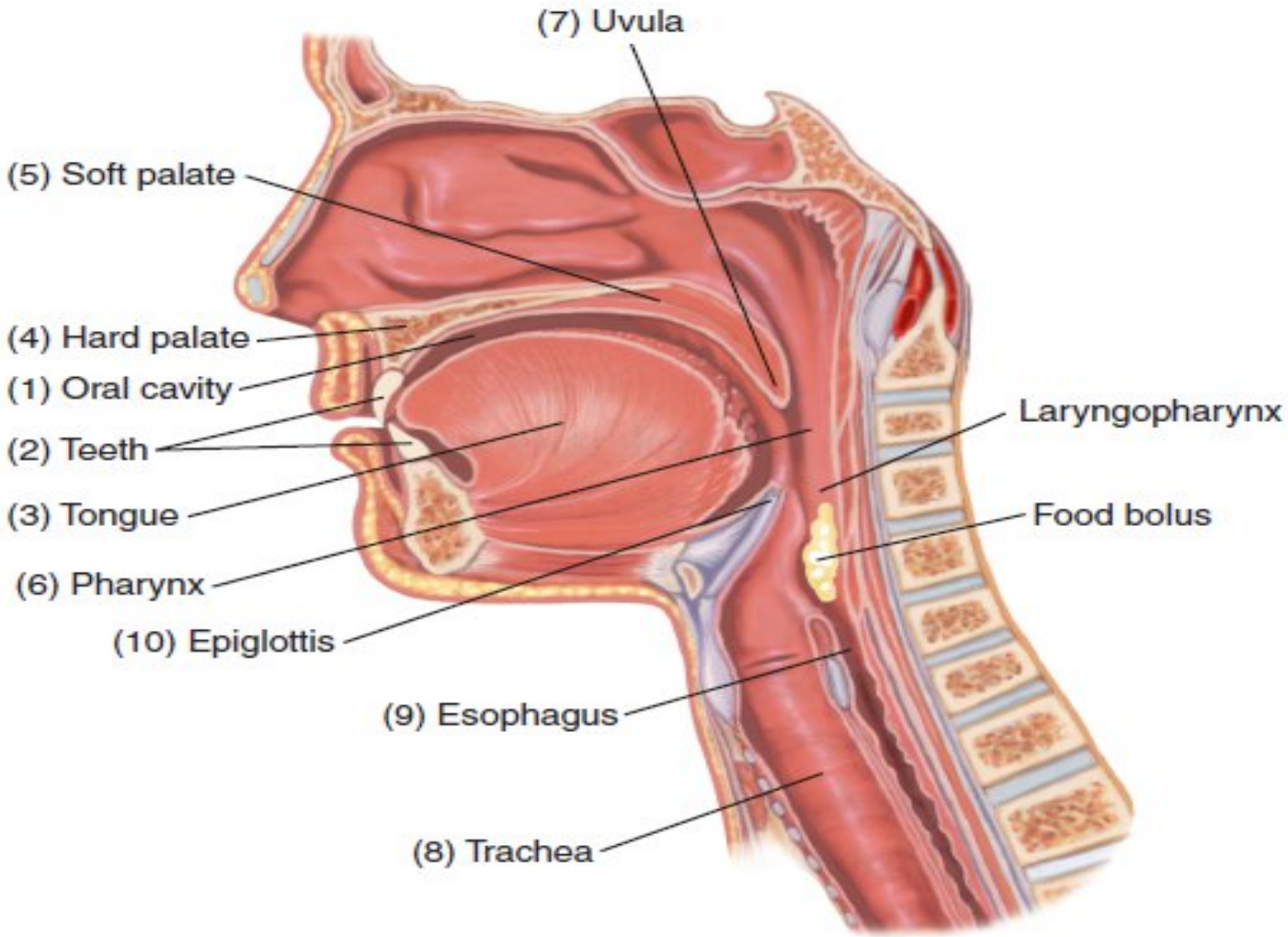


Figure 6-2 Sagittal view of the head showing oral, nasal, and pharyngeal components of the digestive system.

Anatomy and Physiology (Cont.)

3. Tongue

The (3) tongue assists in the chewing process by manipulating the bolus of food during chewing and moving it to the back of the mouth for swallowing (deglutition). The tongue also aids in speech production and taste. Rough projections on the surface of the tongue called papillae contain taste buds.

4. Hard and Soft Palates

The two structures forming the roof of the mouth are the (4) hard palate (anterior portion) and the (5) soft palate (posterior portion). The soft palate, which forms a partition between the mouth and the nasopharynx, is continuous with the hard palate.

Anatomy and Physiology (Cont.)

5. Pharynx, Esophagus, and Stomach

As the tongue pushes the bolus into the (6) pharynx (throat), it is guided by the soft, fleshy, V-shaped structure called the (7) uvula. The funnel-shaped pharynx serves as a passageway to the respiratory and GI tracts and provides a resonating chamber for speech sounds. The lowest portion of the pharynx divides into two tubes: one that leads to the lungs, called the (8) trachea, and one that leads to the stomach, called the (9) esophagus. A small flap of cartilage called the (10) epiglottis folds back to cover the trachea during swallowing, forcing food to enter the esophagus.

The stomach, a saclike structure located in the left upper quadrant (LUQ) of the abdominal cavity, serves as a food reservoir that continues mechanical and chemical digestion.

Anatomy and Physiology (Cont.)

6. Small Intestine

The small intestine is a coiled tube approximately 6 meters long that begins at the pyloric sphincter and ends at the large intestine. (See Fig. 6-4, page 132.) It consists of three parts:

- (1) duodenum, the uppermost segment, which is approximately 25cm long
- (2) jejunum, which is approximately 2-3 meters long
- (3) ileum, which is approximately 3-4 meters long

Digestion is completed in the small intestine with the help of additional enzymes and secretions from the (4) pancreas and (5) liver. At the terminal end of the small intestine, a sphincter muscle called the ileocecal valve allows undigested or unabsorbed material from the small intestine to pass into the large intestine and eventually be excreted from the body.

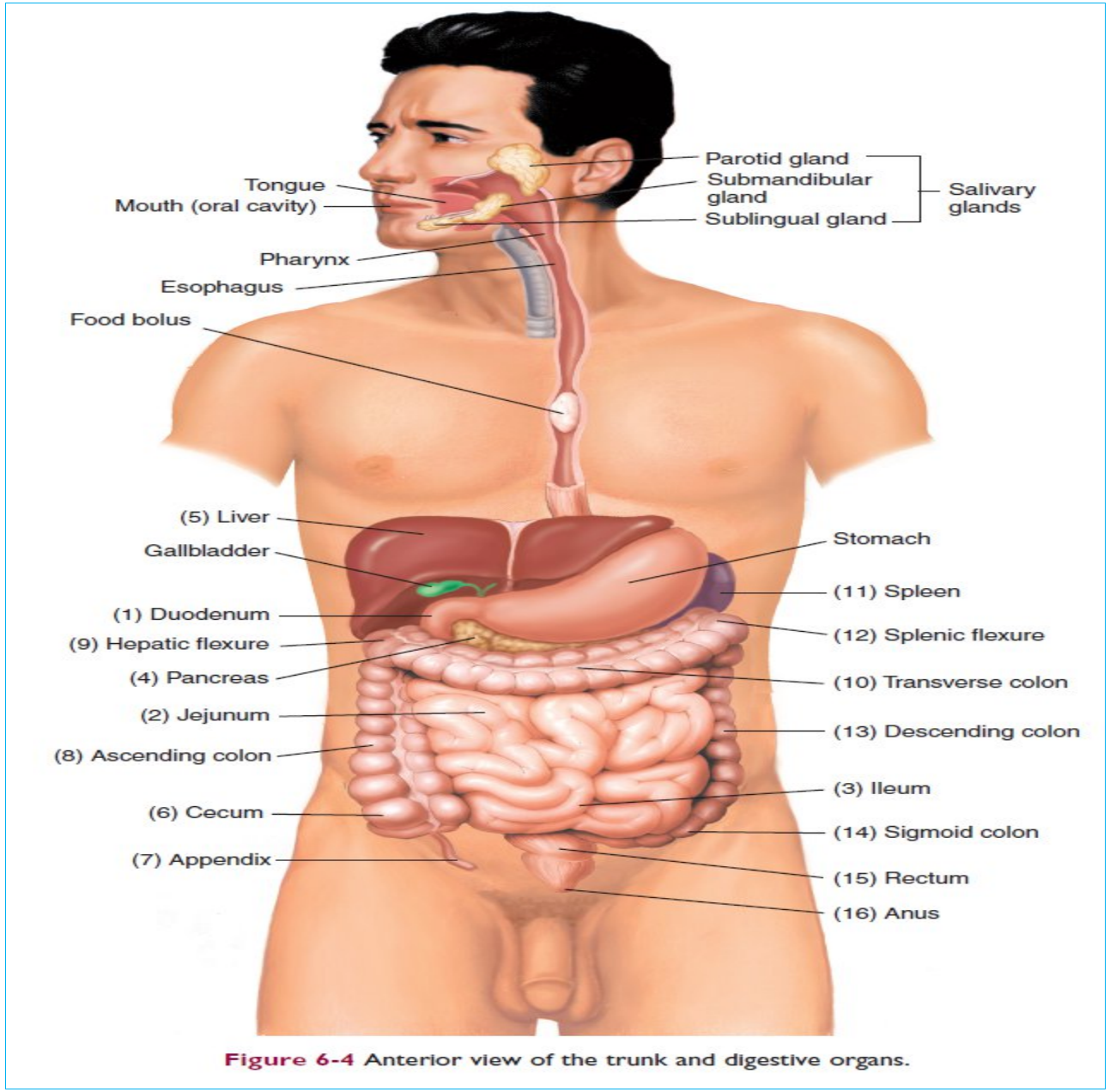


Figure 6-4 Anterior view of the trunk and digestive organs.

Anatomy and Physiology (Cont.)

7. Large Intestine

The large intestine is approximately 1.5 meters long. It begins at the end of the ileum and extends to the anus. No digestion takes place in the large intestine. The only secretion is mucus in the colon, which lubricates fecal material so it can pass from the body. The large intestine has three main components: the cecum, colon, and rectum. The first 2 or 3 inches of the large intestine comprise

** Accessory Organs of Digestion

Although the liver, gallbladder, and pancreas lie outside the GI tract, they play a vital role in the proper digestion and absorption of nutrients. (See Fig. 6-5).

1. Liver

It is located beneath the diaphragm in the right upper quadrant (RUQ) and the left upper quadrant (LUQ) of the abdominal cavity. The liver performs many vital functions, and

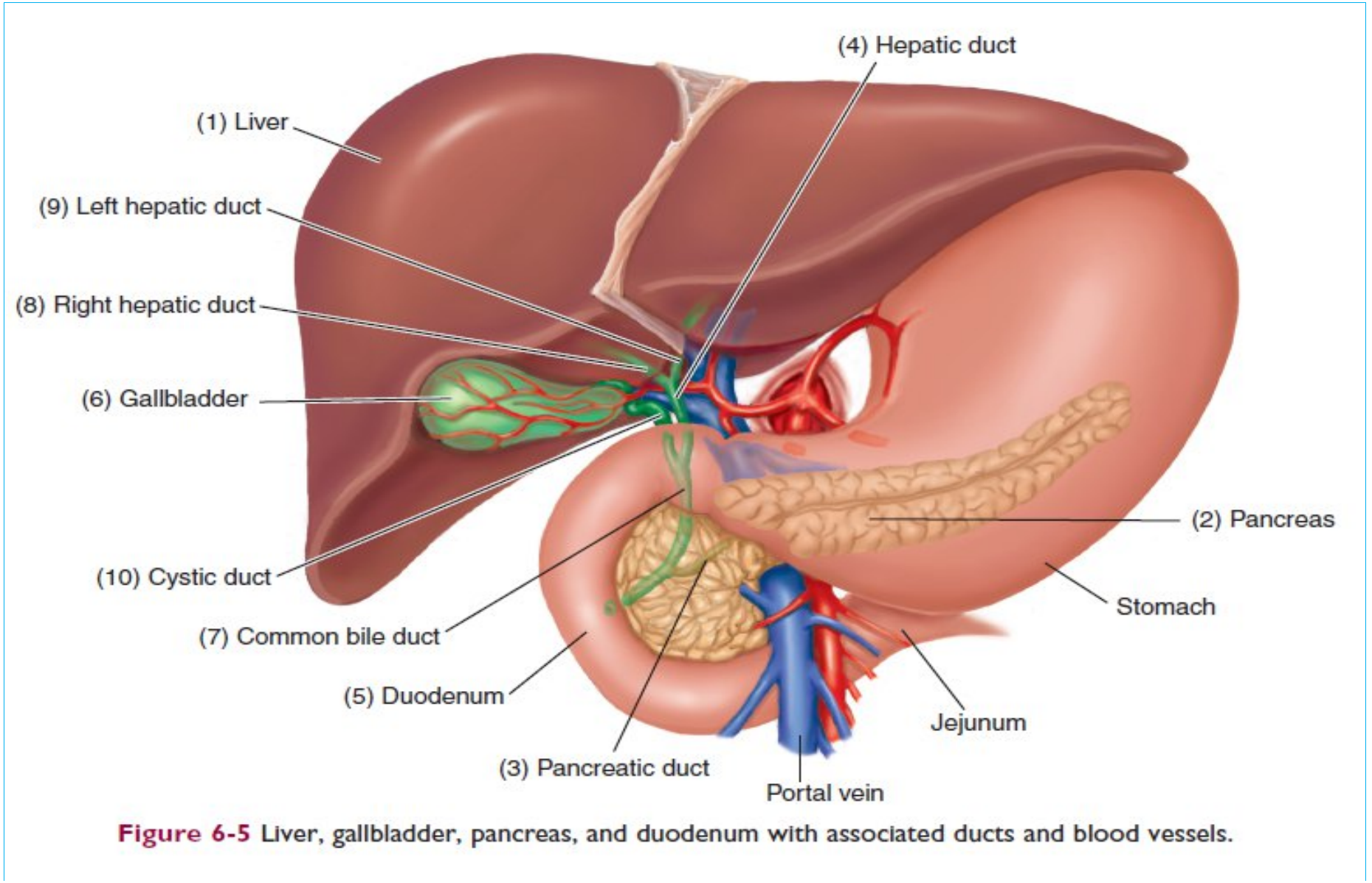


Figure 6-5 Liver, gallbladder, pancreas, and duodenum with associated ducts and blood vessels.

Anatomy and Physiology (Cont.)

death occurs if it ceases to function. Some of its important functions include the following:

- Producing bile, which aids in the digestion of fat
- Removing glucose (sugar) from the blood to synthesize glycogen (starch) and retain it for later use• Storing vitamins, such as B12, A, D, E, and K
- Destroying or transforming toxic products into less harmful compounds
- Maintaining normal glucose levels in the blood
- Destroying old erythrocytes and releasing bilirubin
- Synthesizing proteins that circulate in the blood, such as albumin for fluid balance and prothrombin and fibrinogen for coagulation (blood clotting)

Anatomy and Physiology (Cont.)

2. Pancreas

The (2) pancreas is an elongated, somewhat flattened organ that lies posterior and slightly inferior to the stomach. It performs endocrine and exocrine functions. As an endocrine gland, the pancreas secretes insulin directly into the bloodstream to maintain normal blood glucose levels. As an exocrine gland, the pancreas produces digestive enzymes that pass into the duodenum through the (3) pancreatic duct. The pancreatic duct extends along the pancreas and, together with the (4) hepatic duct from the liver, enters the (5) duodenum.

3. Gallbladder

The (6) gallbladder, a saclike structure on the inferior surface of the liver, serves as a storage area for bile, which is produced by the liver. When bile is needed for digestion, the gallbladder releases it into the duodenum through the (7) common bile duct.

Anatomy and Physiology (Cont.)

Bile is also drained from the liver through the (8) right hepatic duct and the (9) left hepatic duct. These two structures eventually form the hepatic duct. The (10) cystic duct of the gallbladder merges with the hepatic duct to form the common bile duct, which leads into the duodenum. Without bile, fat digestion is not possible.

Medical Word Elements

This section introduces combining forms, suffixes, and prefixes related to the digestive system. Word analyses are also provided. From the information provided, complete the meaning of the medical words in the right-hand column. The first one is completed for you.

Element	Meaning	Word Analysis and Meaning
Combining Forms		
Mouth		
or/o	mouth	**or**/al (OR-ăl): *pertaining to the mouth* *-al:* pertaining to
stomat/o		**stomat**/itis (stŏ-mă-TĪ-tĭs): _____ *-itis:* inflammation
gloss/o	tongue	**gloss**/ectomy (glŏs-ĔK-tō-mē): _____ *-ectomy:* excision, removal
lingu/o		**lingu**/al (LĬN-gwăl): _____ *-al:* pertaining to
bucc/o	cheek	**bucc**/al (BŬK-ăl): _____ *-al:* pertaining to

Medical Word Elements—cont'd

Element	Meaning	Word Analysis and Meaning
cheil/o	lip	**cheil/o**/plasty (KĪ-lō-plăs-tē): _____ *–plasty:* surgical repair
labi/o		**labi**/al (LĀ-bē-ăl): _____ *–al:* pertaining to
dent/o	teeth	**dent**/ist (DĚN-tĭst): _____ *–ist:* specialist
odont/o		orth/**odont**/ist (or-thō-DŎN-tĭst): _____ *orth:* straight *–ist:* specialist *Orthodontists are dentists who specialize in correcting and preventing irregularities of abnormally aligned teeth.*
gingiv/o	gum(s)	**gingiv**/ectomy (jĭn-jĭ-VĚK-tō-mē): _____ *–ectomy:* excision, removal *Gingivectomy is a surgical treatment for periodontal disease.*
sial/o	saliva, salivary gland	**sial/o**/lith (sī-ĂL-ō-lĭth): _____ *–lith:* stone, calculus

Medical Word Elements—cont'd

Element	Meaning	Word Analysis and Meaning
Esophagus, Pharynx, and Stomach		
esophag/o	esophagus	**esophag/o**/scope (ē-SŎF-ă-gō-skōp): _____ *-scope:* instrument for examining
pharyng/o	pharynx (throat)	**pharyng/o**/tonsill/itis (fă-rĭng-gō-tŏn-sĭ-LĪ-tĭs): _____ *tonsill:* tonsils *-itis:* inflammation
gastr/o	stomach	**gastr**/algia (găs-TRĂL-jē-ă): _____ *-algia:* pain *Gastralgia is also called stomachache.*
pylor/o	pylorus	**pylor/o**/spasm (pī-LOR-ō-spăzm): _____ *-spasm:* involuntary contraction, twitching *Pylorospasm occurs in the pyloric sphincter of the stomach.*

Medical Word Elements—cont'd

Element	Meaning	Word Analysis and Meaning
Small Intestine		
duoden/o	duodenum (first part of small intestine)	**duoden/o**/scopy (dū-ŏd-ĕ-NŎS-kō-pē): _____ *-scopy:* visual examination
enter/o	intestine (usually small intestine)	**enter/o**/pathy (ĕn-tĕr-ŎP-ă-thē): _____ *-pathy:* disease *Enteropathy typically occurs in the small intestine.*
jejun/o	jejunum (second part of small intestine)	**jejun/o**/rrhaphy (jĕ-joo-NOR-ă-fē): _____ *-rrhaphy:* suture
ile/o	ileum (third part of small intestine)	**ile/o**/stomy (ĭl-ē-ŎS-tō-mē): _____ *-stomy*:* forming an opening (mouth) *An ileostomy creates an opening on the surface of the abdomen to allow feces to be discharged into a bag worn on the abdomen.*

Medical Word Elements—cont'd

Element	Meaning	Word Analysis and Meaning
Large Intestine		
append/o	appendix	**append**/ectomy (ăp-ĕn-DĔK-tō-mē): _____ *-ectomy:* excision, removal *An appendectomy removes a diseased appendix that is in danger of rupturing.*
appendic/o		**appendic**/itis (ă-pĕn-dĭ-SĪ-tĭs): _____ *-itis:* inflammation
col/o	colon	**col/o**/stomy (kō-LŎS-tō-mē): _____ *-stomy:** forming an opening (mouth) *A colostomy creates a place for fecal matter to exit the body other than through the anus.*
colon/o		**colon/o**/scopy (kō-lŏn-ŎS-kō-pē): _____ *-scopy:* visual examination *Colonoscopy is performed with an elongated flexible endoscope called a colonoscope.*
sigmoid/o	sigmoid colon	**sigmoid/o**/tomy (sĭg-moyd-ŎT-ō-mē): _____ *-tomy:* incision

Medical Word Elements—cont'd

Element	Meaning	Word Analysis and Meaning
Terminal End of Large Intestine		
rect/o	rectum	**rect/o**/cele (RĚK-tŏ-sēl): _____ *-cele:* hernia, swelling *Rectocele is also known as* proctocele.
proct/o	anus, rectum	**proct/o**/logist (prŏk-TŎL-ō-jĭst): _____ *-logist:* specialist in the study of *A proctologist is a physician who specializes in treating disorders of the colon, rectum, and anus.*
an/o	anus	peri/**an**/al (pĕr-ē-Ā-năl): _____ *peri-:* around *-al:* pertaining to

Medical Word Elements—cont'd

Element	Meaning	Word Analysis and Meaning
cholecyst/o	gallbladder	**cholecyst**/ectomy (kō-lē-sĭs-TĔK-tō-mē): _____ *-ectomy:* excision, removal *Cholecystectomy is performed by laparoscopic or open surgery.*
choledoch/o	bile duct	**choledoch/o**/plasty (kō-LĔD-ō-kō-plăs-tē): _____ *-plasty:* surgical repair
Suffixes		
-emesis	vomit	hyper/**emesis** (hī-pĕr-ĔM-ĕ-sĭs): _____ *hyper-:* excessive, above normal
-iasis	abnormal condition (produced by something specified)	chol/e/lith/**iasis** (kō-lē-lĭ-THĪ-ă-sĭs): _____ *chol/e:* bile, gall *lith:* stone, calculus *When gallstones form in the common bile duct, the condition is called* **choledocholithiasis.**

Medical Word Elements—cont'd

Element	Meaning	Word Analysis and Meaning
Suffixes		
-megaly	enlargement	hepat/o/**megaly** (hĕp-ă-tō-MĔG-ă-lē): _____ *hepat/o:* liver *Hepatomegaly may be caused by hepatitis or infection, fatty infiltration (as in alcoholism), biliary obstruction, or malignancy.*
-orexia	appetite	an/**orexia** (ăn-ō-RĔK-sē-ă): _____ *an-:* without, not *Anorexia can result from various conditions, such as adverse effects of drugs or various physical or psychological causes.*
-pepsia	digestion	dys/**pepsia** (dĭs-PĔP-sē-ă): _____ *dys-:* bad; painful; difficult *Dyspepsia, also called* **indigestion**, *is an epigastric discomfort felt after eating.*
-phagia	swallowing, eating	aer/o/**phagia** (ĕr-ō-FĀ-jē-ă): _____ *aer/o:* air
-prandial	meal	post/**prandial** (pōst-PRĂN-dē-ăl): _____ *post-:* after, behind
-rrhea	discharge, flow	steat/o/**rrhea** (stē-ă-tō-RĒ-ă): _____ *steat/o:* fat

Medical Word Elements—cont'd

Element	Meaning	Word Analysis and Meaning
Prefixes		
dia-	through, across	**dia**/rrhea (dī-ă-RĒ-ă): _____ *-rrhea:* discharge, flow *Diarrhea is a discharge or flow of fluid fecal matter through the bowel.*
peri-	around	**peri**/odont/itis (pĕr-ē-ō-dŏn-TĪ-tĭs): _____ *odont-:* tooth *-itis:* inflammation
sub-	under, below	**sub**/lingu/al (sŭb-LĬN-gwăl): _____ *lingu:* tongue *-al:* pertaining to

It is time to review medical word elements by completing Learning Activities 6-1, 6-2, and 6-3.

Learning Activities

The activities that follow provide a review of the a review of the cardiovascular system terms introduced in this chapter.

Complete each activity from 6-1, 6-2, and 6-3 and review your answers to evaluate your understandir of chapter 10.

